Deployment of 3-Tier Applications Using AWS Services

A Traditional Cloud Deployment for a Web Application with Database

Backend
Domain: Cloud Computing and DevOps

Project Overview:
This project demonstrates a manually deployed 3-tier architecture on AWS for a web application where users can submit data, which is then stored in a backend database. The setup does not use DevOps tools (like Jenkins, Docker, or Kubernetes) and relies on AWS native services for hosting, networking, and storage.

GitHub URL:
https://github.com/Deepanshu-Sehgal/3TierArchitectureApp.git

Key Components:
- Presentation Tier (Frontend) – Hosted on AWS S3 (Static Website).
- Application Tier (Backend) – Runs on AWS EC2 (Web Server).
- Database Tier – Managed by AWS RDS (MySQL/PostgreSQL).
- Networking & Security – VPC, IAM, Route 53 for access control and DNS.

Business Requirements:
Users should be able to submit data via a web form.
Data should be stored securely in a backend database.
The application should be accessible via a custom domain (using Route 53).Security best practices should be followed (IAM roles, VPC isolation).

1. Presentation Tier (Frontend - S3):
    a. A static website (HTML, CSS, JavaScript) hosted on AWS S3.
    b. Users interact with a web form that sends data to the EC2 backend.
    c. S3 is configured for static web hosting with public read access.

2. Application Tier (Backend - EC2):
    a. An EC2 instance running a web server (Apache/Nginx) with a backend application (Node.js/Python/Java).
    b. The application processes user requests and stores data in RDS.
    c. Security Groups restrict traffic only from S3 (frontend) and RDS (database).
3. Database Tier (RDS):
    a. A managed RDS instance (MySQL/PostgreSQL) for storing user data.
    b. Deployed in a private subnet for security.
    c. Accessible only by the EC2 application server.
4. Networking & Security (VPC, IAM, Route 53):
    a. VPC Setup:
        i. Public Subnet (EC2, S3)
        ii. Private Subnet (RDS)
    b. IAM Roles:
        i. EC2 instance has permissions to access RDS.
        ii. S3 has restricted access policies.
    c. Route 53:
        i. Custom domain name

Step-by-Step Manual Deployment:

1. Set Up VPC & Subnets:
    a. Create a VPC with:
    b. Public Subnet (for EC2 & S3)
    c. Private Subnet (for RDS)
    d. Configure Internet Gateway for public access.
    e. Set up NAT Gateway (if needed for private subnet outbound traffic).
2. Deploy S3 Static Website (Frontend):
    a. Create an S3 bucket with static website hosting enabled.
    b. Upload HTML, CSS, and JavaScript files.
    c. Set bucket policy for public read access.
3. Launch EC2 Instance (Backend):

    a. Choose an Amazon Linux/Ubuntu AMI.
    b. Install web server (Apache/Nginx) and backend runtime (Node.js/Python).
    c. Configure Security Group to allow HTTP/HTTPS traffic from S3.
    d. Assign an IAM role with RDS access permissions.
4. Set Up RDS Database:
    a. Launch an RDS instance (MySQL/PostgreSQL) in the private subnet.
    b. Configure Security Group to allow only EC2 connections.
    c. Note down endpoint, username, and password.
5. Connect Frontend to Backend:
    a. Modify the frontend (S3 JavaScript) to send API requests to the EC2 backend.
    b. Example: Javascript:
        i. fetch ("http://<EC2-Public-IP>/submit-data", { method: "POST", body: JSON.stringify(data) });
6. Configure Route 53 (Custom Domain):
    a. Register a domain (or use an existing one).
    b. Create an A record pointing to the S3 static website endpoint.
    c. (Optional) Set up HTTPS using ACM (AWS Certificate Manager).
7. Test the Application:
    a. Users access the website via the Route 53 domain.
    b. Data entered in the form is sent to EC2, which stores it in RDS.
    c. Verify data persistence in the RDS database.

Project Overview

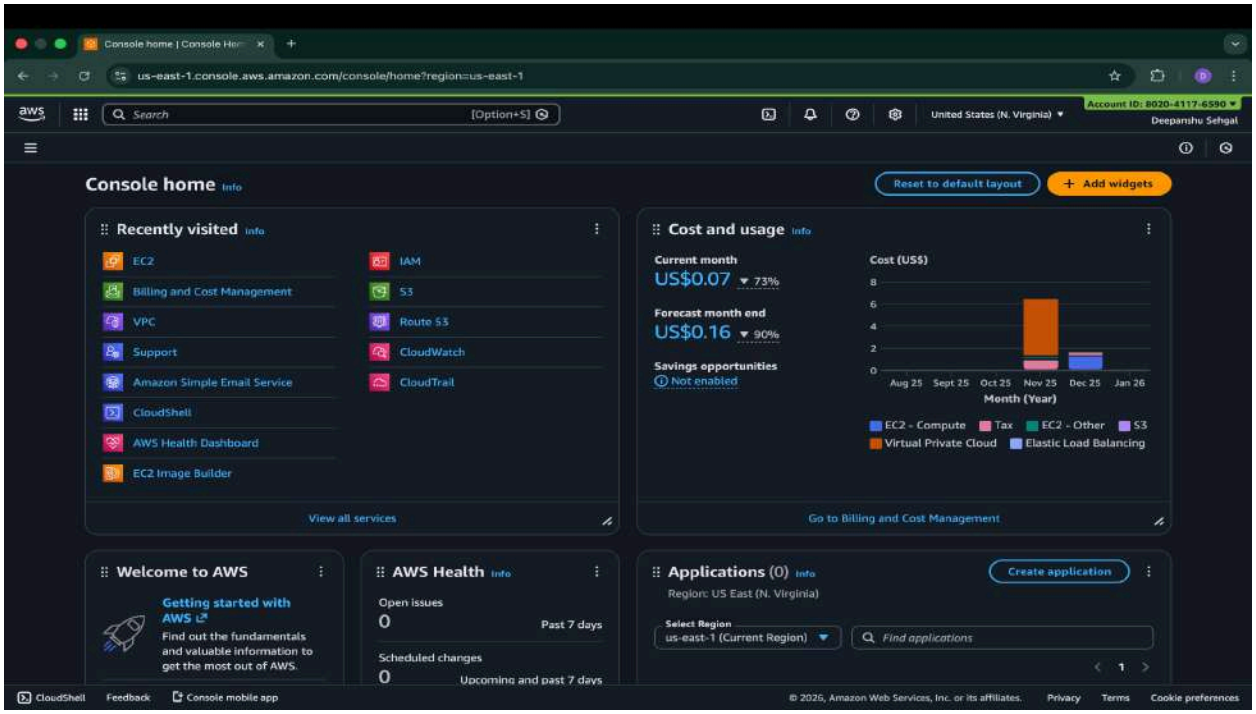
The goal of this project is to design and deploy a scalable and secure 3-tier architecture on AWS. The architecture separates the Web Tier, Application Tier, and Database Tier to improve security, scalability, and manageability.

1. **VPC Setup (Networking Foundation):** A dedicated Virtual Private Cloud (VPC) was created to serve as the foundation for the infrastructure.
    a. Key Configuration
        i. **VPC Name:** project-one-vpc
        ii. **CIDR Block:** 192.168.0.0/22
        iii. **IPv6:** Disabled
        iv. **Availability Zones:** 2

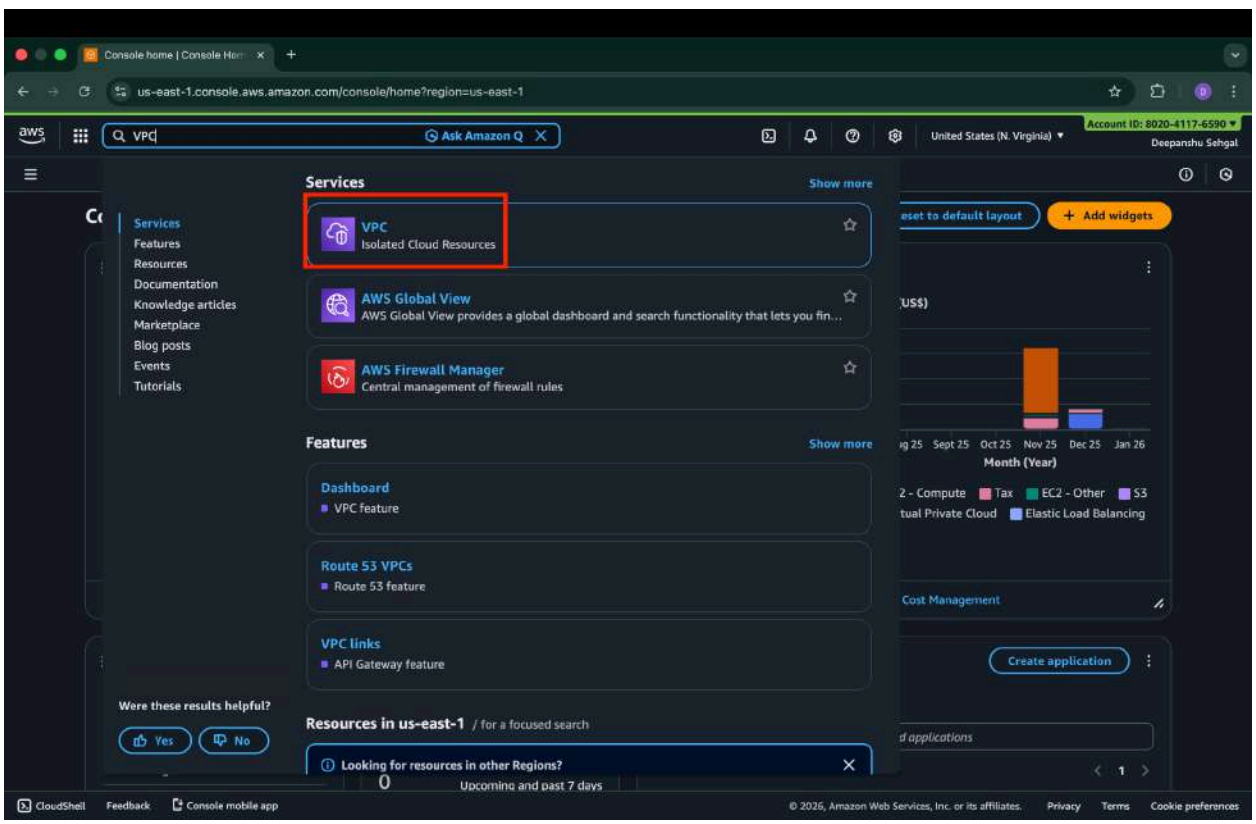
    b. Subnet Design
        i. **2 Public Subnets** → For internet-facing resources (Load Balancer)
        ii. **4 Private Subnets** → For application servers and databases

Networking Components

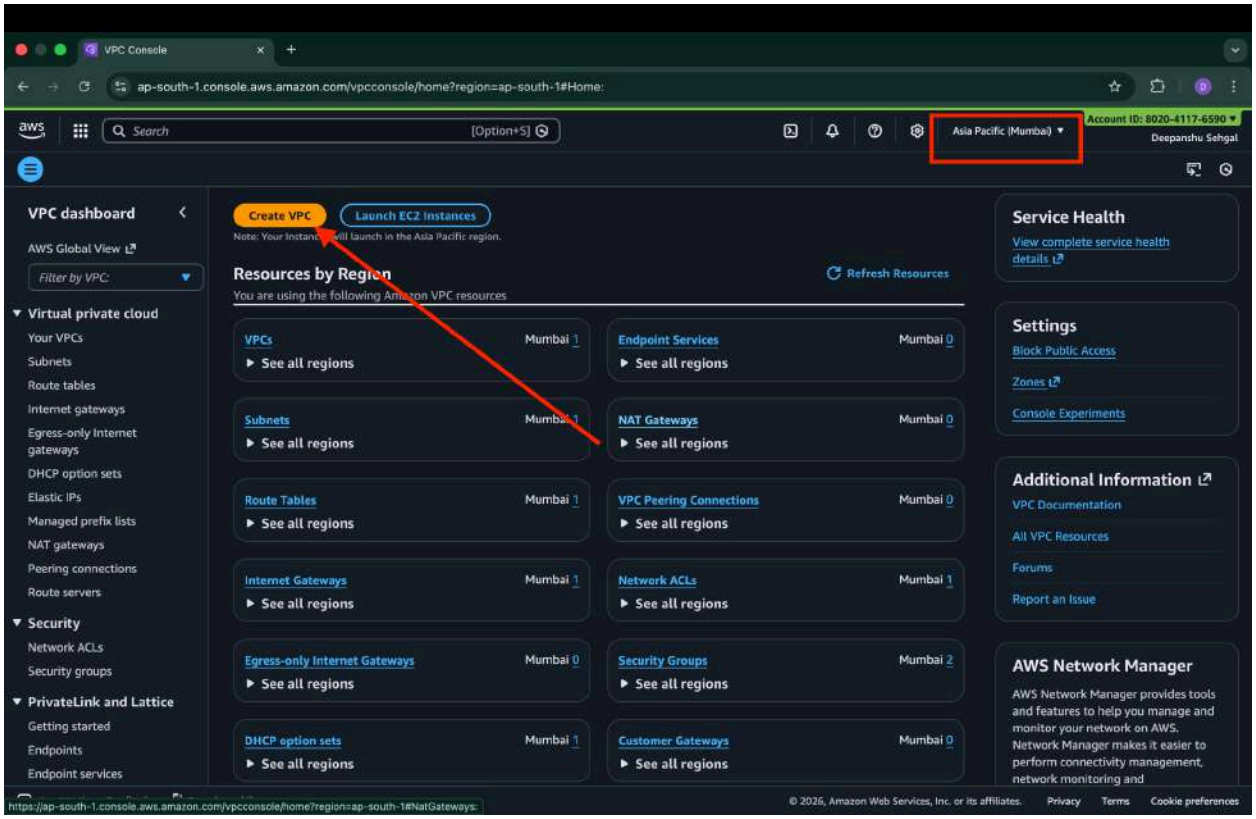
- Internet Gateway for public connectivity
- NAT Gateway for outbound access from private subnets
- Route tables configured for proper traffic routing



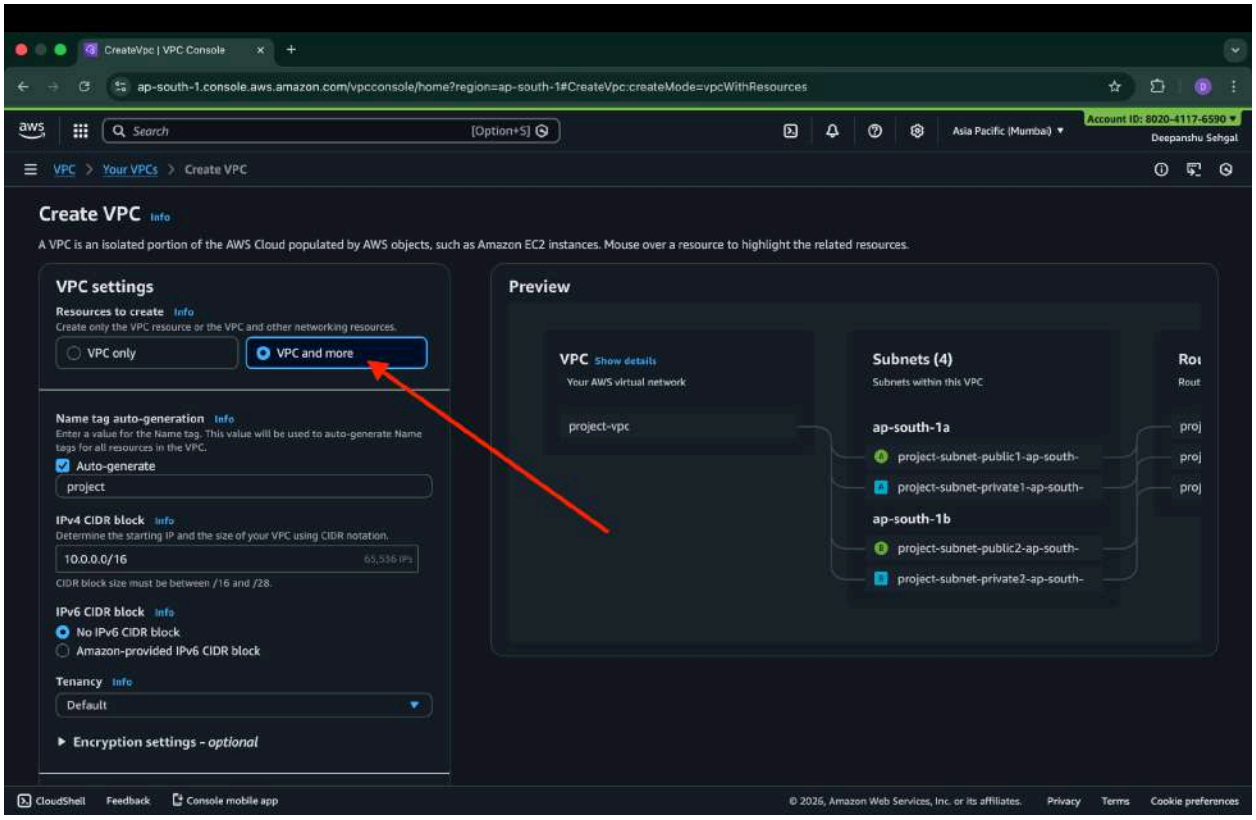
- Click on VPC to create VPC



- Click On Create VPC



- We will go with VPC and More to create everything like VPS, SUbnets, route tables, etc.



VPC Name: project-one-vpc
IPv4 CIDR Block: 192.168.0.0/22
IPv6: Not required (disabled)
Tenancy: Default
Number of Availability Zones: 2

VPC > Your VPCs > Create VPC

Name tag auto-generation Info

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☑ Auto-generate

project-one-vpc

IPv4 CIDR block Info

Determine the starting IP and the size of your VPC using CIDR notation.

192.168.0.0/22 1,024 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

◉ No IPv6 CIDR block

○ Amazon-provided IPv6 CIDR block

Tenancy Info

Default

▶ **Encryption settings - *optional***

Number of Availability Zones (AZs) Info

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

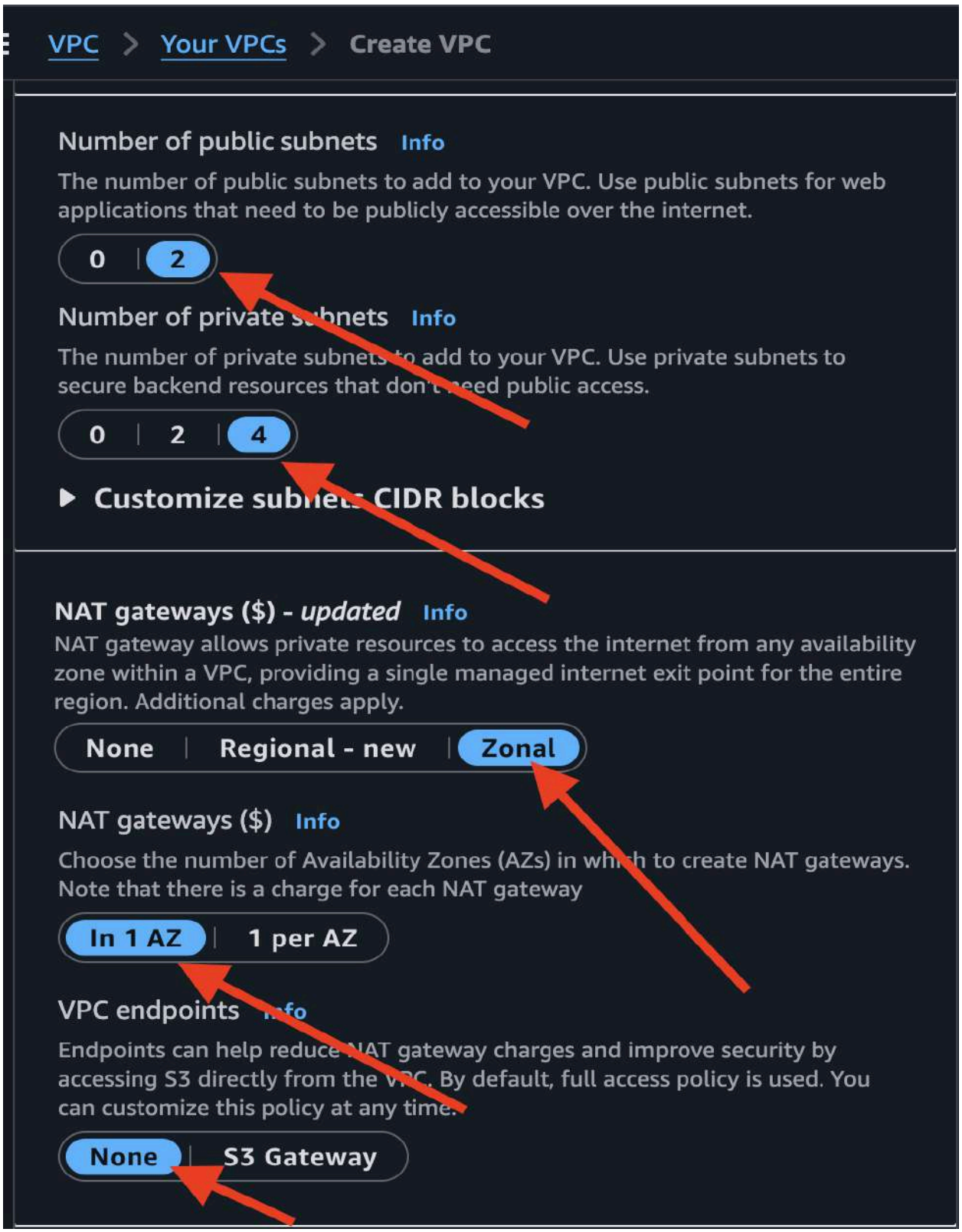
1 | 2 | 3

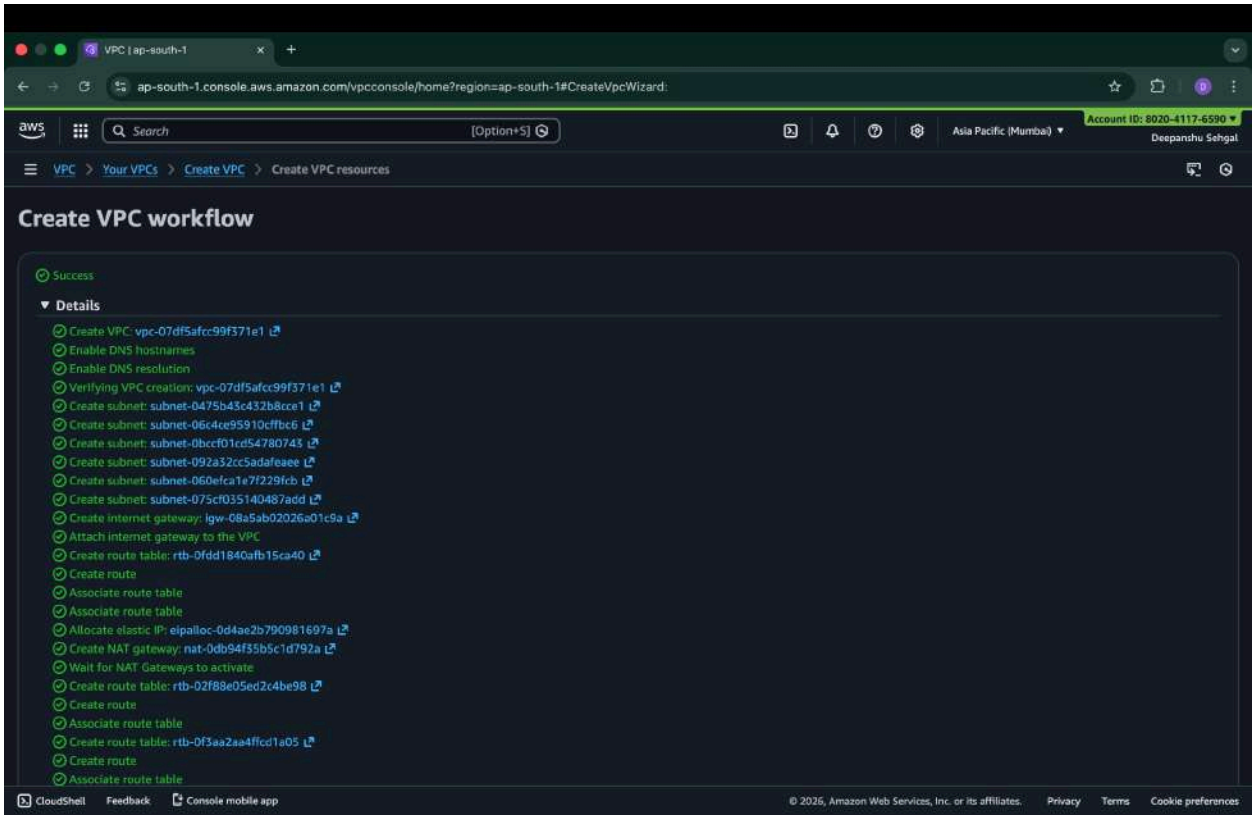
▶ **Customize AZs**

Number of public subnets Info

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

Public Subnets: 2 (for Internet Gateway and public-facing resources)
Private Subnets: 4 (for Application Tier and Database layer)




VPC | ap-south-1
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateVpcWizard:
Account ID: 8020-4117-6590
Deepanshu Sehgal
Asia Pacific (Mumbai)
VPC > Your VPCs > Create VPC > Create VPC resources
Create VPC workflow
Success
Details
Create VPC: vpc-07df5afcc99f371e1
Enable DNS hostnames
Enable DNS resolution
Verifying VPC creation: vpc-07df5afcc99f371e1
Create subnet: subnet-0475b43c432b8cce1
Create subnet: subnet-06c4ce95910cffbc6
Create subnet: subnet-0bccf01cd54780743
Create subnet: subnet-092a32cc5adafeaee
Create subnet: subnet-060efca1e7f229fcb
Create subnet: subnet-075cf035140487add
Create internet gateway: igw-08a5ab02026a01c9a
Attach internet gateway to the VPC
Create route table: rtb-0fdd1840afb15ca40
Create route
Associate route table
Associate route table
Allocate elastic IP: eipalloc-0d4ae2b790981697a
Create NAT gateway: nat-0db94f35b5c1d792a
Wait for NAT Gateways to activate
Create route table: rtb-02f88e05ed2c4be98
Create route
Associate route table
Create route table: rtb-0f3aa2aa4ffcd1a05
Create route
Associate route table
CloudShell
Feedback
Console mobile app
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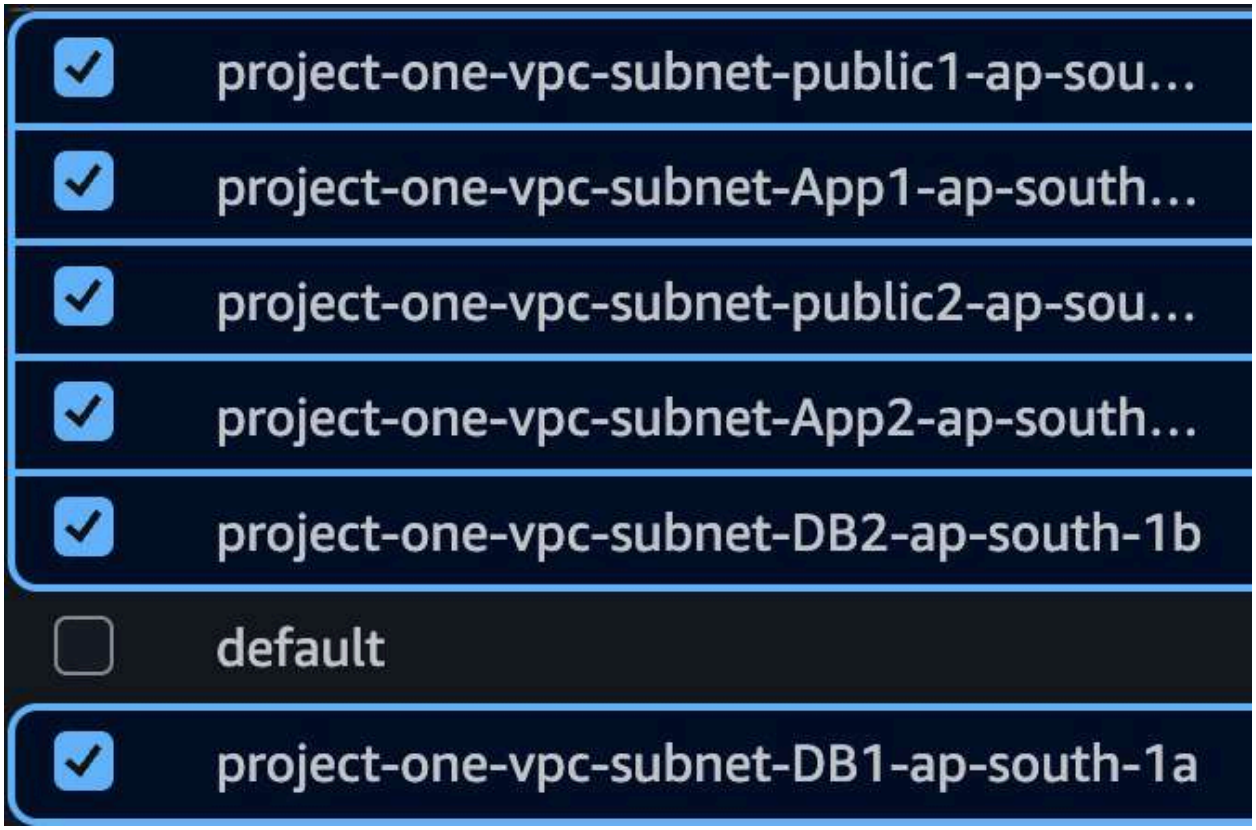
VPC > Subnets
VPC dashboard
Subnets (6/7) Info
Now we have 4 private subnets and 2 public subnets now change the name
Filter by VPC:
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Name
Subnet ID
project-one-vpc-subnet-public1-ap-sou...
subnet-0475b43c432b8cce1
project-one-vpc-subnet-private1-ap-so...
subnet-0bccf01cd54780743
project-one-vpc-subnet-public2-ap-sou...
subnet-06c4ce95910cffbc6
project-one-vpc-subnet-private2-ap-so...
subnet-092a32cc5adafeaee
project-one-vpc-subnet-private4-ap-so...
subnet-075cf035140487add
default
subnet-003ea9b1c8e5f7522
project-one-vpc-subnet-private3-ap-so...
subnet-060efca1e7f229fcb



2. **Security Groups Configuration:** Separate security groups were created for each layer to enforce strict access control.
    a. **Web ALB Security Group:** Allows HTTP/HTTPS traffic from the internet
    b. **Web Tier Security Group:** Accepts traffic only from the Load Balancer
    c. **Application Tier Security Group:** Accepts traffic only from the Web Tier
    d. **Database Security Group:** Database access restricted to the Application Tier only

This ensures layered security across the architecture.

VPC | ap-south-1

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

VPC > Security Groups > Create security group

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

Web-ALB-SG

Name cannot be edited after creation.

Description Info

Web-ALB-SG

VPC Info

vpc-07df5afcc99f371e1 (project-one-vpc)

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
HTTP	TCP	80	Anyw... 0.0.0.0/0		Delete

Add rule

⚠ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules Info

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info

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VPC > Security Groups > Create security group

Create security group Info

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Basic details

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Web-ALB-SG

Name cannot be edited after creation.

Description Info

Web-ALB-SG

VPC Info

vpc-07df5afcc99f371e1 (project-one-vpc)

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
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Add rule

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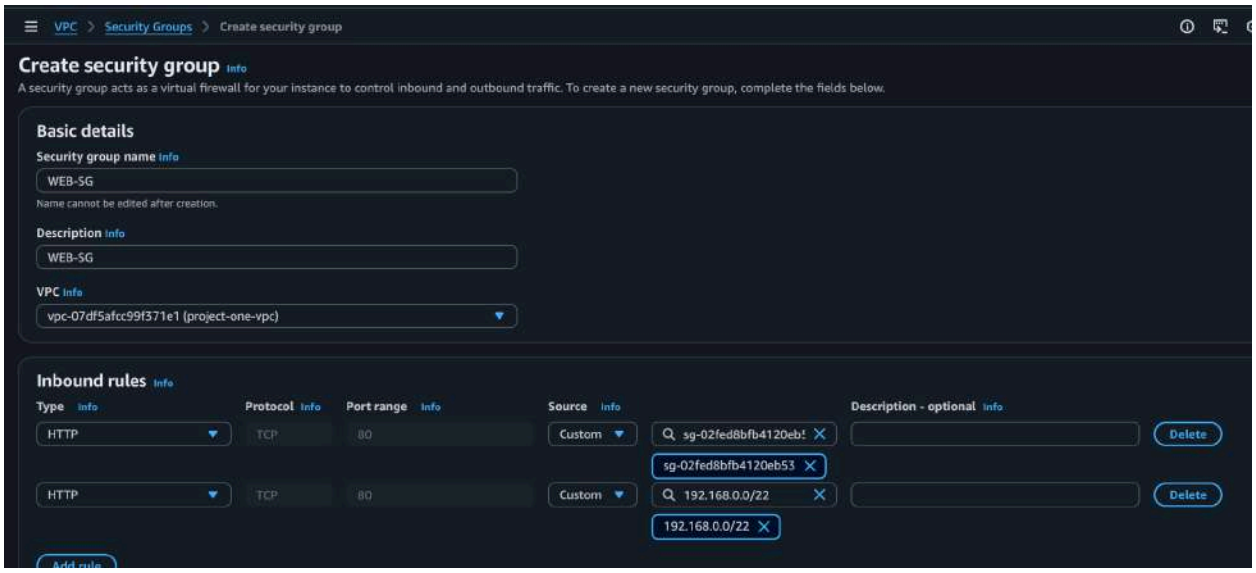
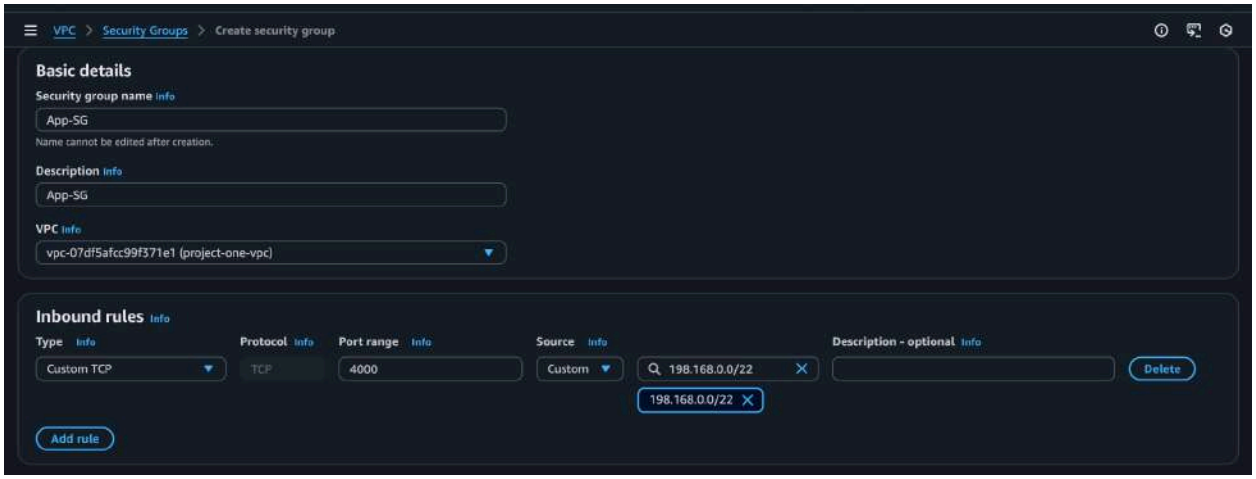
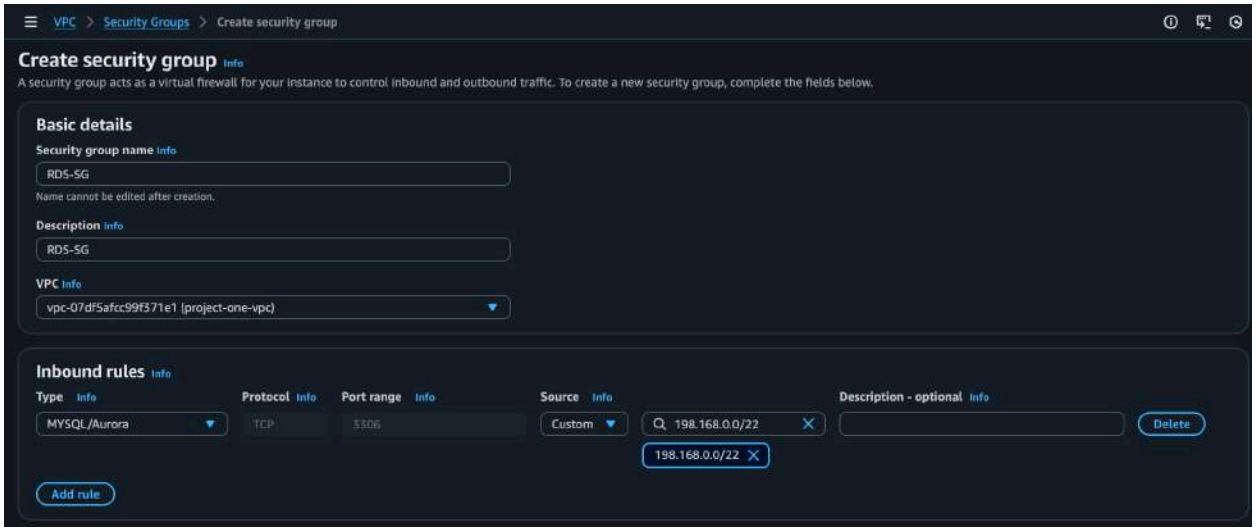
- **Create a Security Group for the Web Tier**

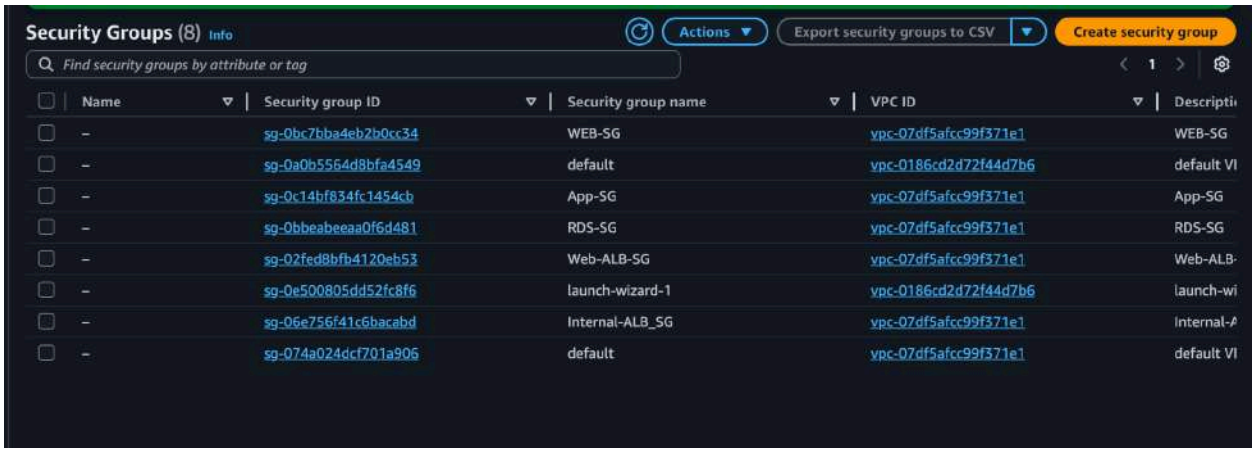


- **Create a Security Group for the Application Tier**

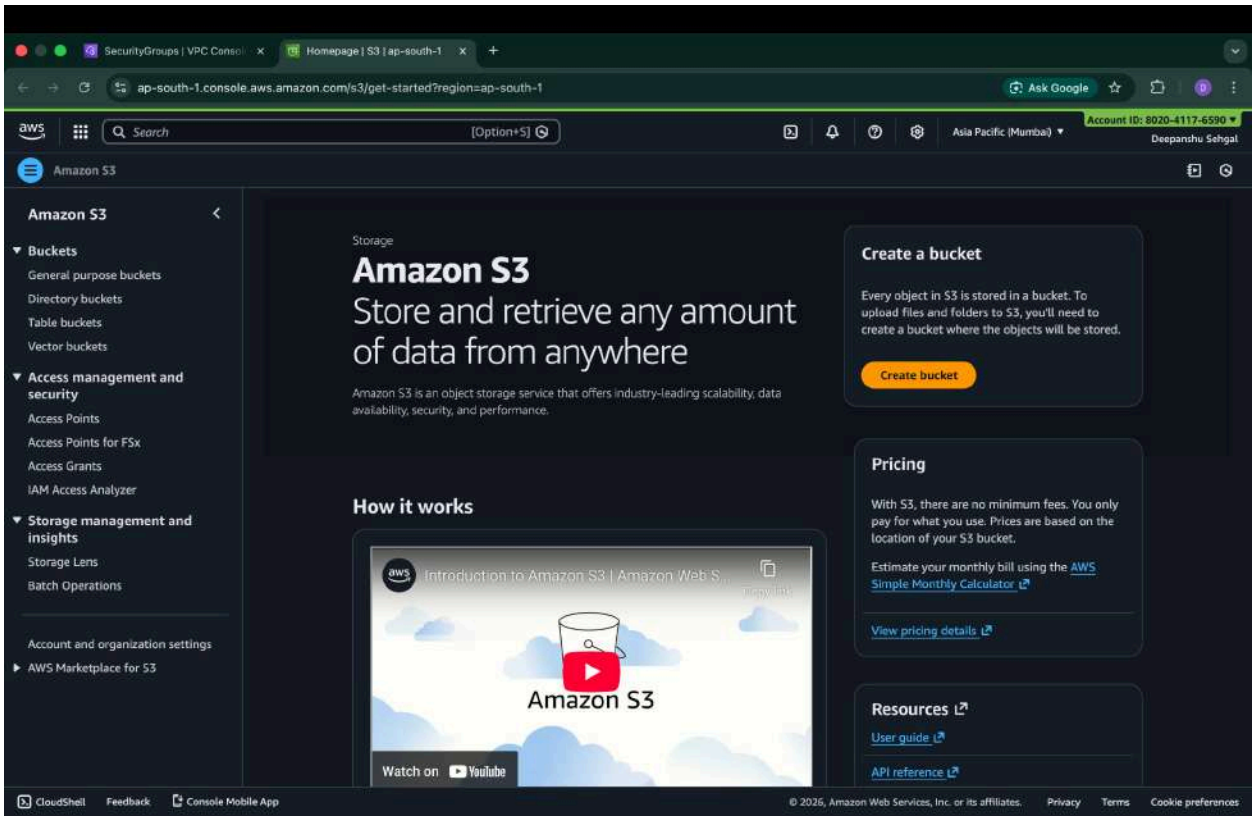


- **Create a Security Group for the Database Layer**





3. **S3 Bucket and IAM Role Setup**
    a. **S3 Bucket**
        i. Created to store application code
        ii. Project files uploaded for deployment
    b. **IAM Role**
        i. Created for EC2 instances
        ii. Attached policy: **AmazonEC2RoleforSSM**
        iii. Enables secure access without using SSH keys



SecurityGroups | VPC Consol × | Create S3 bucket | S3 | ap-s ×

ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1

Ask Google

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

Amazon S3 > Buckets > Create bucket

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Asia Pacific (Mumbai) ap-south-1

Bucket type | Info

- General purpose
  Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- Directory
  Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name | Info

demo-sai-3-tier-project

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more

Copy settings from existing bucket - *optional*

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

- ACLs disabled (recommended)
  All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
- ACLs enabled
  Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

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SecurityGroups | VPC Consol × | Create S3 bucket | S3 | ap-s ×

ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1

Ask Google

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

Amazon S3 > Buckets > Create bucket

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

- ACLs disabled (recommended)
  All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
- ACLs enabled
  Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

- [x] **Block *all* public access**
  Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
  - [x] **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
    S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
  - [x] **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
    S3 will ignore all ACLs that grant public access to buckets and objects.
  - [x] **Block public access to buckets and objects granted through *new* public bucket or access point policies**
    S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
  - [x] **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
    S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

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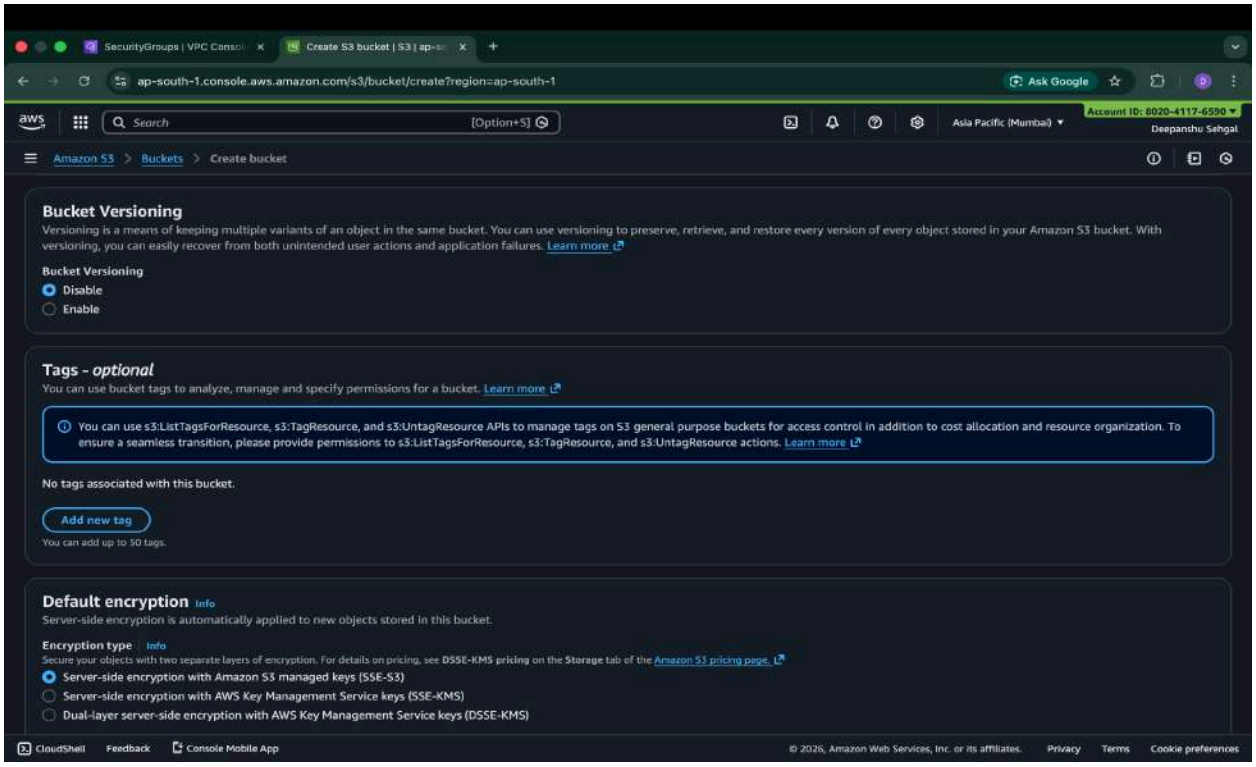
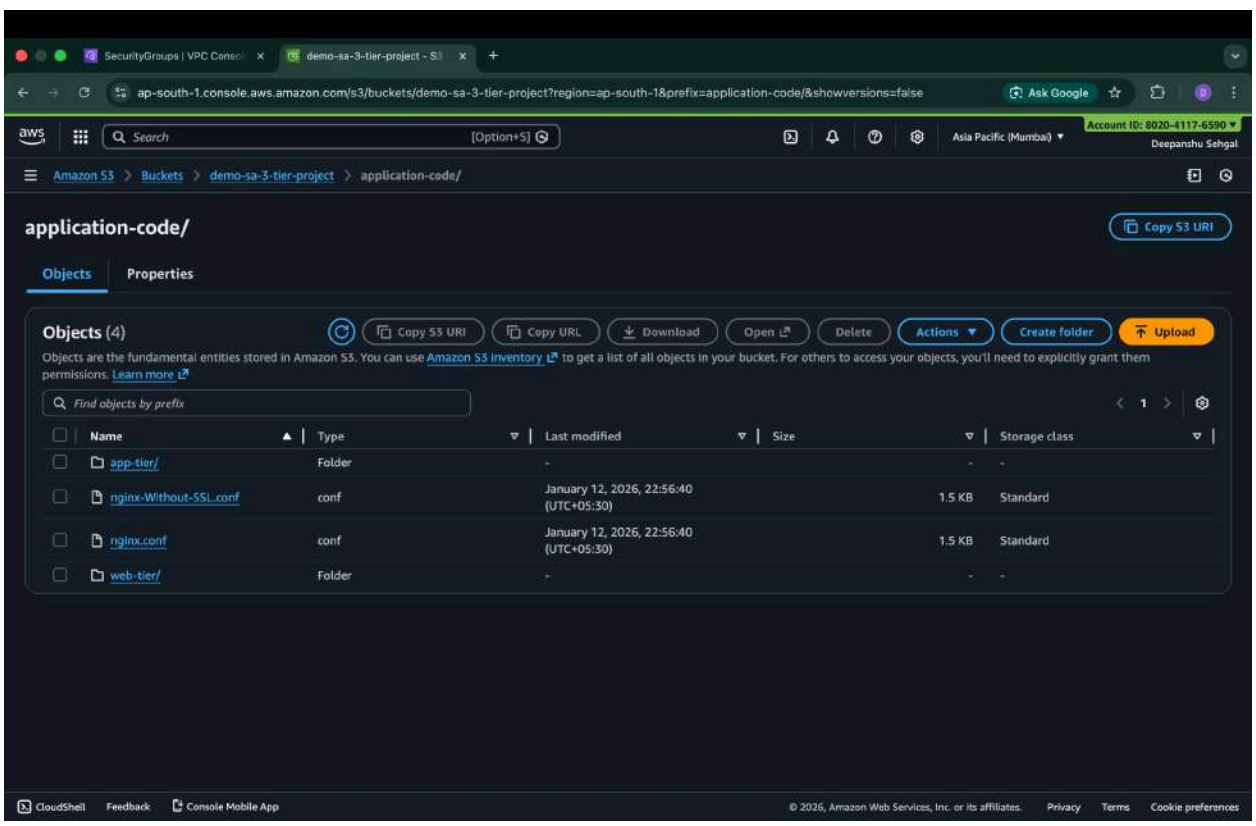
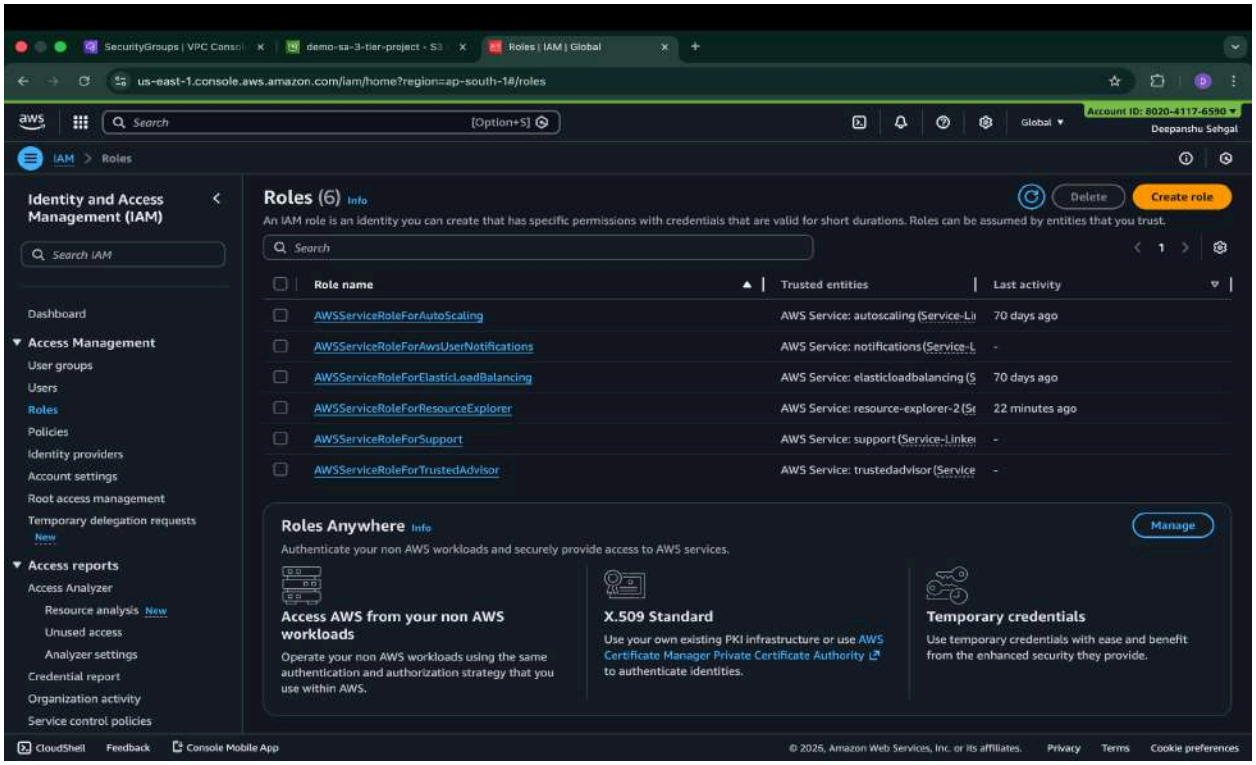
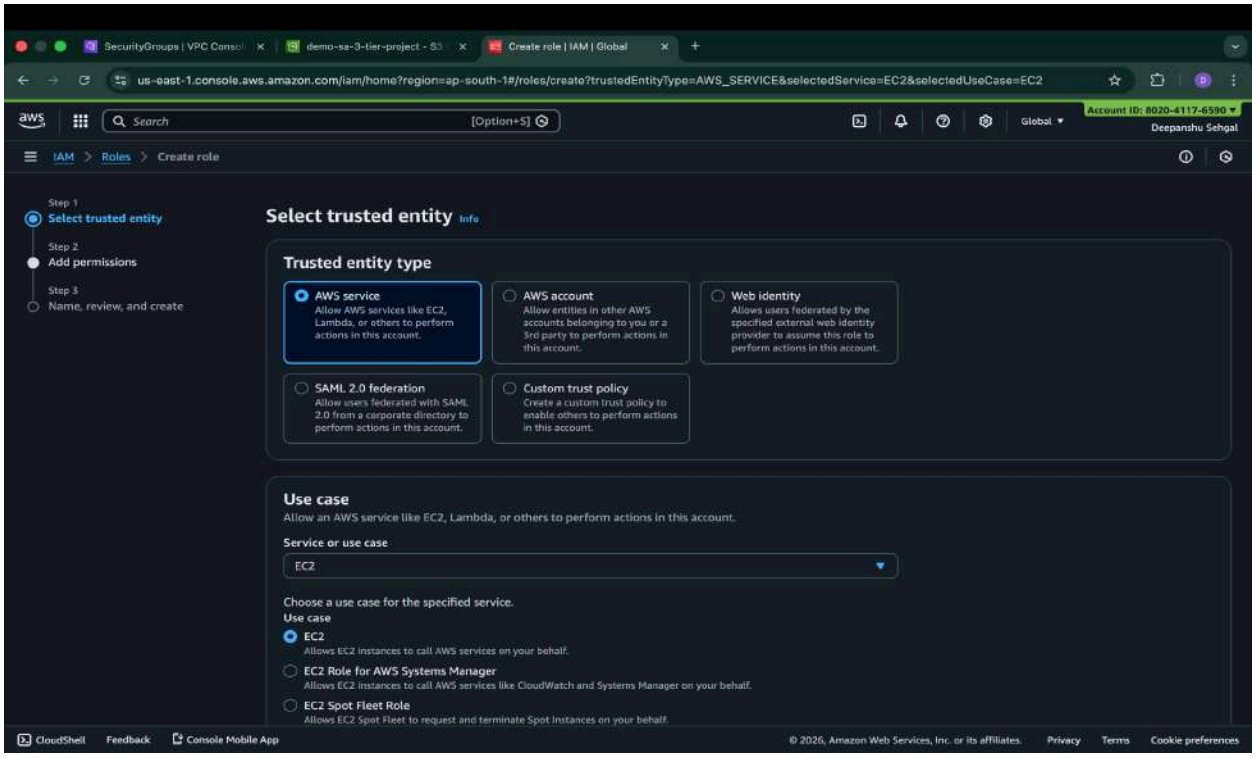


After creating the bucket, upload the project folder to the S3 bucket.

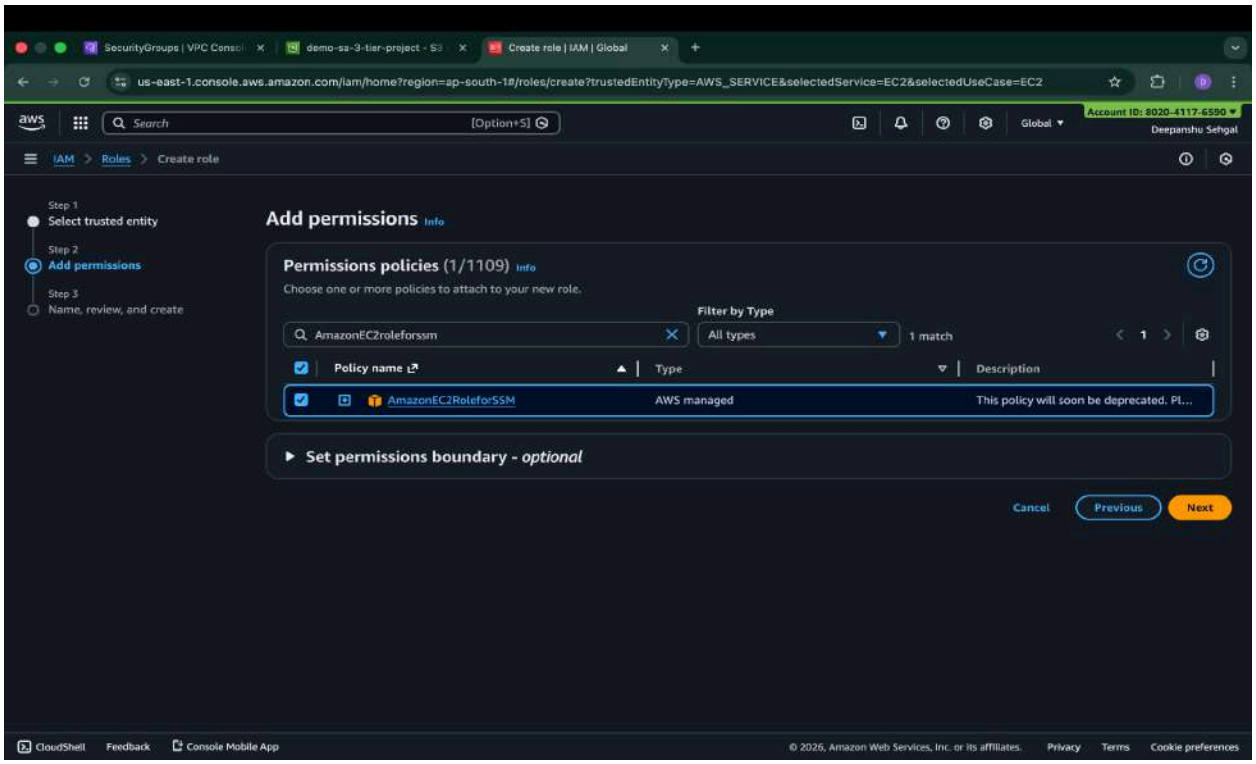


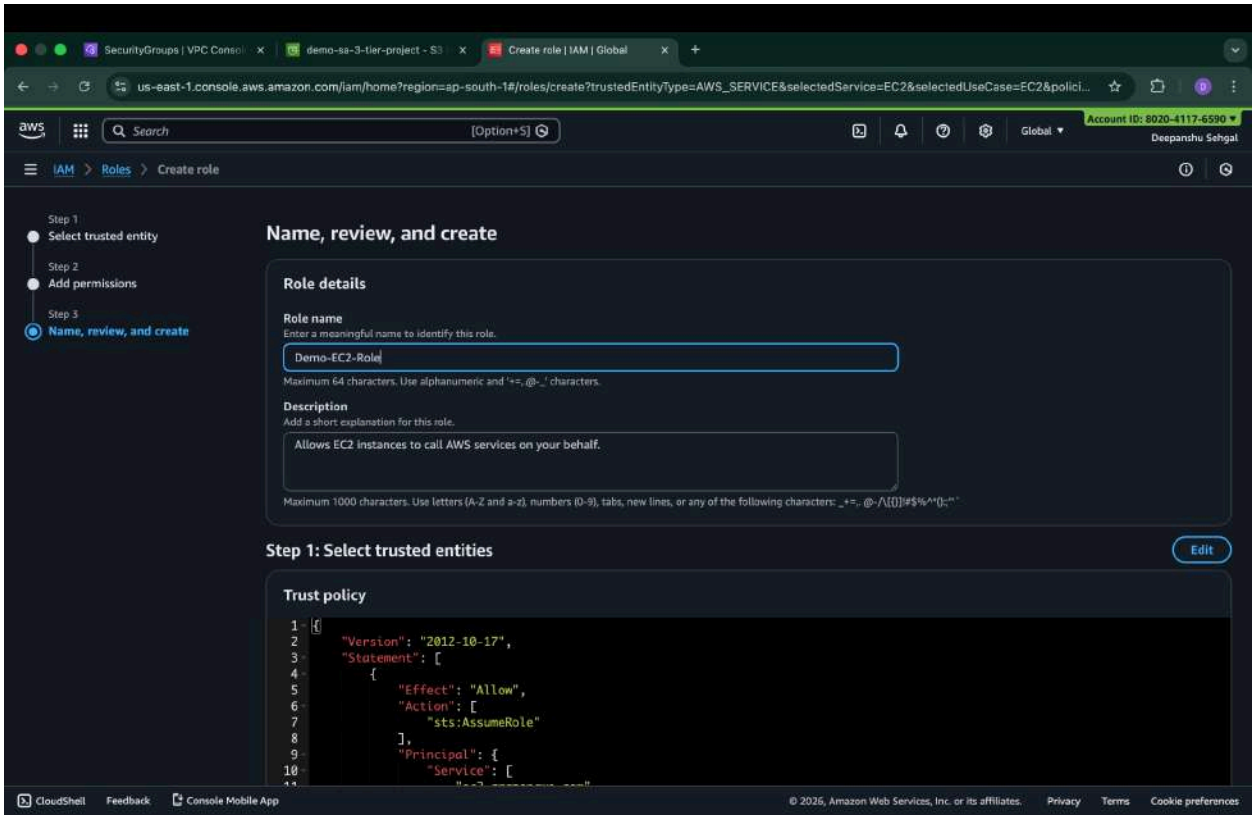
Create an IAM role for the EC2 instance.

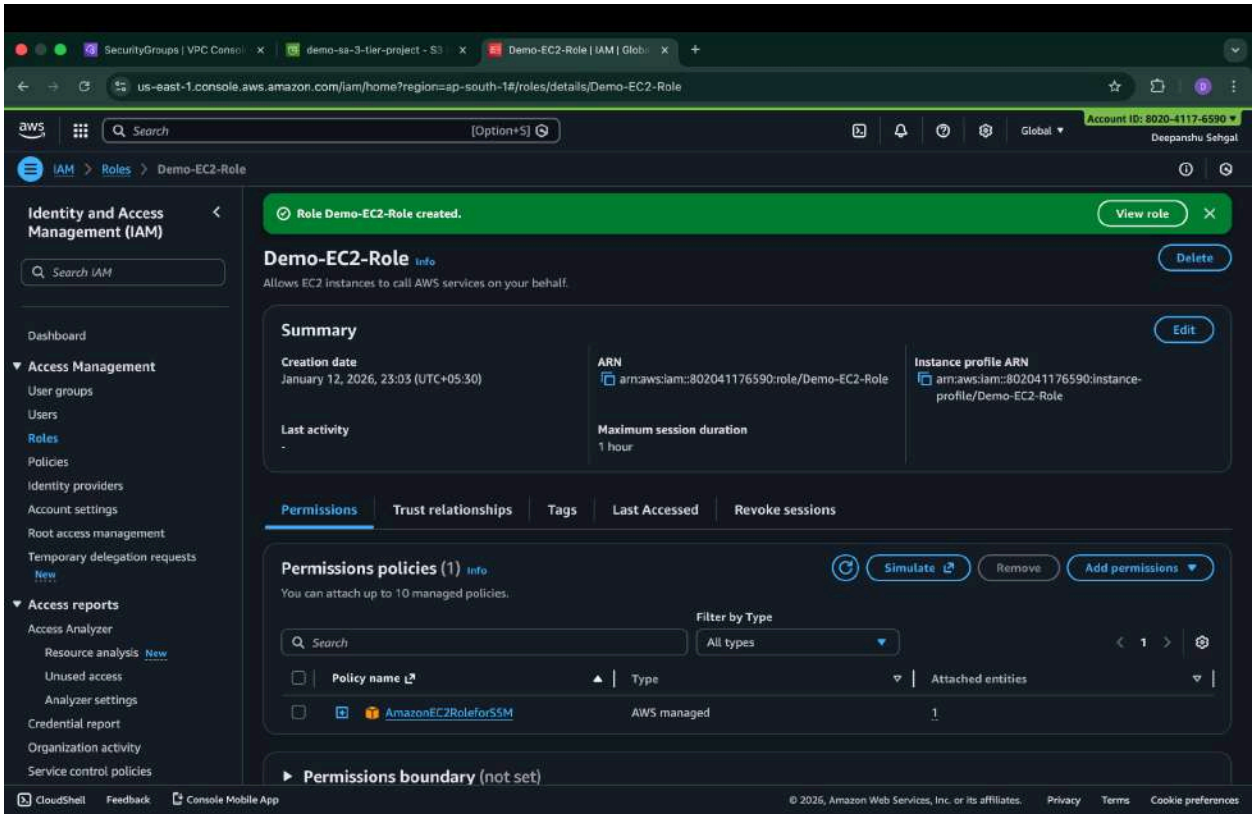




Attach the policy: AmazonEC2RoleforSSM.



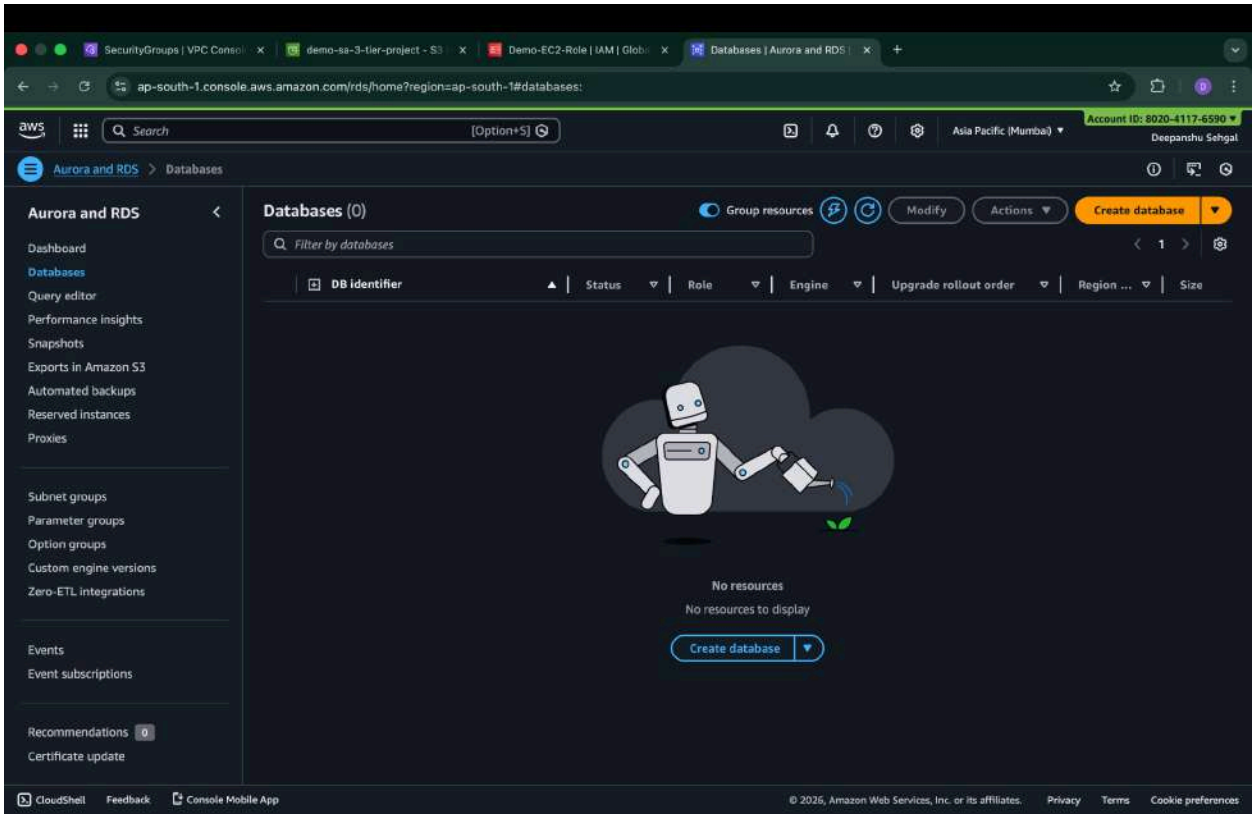




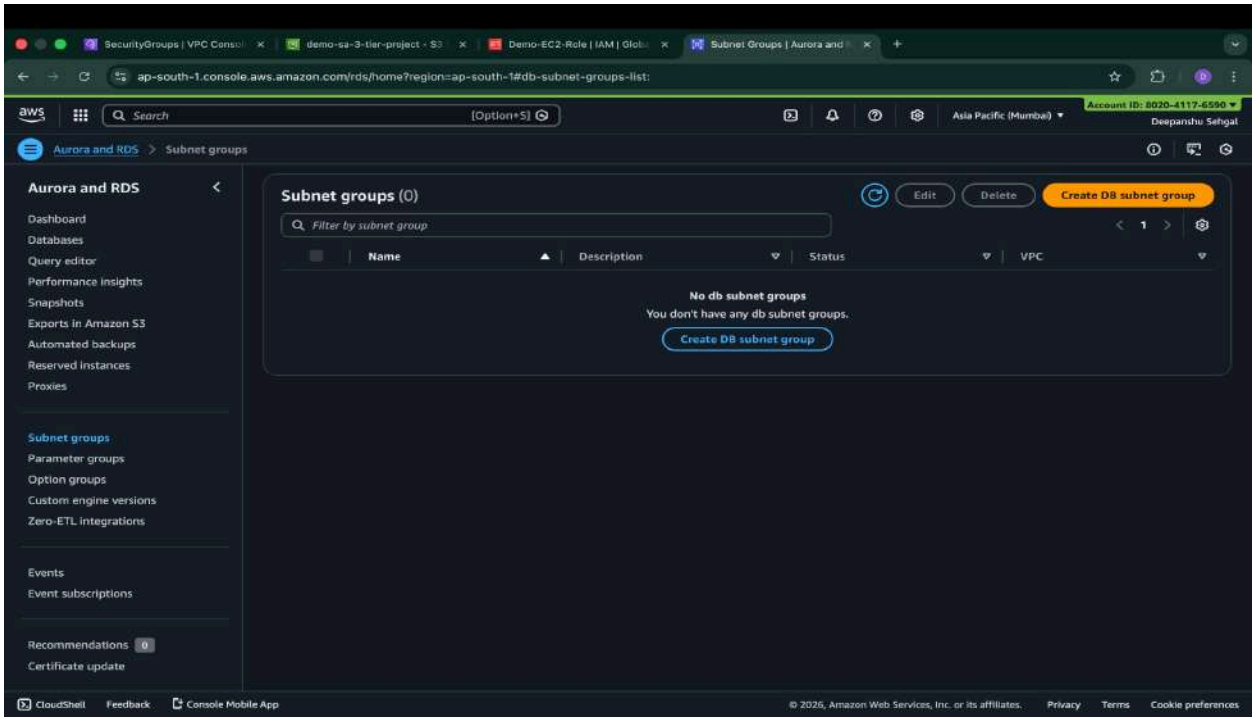
4. **Database Configuration (RDS):**An RDS instance was launched to serve as the backend database.

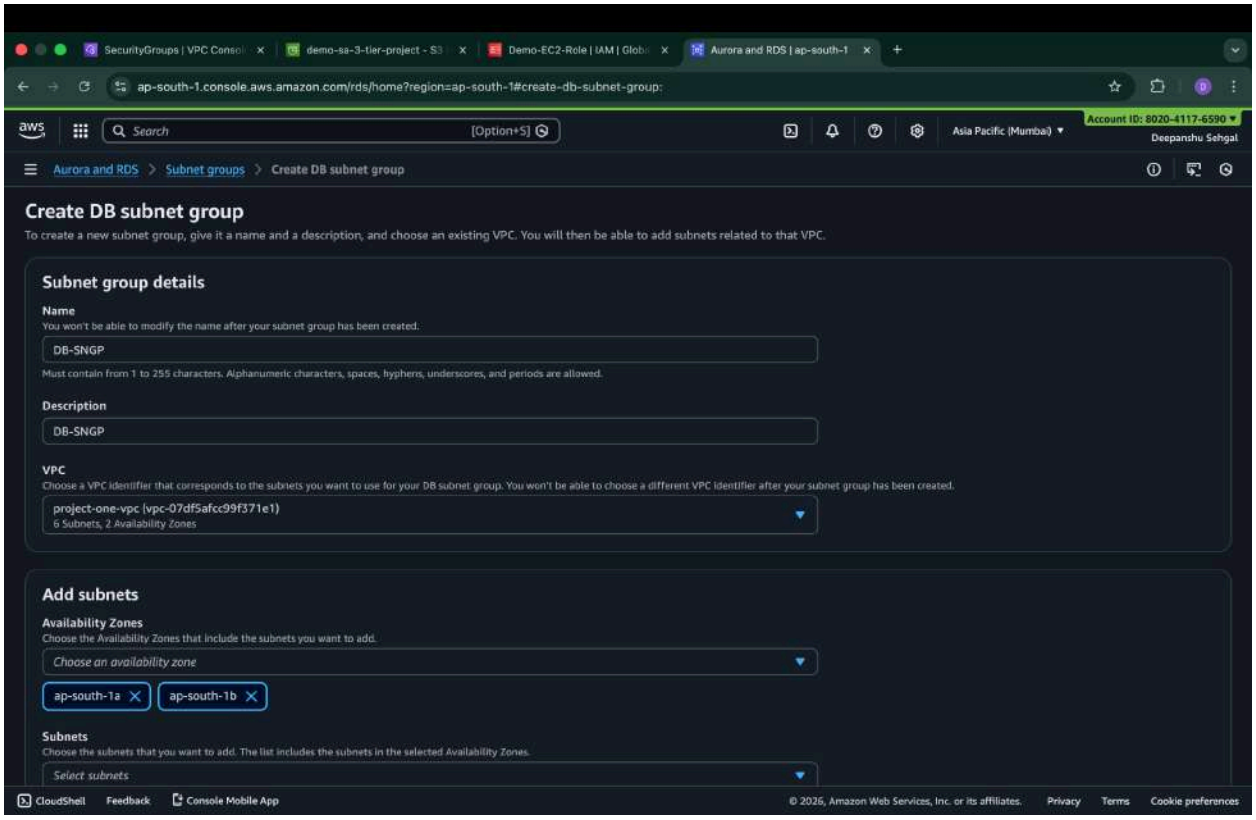
Steps

- Created a DB subnet group in ap-south-1a and ap-south-1b
- Launched an RDS instance using MySQL
- Created database: **webappdb**
- Created a transactions table
- Inserted dummy data for testing



- **Create a database subnet group across two Availability Zones: ap-south-1a and ap-south-1b.**




SecurityGroups | VPC Consol
demo-sa-3-tier-project - S3
Demo-EC2-Role | IAM | Glob
Aurora and RDS | ap-south-1
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#create-db-subnet-group:
Search
[Option+S]
Asia Pacific (Mumbai)
Account ID: 8020-4117-6590
Deepanshu Sehgal
Aurora and RDS > Subnet groups > Create DB subnet group
Create DB subnet group
To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.
Subnet group details
Name
You won't be able to modify the name after your subnet group has been created.
DB-SNGP
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.
Description
DB-SNGP
VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
project-one-vpc (vpc-07df5afcc99f371e1)
6 Subnets, 2 Availability Zones
Add subnets
Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
ap-south-1a
ap-south-1b
Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets
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Feedback
Console Mobile App
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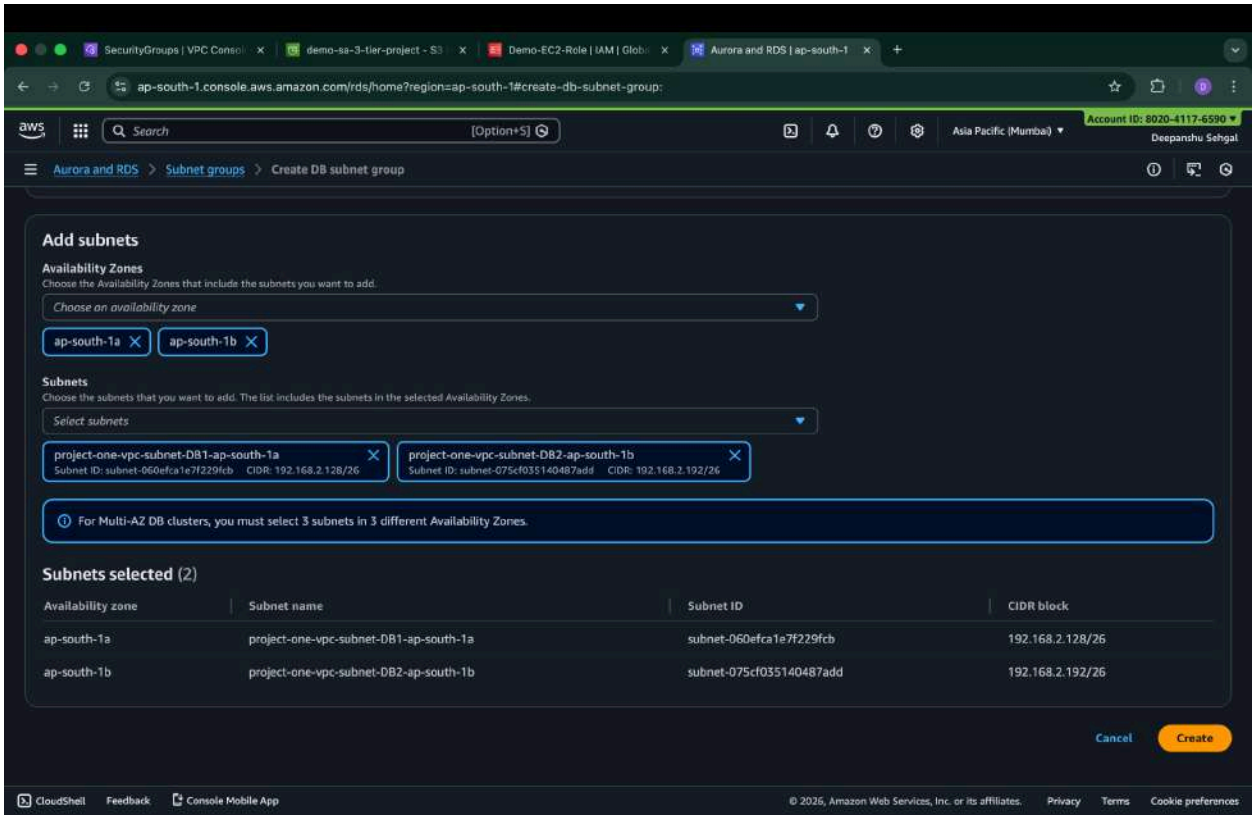
SecurityGroups | VPC Consol
demo-sa-3-tier-project - S3
Demo-EC2-Role | IAM | Glob
Aurora and RDS | ap-south-1
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#create-db-subnet-group:
Search
[Option+S]
Asia Pacific (Mumbai)
Account ID: 8020-4117-6590
Deepanshu Sehgal
Aurora and RDS > Subnet groups > Create DB subnet group
Add subnets
Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
ap-south-1a
ap-south-1b
Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets
project-one-vpc-subnet-DB1-ap-south-1a
Subnet ID: subnet-060efca1e7f229fcb CIDR: 192.168.2.128/26
project-one-vpc-subnet-DB2-ap-south-1b
Subnet ID: subnet-075cf035140487add CIDR: 192.168.2.192/26
For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.
Subnets selected (2)
Availability zone
Subnet name
Subnet ID
CIDR block
ap-south-1a
project-one-vpc-subnet-DB1-ap-south-1a
subnet-060efca1e7f229fcb
192.168.2.128/26
ap-south-1b
project-one-vpc-subnet-DB2-ap-south-1b
subnet-075cf035140487add
192.168.2.192/26
Cancel
Create
CloudShell
Feedback
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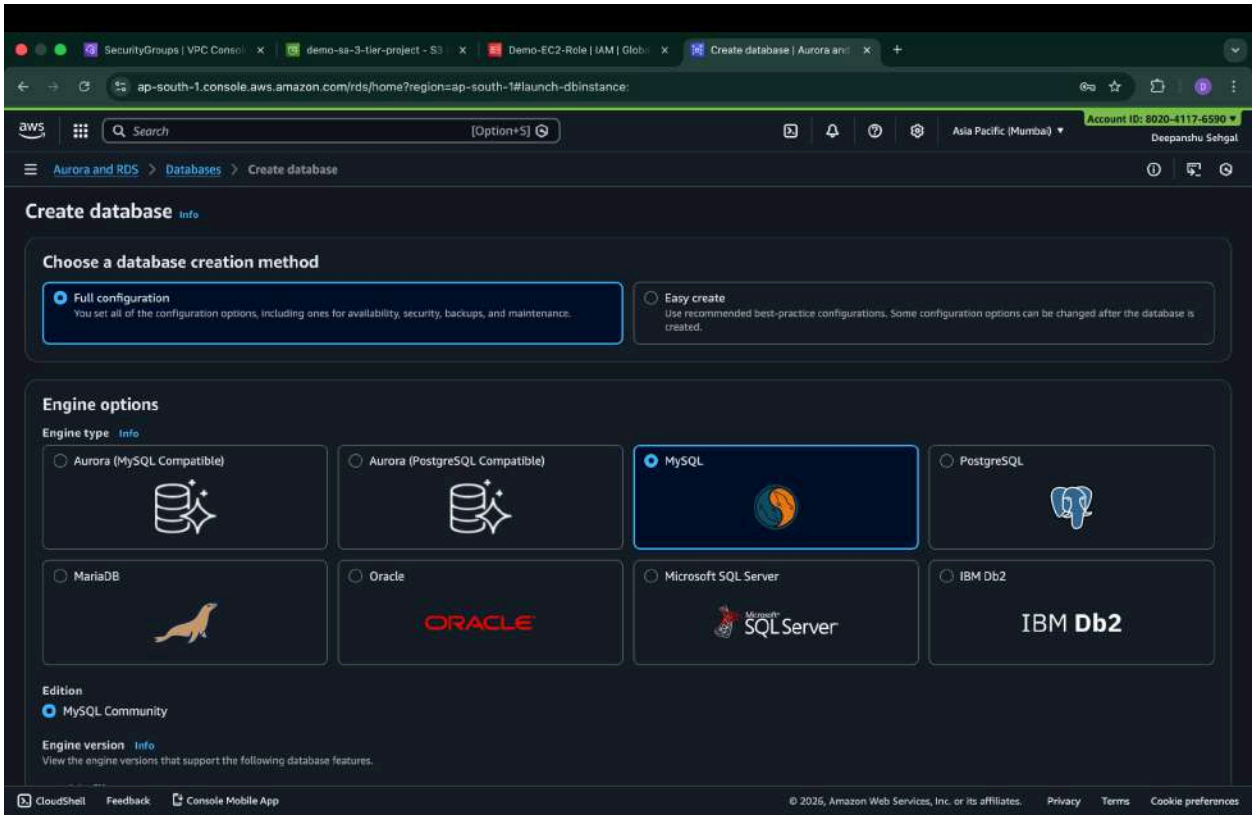
SecurityGroups | VPC Consol
demo-sa-3-tier-project - S3
Demo-EC2-Role | IAM | Glob
Create database | Aurora and
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:
Search
[Option+S]
Asia Pacific (Mumbai)
Account ID: 8020-4117-6590
Deepanshu Sehgal
Aurora and RDS > Databases > Create database
Create database Info
Choose a database creation method
Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.
Engine options
Engine type Info
Aurora (MySQL Compatible)
Aurora (PostgreSQL Compatible)
MySQL
PostgreSQL
MariaDB
Oracle
ORACLE
Microsoft SQL Server
SQL Server
IBM Db2
IBM Db2
Edition
MySQL Community
Engine version Info
View the engine versions that support the following database features.
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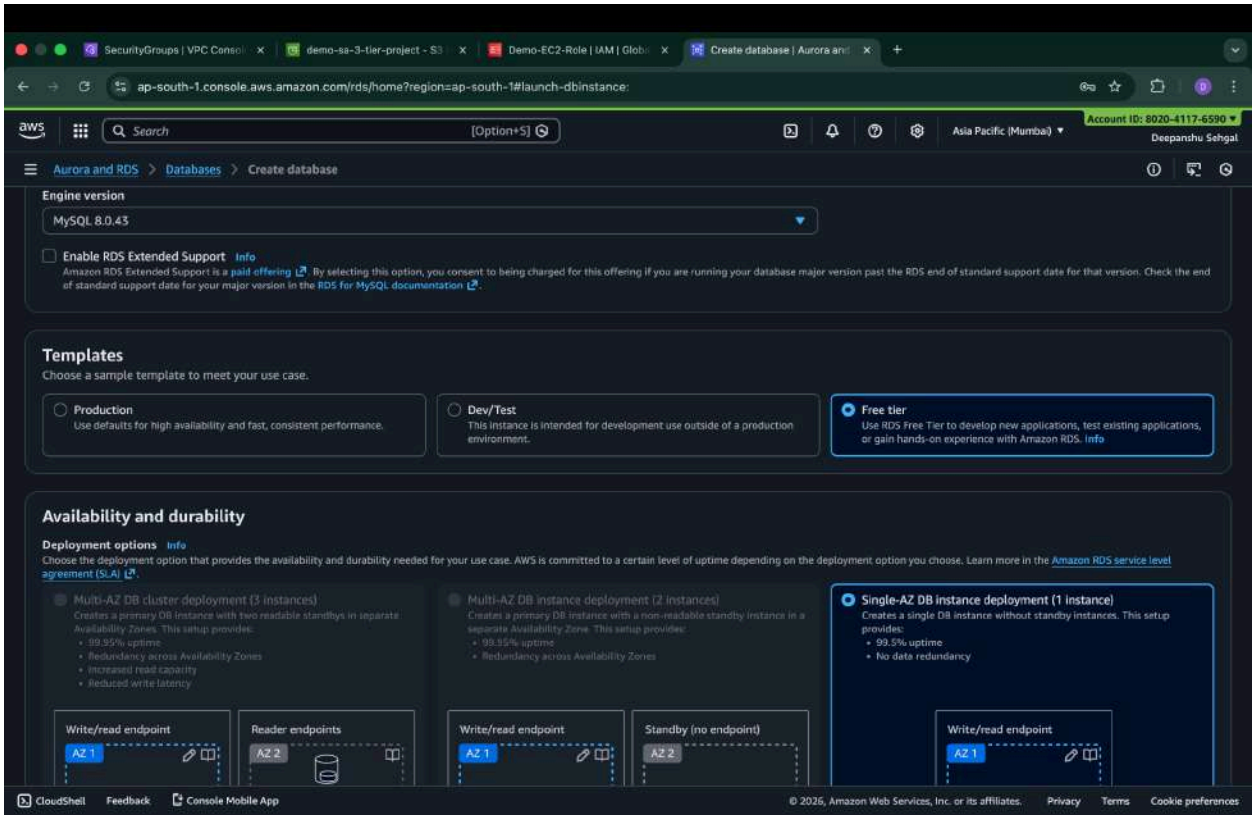
SecurityGroups | VPC Consol
demo-sa-3-tier-project - S3
Demo-EC2-Role | IAM | Glob
Create database | Aurora and
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:
Search
[Option+S]
Asia Pacific (Mumbai)
Account ID: 8020-4117-6590
Deepanshu Sehgal
Aurora and RDS > Databases > Create database
Engine version
MySQL 8.0.43
Enable RDS Extended Support Info
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the RDS for MySQL documentation.
Templates
Choose a sample template to meet your use case.
Production
Use defaults for high availability and fast, consistent performance.
Dev/Test
This instance is intended for development use outside of a production environment.
Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. Info
Availability and durability
Deployment options Info
Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the Amazon RDS service level agreement (SLA).
Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
• 99.95% uptime
• Redundancy across Availability Zones
• Increased read capacity
• Reduced write latency
Write/read endpoint
AZ 1
Reader endpoints
AZ 2
Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
• 99.95% uptime
• Redundancy across Availability Zones
Write/read endpoint
AZ 1
Standby (no endpoint)
AZ 2
Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:
• 99.5% uptime
• No data redundancy
Write/read endpoint
AZ 1
CloudShell
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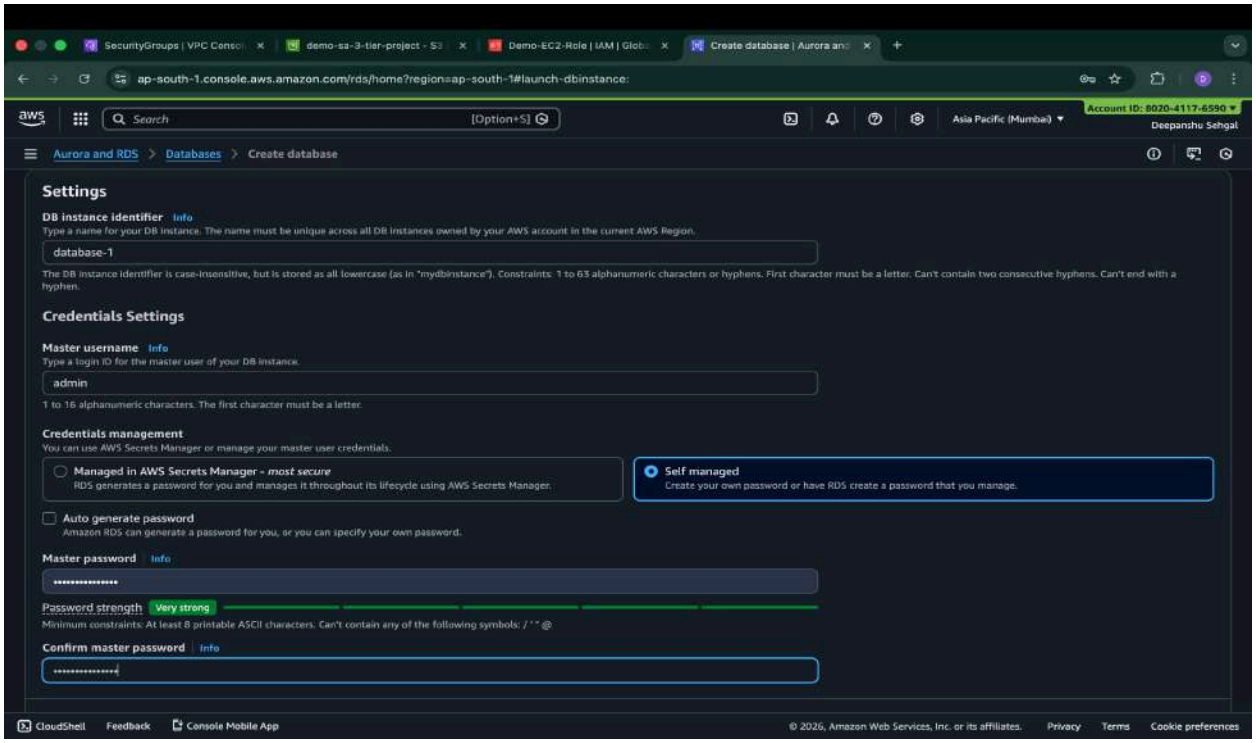
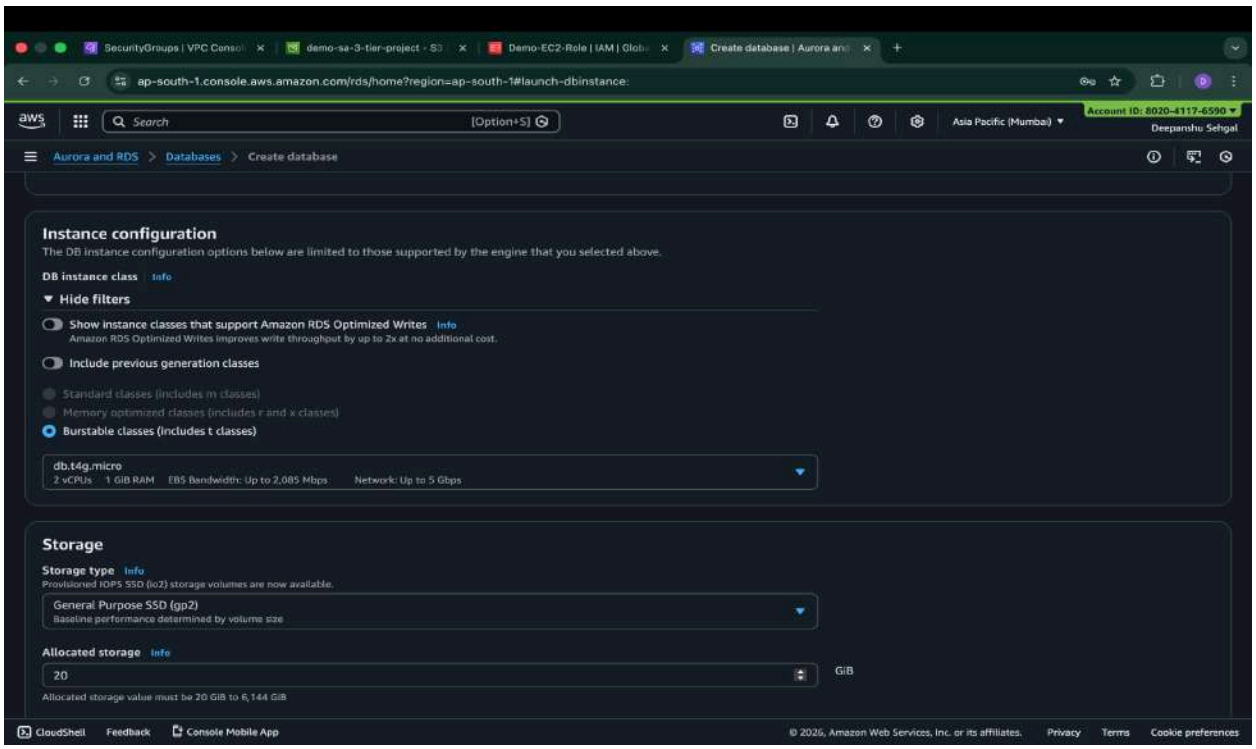
- **Now create the database for the application.**





SecurityGroups | VPC Consol × | demo-sa-3-tier-project - S3 × | Demo-EC2-Role | IAM | Glob × | Create database | Aurora and × +

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

Aurora and RDS > Databases > Create database

Connectivity Info

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- (selected) **Don't connect to an EC2 compute resource**
  Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- **Connect to an EC2 compute resource**
  Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

project-one-vpc (vpc-07df5afcc99f371e1)
6 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

db-sngp
2 Subnets, 2 Availability Zones

Public access Info

- Yes
  RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- (selected) No
  RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

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SecurityGroups | VPC Consol × | demo-sa-3-tier-project - S3 × | Demo-EC2-Role | IAM | Glob × | Create database | Aurora and × +

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

Aurora and RDS > Databases > Create database

Public access Info

- Yes
  RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- (selected) No
  RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- (selected) **Choose existing**
  Choose existing VPC security groups
- **Create new**
  Create new VPC security group

Existing VPC security groups

Choose one or more options

RDS-SG ×

Availability Zone Info

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** Info

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.

Certificate authority - *optional* Info

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061

If you don't select a certificate authority, RDS chooses one for you.

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SecurityGroups | VPC Consol × | demo-sa-3-tier-project - S3 × | Demo-EC2-Role | IAM | Glob × | Create database | Aurora and × +

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

Aurora and RDS > Databases > Create database

Creates a point-in-time snapshot of your database

Backup tags Info

Copies tags from the source database to the automated backup and snapshots respectively.

- [ ] Copy tags to automated backup
  This is a one-time setting. Future tag modifications need manual updates.

- [x] Enable encryption
  Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. Info

AWS KMS key Info

(default) aws/rds

Account

802041176590

KMS key ID

alias/aws/rds

Maintenance

Auto minor version upgrade Info

- [ ] Enable auto minor version upgrade
  Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version documentation.

Maintenance window Info

Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

- ( ) Choose a window
- (•) No preference

- [ ] Enable deletion protection
  Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

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SecurityGroups | VPC Consol × | demo-sa-3-tier-project - S3 × | Demo-EC2-Role | IAM | Glob × | database-1 - Database Detai × +

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#database:id=database-1;is-cluster=false

aws | Search [Option+S] | Asia Pacific (Mumbai) | Account ID: 8020-4117-6590 | Deepanshu Sehgal

Aurora and RDS > Databases > database-1

To exit full screen, press and hold esc

Aurora and RDS

- Dashboard
- Databases
- Query editor
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies

- Subnet groups
- Parameter groups
- Option groups
- Custom engine versions
- Zero-ETL integrations

- Events
- Event subscriptions

- Recommendations 0
- Certificate update

Successfully created database database-1 | View connection details

You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

database-1

Modify | Actions

Summary

DB identifier	Status	Role	Engine	Recommendations
database-1	Available	Instance	MySQL Community	
CPU	**Class**	**Current activity**	**Region & AZ**	
24.48%	db.t4g.micro		ap-south-1a	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups

Connectivity & security

Endpoint & port

Endpoint
database-1.cb6sa0k2y14p.ap-south-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
ap-south-1a

VPC
project-one-vpc (vpc-07df5afcc99f371e1)

Subnet group

Security

VPC security groups
RDS-SG (sg-0bbeabeeaa0f6d481)
Active

Publicly accessible
No

Certificate authority Info

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5. **Application Tier Setup**
    a. **EC2 Deployment**
        i. Application EC2 instances launched in private subnets
        ii. No public IP assigned
    b. **Access Method**
        i. Connected via AWS Session Manager

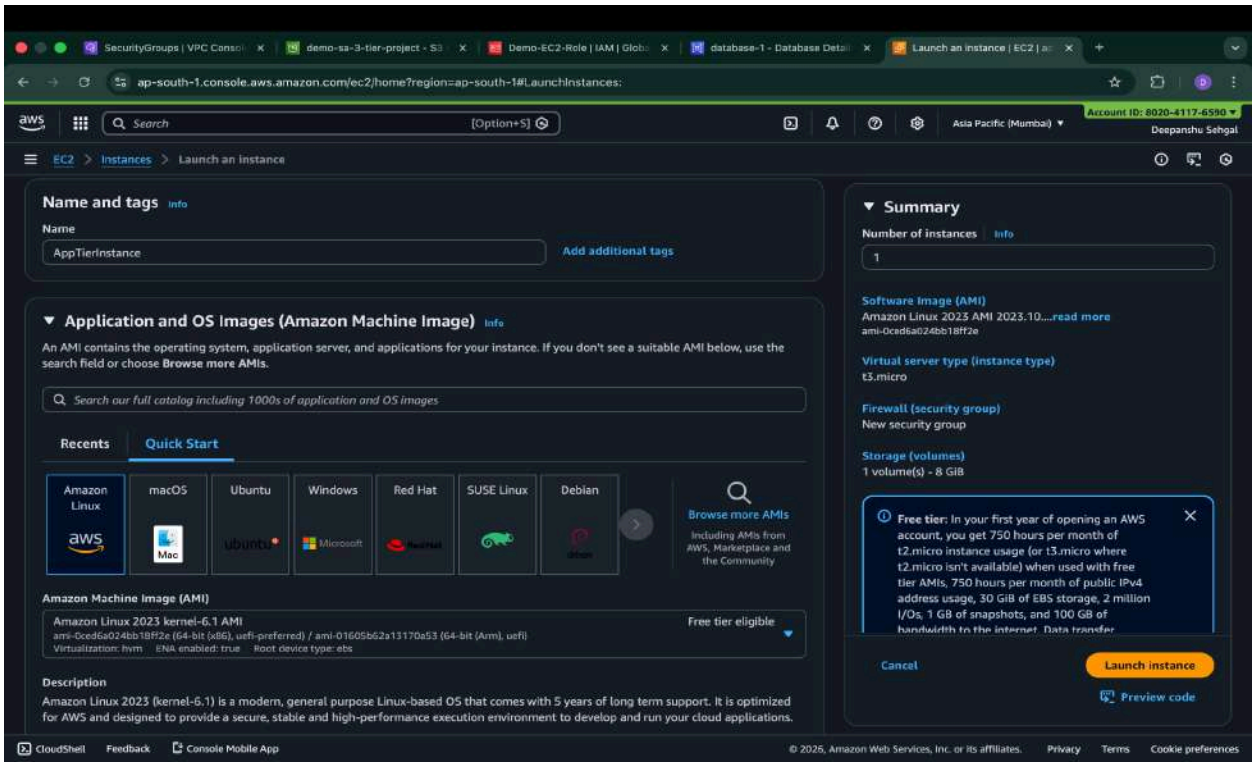
Application Configuration

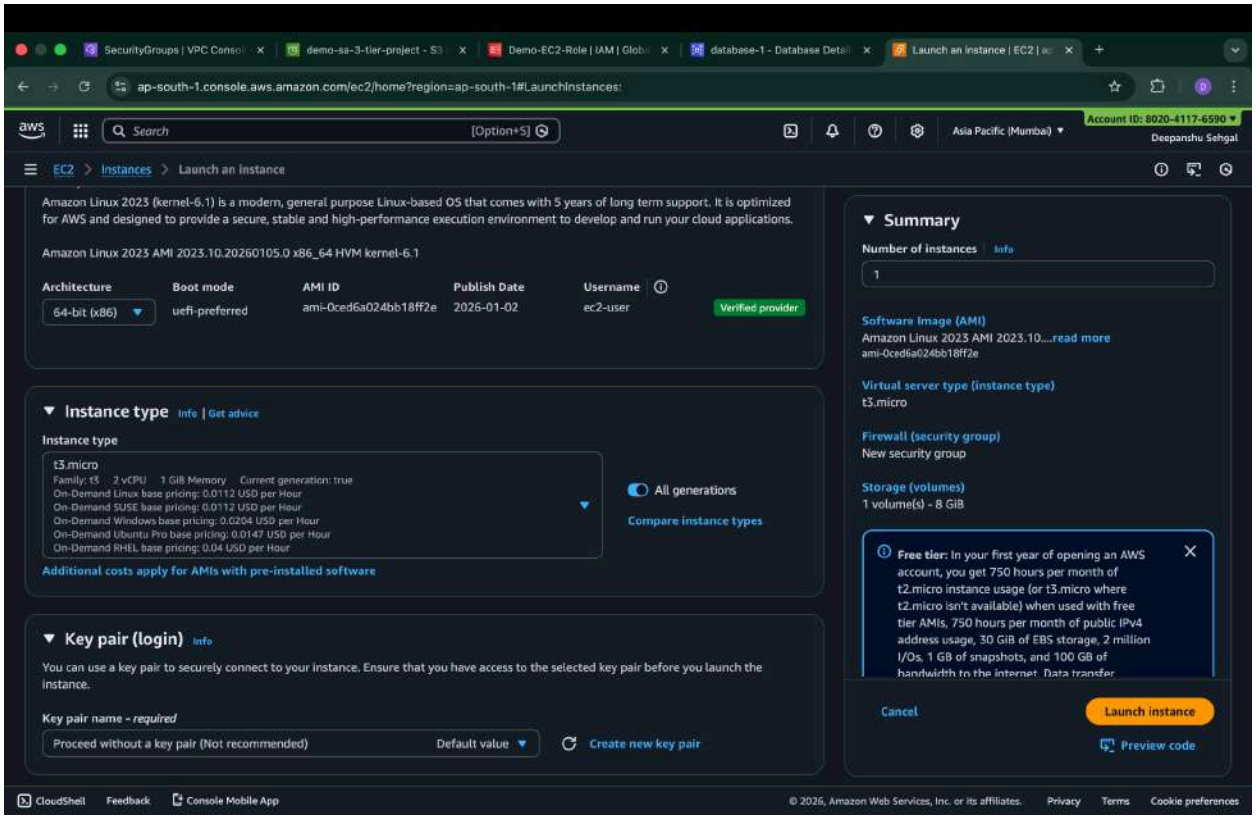
- Installed MySQL client
- Installed NVM (Node Version Manager)
- Installed PM2 for process management
- Downloaded code from S3
- Ran `npm install`
- Started application using PM2
- Configured PM2 startup and process saving for auto-restart

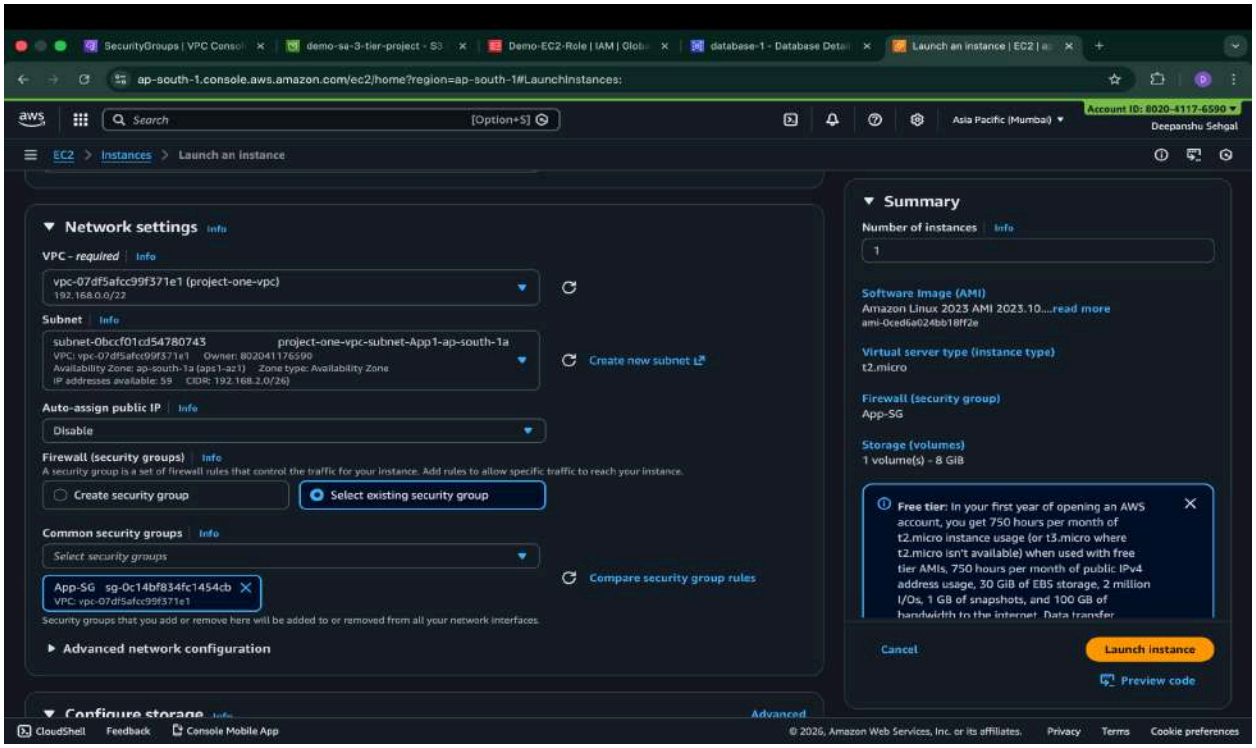
Internal Load Balancer

- Created a target group for health checks
- Configured an internal load balancer for app tier traffic

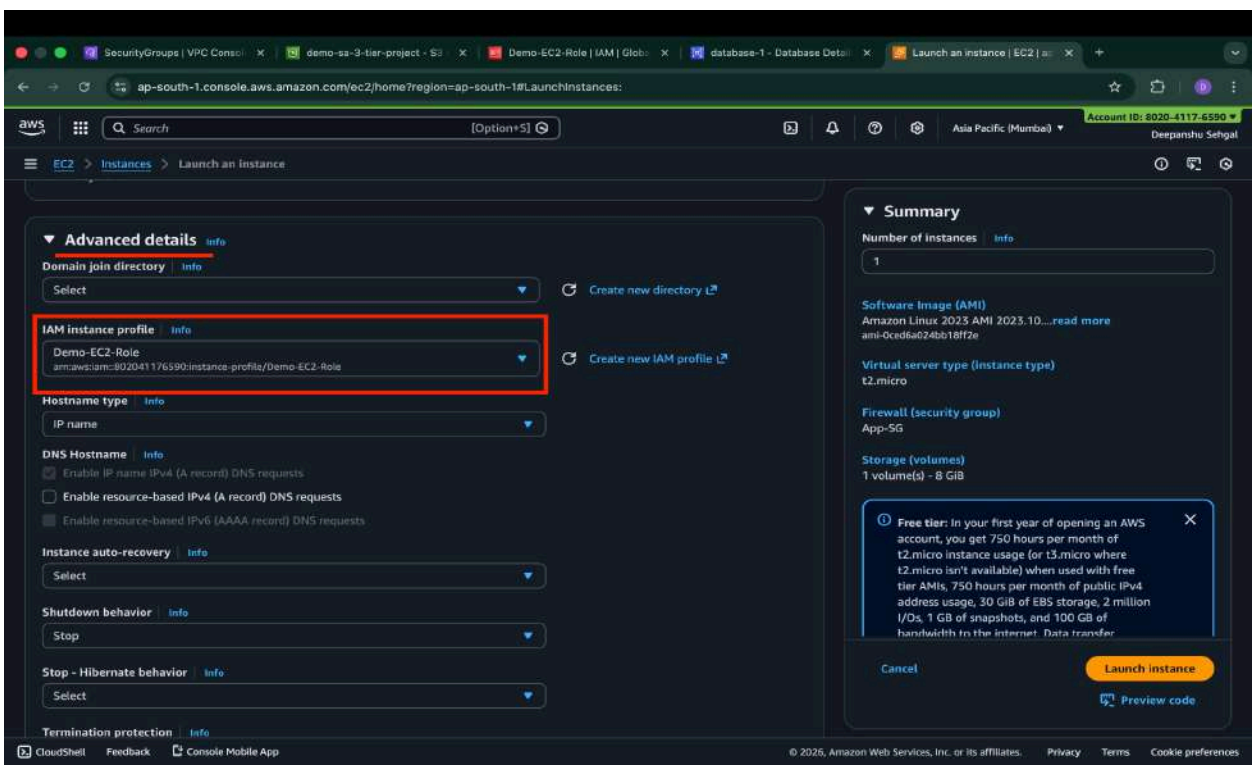
- **Now launch an EC2 instance for the Application Tier.**

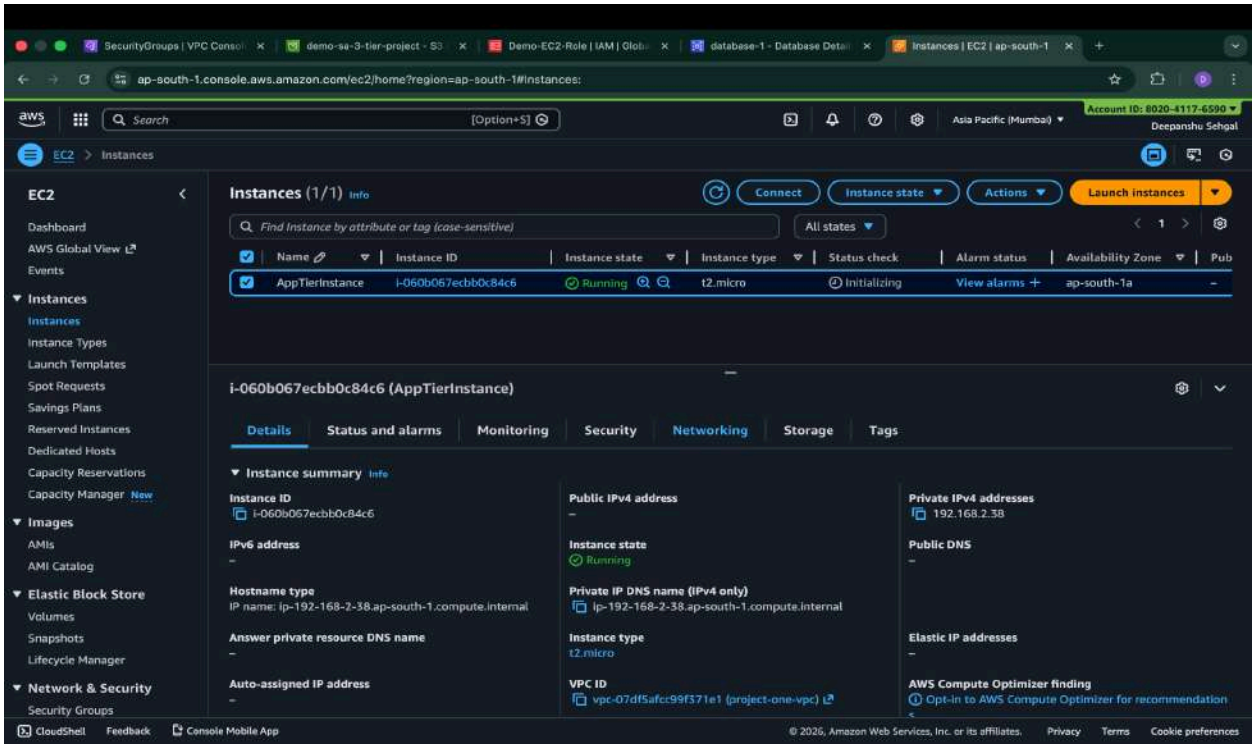




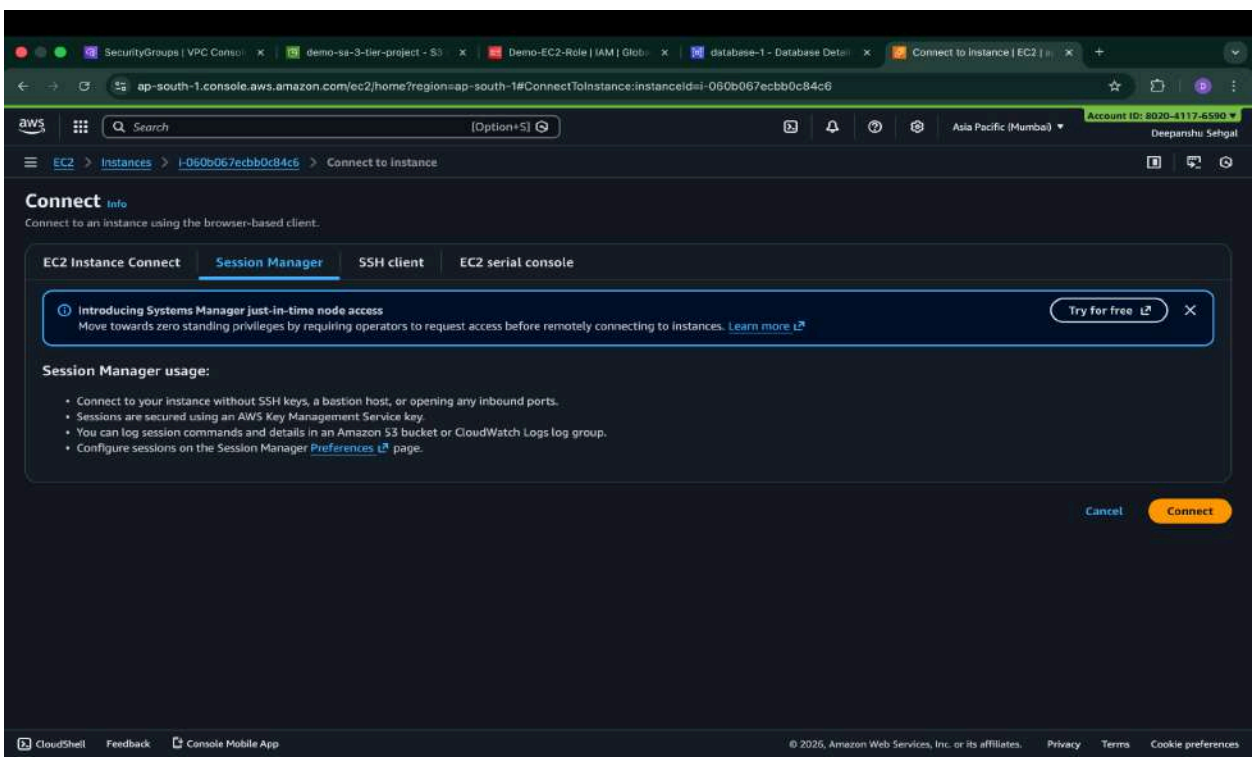


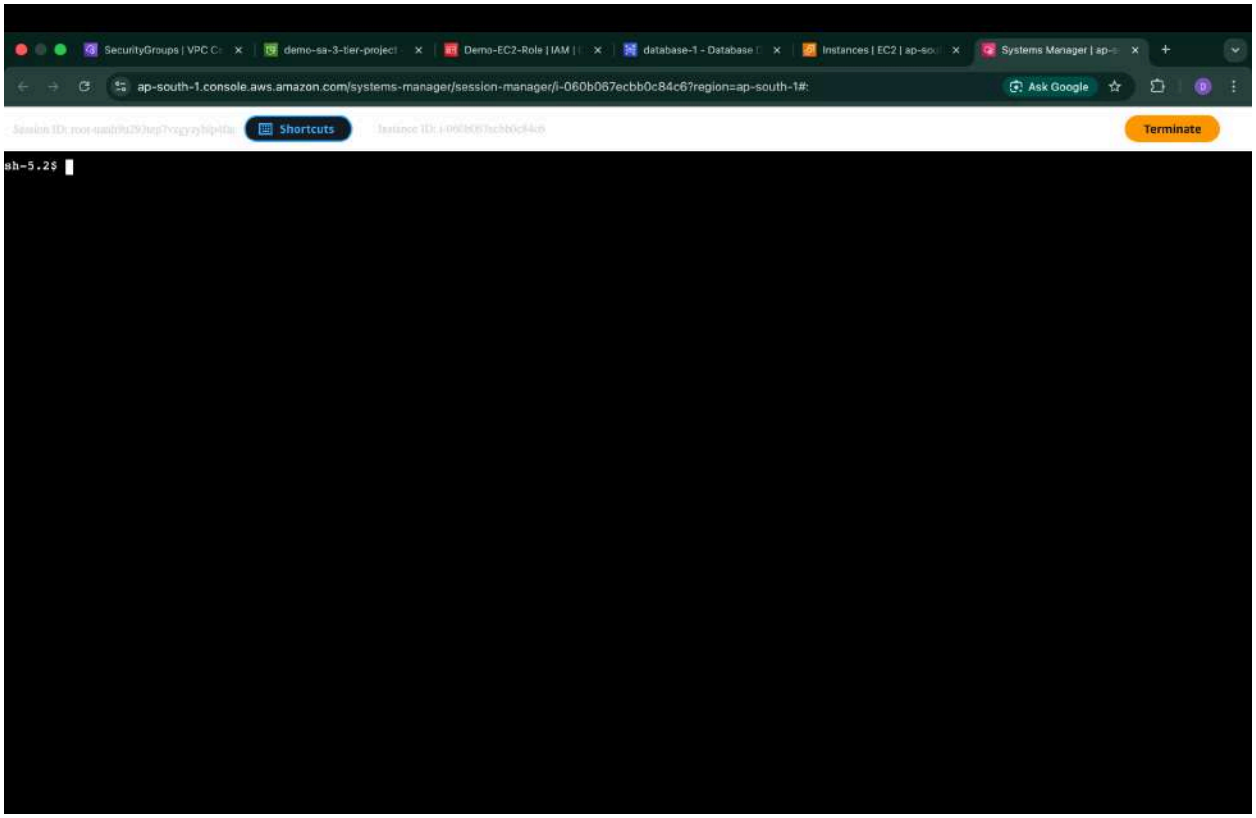
- **Attach the IAM role to the EC2 instance that was created in the previous steps.**





- **Connect through Session Manager, as the instance does not have internet access and therefore no public IP.**





- **Switch to the root user using sudo su, then install the MySQL database.**

```
[root@ip-192-168-2-55 ec2-user]# mysql -h database-1.cb6sa0k2y14p.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 109
Server version: 8.0.43 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

- **Create a MySQL database named webappdb.**

```
MySQL [(none)]> CREATE DATABASE webappdb;
Query OK, 1 row affected (0.034 sec)

MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| webappdb           |
+--------------------+
5 rows in set (0.014 sec)

MySQL [(none)]>
```

- **Use the database that was created.**

```
MySQL [(none)]> USE webappdb;
Database changed
MySQL [webappdb]>
```

- **Create a `transactions` table with columns for id, amount, and description.**

```
MySQL [webappdb]> CREATE TABLE IF NOT EXISTS transactions(
    ->   id INT NOT NULL AUTO_INCREMENT,
    ->   amount DECIMAL(10,2),
    ->   description VARCHAR(100),
    ->   PRIMARY KEY(id)
    -> );
Query OK, 0 rows affected (0.084 sec)

MySQL [webappdb]> SHOW TABLES;
+--------------------+
| Tables_in_webappdb |
+--------------------+
| transactions       |
+--------------------+
1 row in set (0.003 sec)

MySQL [webappdb]>
```

Insert dummy data into the database.

```
MySQL [webappdb]> INSERT INTO transactions (amount, description) VALUES ('400', 'groceries');
Query OK, 1 row affected (0.004 sec)

MySQL [webappdb]> SELECT * FROM transactions;
+----+--------+-------------+
| id | amount | description |
+----+--------+-------------+
|  1 | 400.00 | groceries   |
+----+--------+-------------+
1 row in set (0.002 sec)

MySQL [webappdb]>
```

- **Install NVM (Node Version Manager).**

```
[root@ip-192-168-2-55 ec2-user]# curl -o- https://raw.githubusercontent.com/avizway1/aws_3tier_architecture/main/install.sh | bash
```

- **Install PM2, a process manager for Node.js applications.**

```
Shortcuts                                                                    Terminate
[root@ip-192-168-2-55 ec2-user]# curl -o- https://raw.githubusercontent.com/avizway1/aws_3tier_architecture/main/install.sh | bash
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 14926  100 14926    0     0  50407      0 --:--:-- --:--:-- --:--:-- 50425
=> Downloading nvm as script to '/root/.nvm'

=> Appending nvm source string to /root/.bashrc
=> Appending bash_completion source string to /root/.bashrc
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"  # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"  # This loads nvm bash_completion
[root@ip-192-168-2-55 ec2-user]# source ~/.bashrc
[root@ip-192-168-2-55 ec2-user]# nvm install 16
Downloading and installing node v16.20.2...
Downloading https://nodejs.org/dist/v16.20.2/node-v16.20.2-linux-x64.tar.xz...
################################################################################################################################################################ 100.0%
Computing checksum with sha256sum
Checksums matched!
Now using node v16.20.2 (npm v8.19.4)
Creating default alias: default -> 16 (-> v16.20.2)
[root@ip-192-168-2-55 ec2-user]# nvm use 16
Now using node v16.20.2 (npm v8.19.4)
[root@ip-192-168-2-55 ec2-user]#
```

```
[root@ip-192-168-2-55 ec2-user]# npm install -g pm2

changed 134 packages, and audited 134 packages in 7s

13 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
[root@ip-192-168-2-55 ec2-user]#
```

- **Download the application code from the S3 bucket.**

```
[root@ip-192-168-2-55 ~]# sudo aws s3 cp s3://demo-sa-3-tier-project/application-code/app-tier/ app-tier --recursive
download: s3://demo-sa-3-tier-project/application-code/app-tier/DbConfig.js to app-tier/DbConfig.js
download: s3://demo-sa-3-tier-project/application-code/app-tier/package.json to app-tier/package.json
download: s3://demo-sa-3-tier-project/application-code/app-tier/README.md to app-tier/README.md
download: s3://demo-sa-3-tier-project/application-code/app-tier/index.js to app-tier/index.js
download: s3://demo-sa-3-tier-project/application-code/app-tier/package-lock.json to app-tier/package-lock.json
download: s3://demo-sa-3-tier-project/application-code/app-tier/TransactionService.js to app-tier/TransactionService.js
[root@ip-192-168-2-55 ~]# ls
app-tier
[root@ip-192-168-2-55 ~]#
```

- **Switch to the application directory in the app tier.**

```
[root@ip-192-168-2-55 app-tier]# ls
DbConfig.js  README.md  TransactionService.js  index.js  package-lock.json  package.json
[root@ip-192-168-2-55 app-tier]# npm install
```

- **Install the required packages by running `npm install`.**

```
[root@ip-192-168-2-55 app-tier]# ls
DbConfig.js  README.md  TransactionService.js  index.js  package-lock.json  package.json
[root@ip-192-168-2-55 app-tier]# npm install

added 68 packages, and audited 69 packages in 2s

2 packages are looking for funding
  run `npm fund` for details

7 vulnerabilities (3 low, 4 high)

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
[root@ip-192-168-2-55 app-tier]# pm2 start index.js
```

- **Start the application using PM2.**

```
[PM2] Spawning PM2 daemon with pm2_home=/root/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /root/app-tier/index.js in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	index	default	1.0.0	fork	28285	0s	0	online	0%	25.6mb	root	disabled

```
[root@ip-192-168-2-55 app-tier]#
```

Shortcuts

```
[root@ip-192-168-2-55 app-tier]# pm2 list
```

id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	index	default	1.0.0	fork	28285	57s	0	online	0%	51.2mb	root	disabled

```
[root@ip-192-168-2-55 app-tier]#
```

- **Check the application logs using PM2.**

```
[root@ip-192-168-2-55 app-tier]# pm2 logs
[TAILING] Tailing last 15 lines for [all] processes (change the value with --lines option)
/root/.pm2/pm2.log last 15 lines:
PM2        | 2026-01-12T22:28:42: PM2 log: Node.js version      : 16.20.2
PM2        | 2026-01-12T22:28:42: PM2 log: Current arch         : x64
PM2        | 2026-01-12T22:28:42: PM2 log: PM2 home             : /root/.pm2
PM2        | 2026-01-12T22:28:42: PM2 log: PM2 PID file         : /root/.pm2/pm2.pid
PM2        | 2026-01-12T22:28:42: PM2 log: RPC socket file      : /root/.pm2/rpc.sock
PM2        | 2026-01-12T22:28:42: PM2 log: BUS socket file      : /root/.pm2/pub.sock
PM2        | 2026-01-12T22:28:42: PM2 log: Application log path : /root/.pm2/logs
PM2        | 2026-01-12T22:28:42: PM2 log: Worker Interval      : 30000
PM2        | 2026-01-12T22:28:42: PM2 log: Process dump file    : /root/.pm2/dump.pm2
PM2        | 2026-01-12T22:28:42: PM2 log: Concurrent actions   : 2
PM2        | 2026-01-12T22:28:42: PM2 log: SIGTERM timeout      : 1600
PM2        | 2026-01-12T22:28:42: PM2 log: Runtime Binary       : /root/.nvm/versions/node/v16.20.2/bin/node
PM2        | 2026-01-12T22:28:42: PM2 log: ============================================================================
PM2        | 2026-01-12T22:28:42: PM2 log: App [index:0] starting in -fork mode-
PM2        | 2026-01-12T22:28:42: PM2 log: App [index:0] online

/root/.pm2/logs/index-error.log last 15 lines:
/root/.pm2/logs/index-out.log last 15 lines:
0|index     | AB3 backend app listening at http://localhost:4000
```

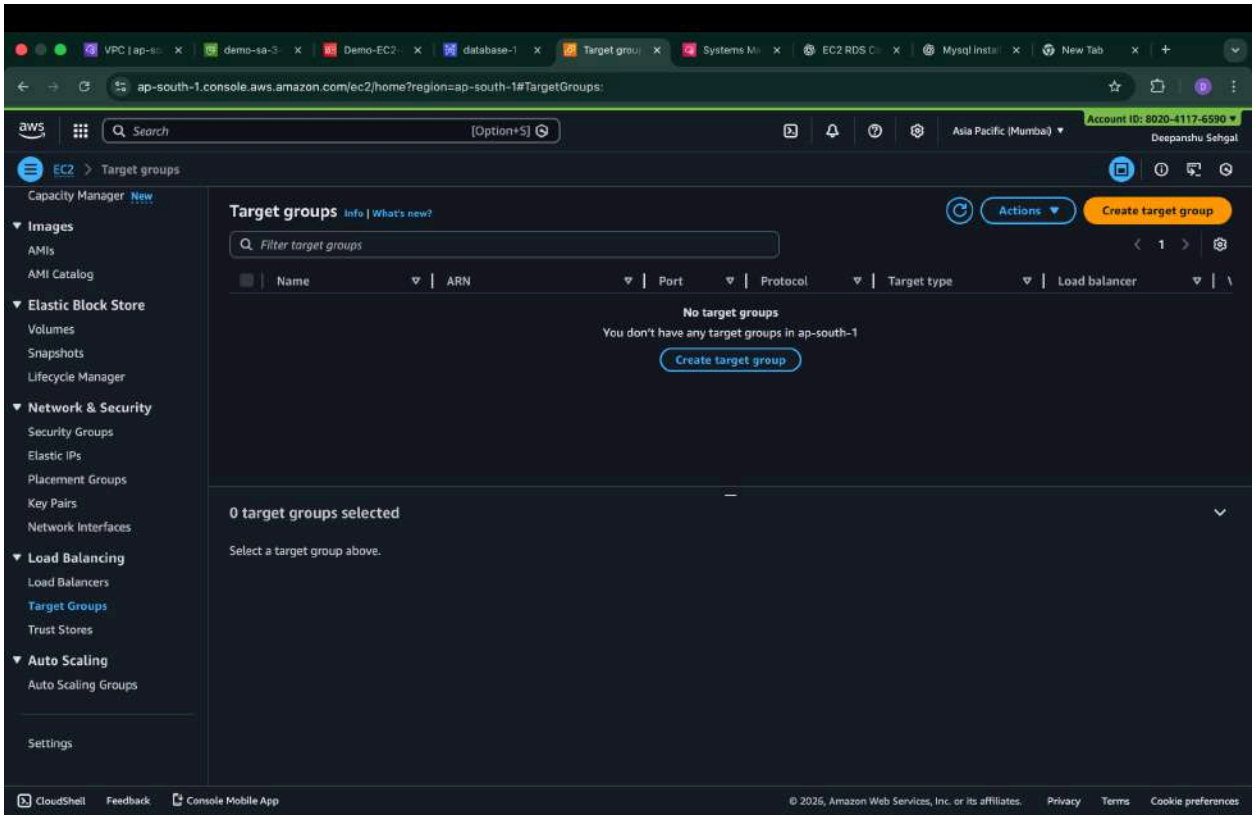
- **Run `pm2 startup` to generate and configure a system-specific startup script for automatic restarts, then run `pm2 save` to save the PM2 processes.**

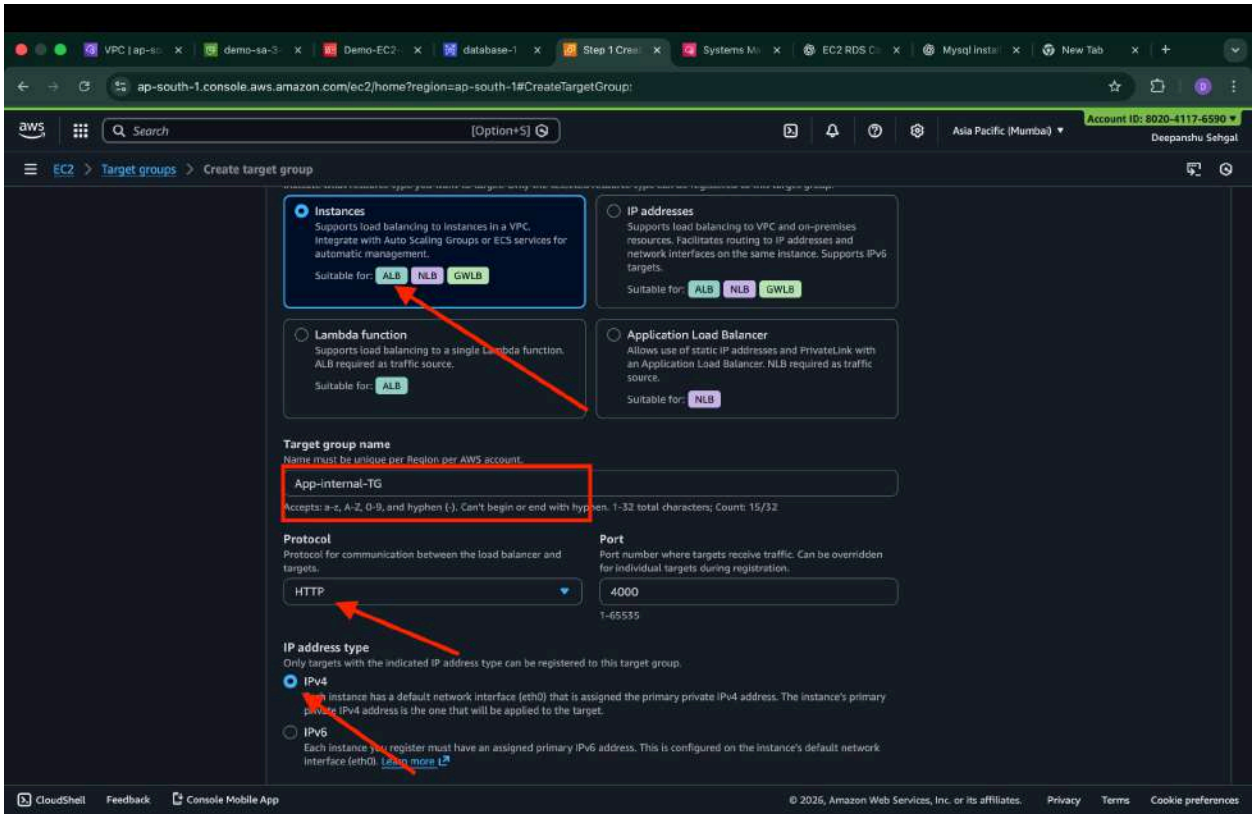
```
[root@ip-192-168-2-55 app-tier]# pm2 startup
[PM2] Init System found: systemd
Platform systemd
Template
[Unit]
Description=PM2 process manager
Documentation=https://pm2.keymetrics.io/
After=network.target

[Service]
Type=forking
User=root
LimitNOFILE=infinity
LimitNPROC=infinity
LimitCORE=infinity
Environment=PATH=/root/.nvm/versions/node/v16.20.2/bin:/root/.local/bin:/root/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/var/lib/snapd/sna
p/bin:/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
Environment=PM2_HOME=/root/.pm2
PIDFile=/root/.pm2/pm2.pid
Restart=on-failure

ExecStart=/root/.nvm/versions/node/v16.20.2/lib/node_modules/pm2/bin/pm2 resurrect
ExecReload=/root/.nvm/versions/node/v16.20.2/lib/node_modules/pm2/bin/pm2 reload all
ExecStop=/root/.nvm/versions/node/v16.20.2/lib/node_modules/pm2/bin/pm2 kill

[Install]
WantedBy=multi-user.target

Target path
/etc/systemd/system/pm2-root.service
Command list
[ 'systemctl enable pm2-root' ]
[PM2] Writing init configuration in /etc/systemd/system/pm2-root.service
[PM2] Making script booting at startup...
[PM2] [-] Executing: systemctl enable pm2-root...
Created symlink /etc/systemd/system/multi-user.target.wants/pm2-root.service → /etc/systemd/system/pm2-root.service.
[PM2] [v] Command successfully executed.
+---------------------------------------+
[PM2] Freeze a process list on reboot via:
$ pm2 save

[PM2] Remove init script via:
```

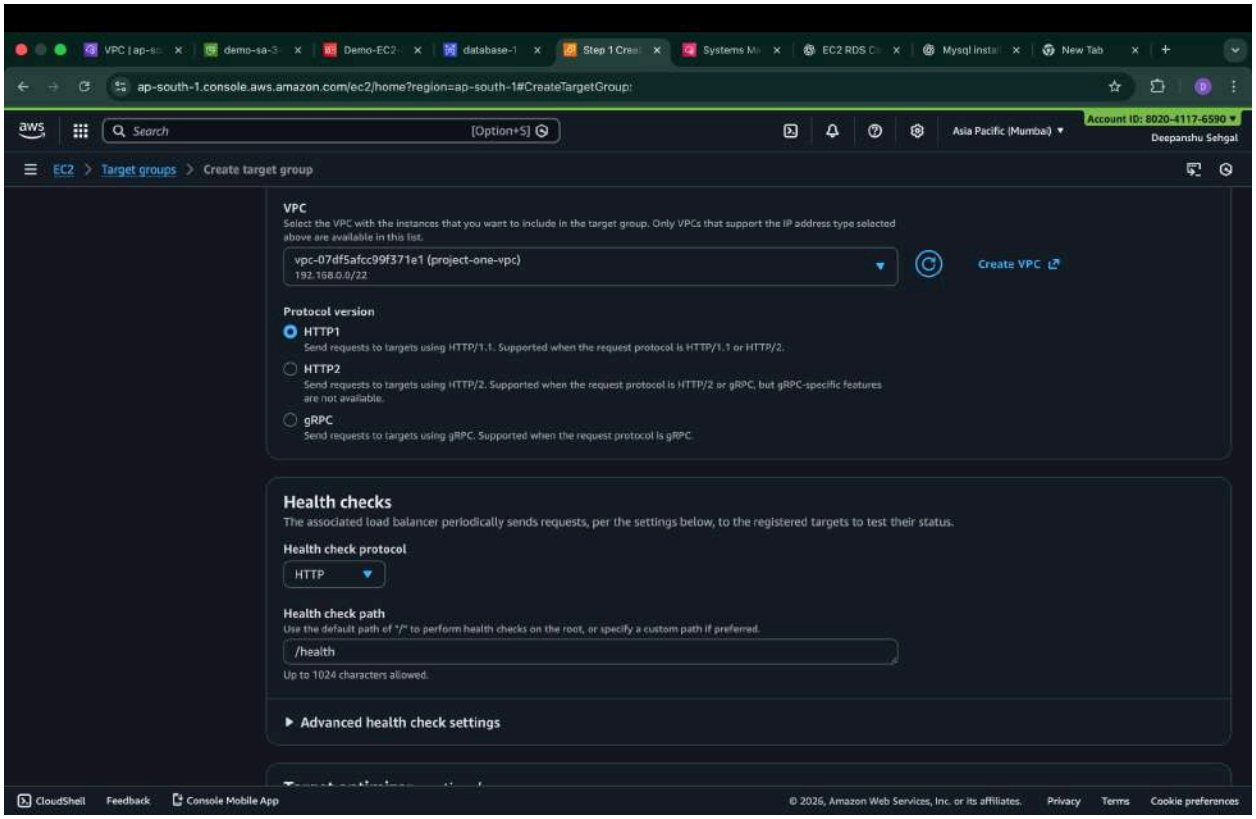
Session ID: root-c7ugcl3lkxlnn4lvk3fnjip98e Shortcuts Instance ID: i-0ab40b2eb6b903a69

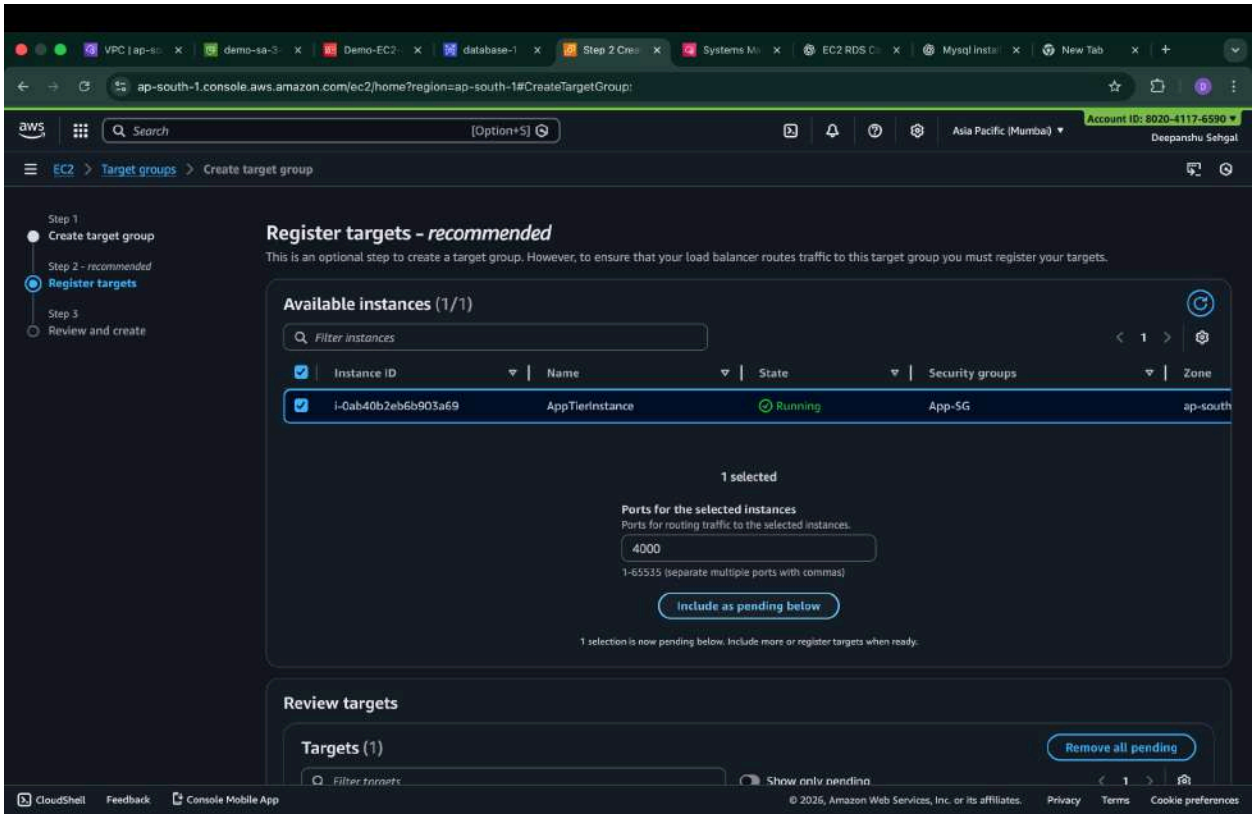
```
[root@ip-192-168-2-55 app-tier]# curl http://localhost:4000/health
"This is the health check"[root@ip-192-168-2-55 app-tier]#
```

- **Create a target group for health checks.**




ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:
Account ID: 8020-4117-6590
Deepanshu Sehgal
Asia Pacific (Mumbai)
EC2 > Target groups > Create target group
Instances
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB
IP addresses
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB
Lambda function
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB
Application Load Balancer
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB
Target group name
Name must be unique per Region per AWS account.
App-internal-TG
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 15/32
Protocol
Protocol for communication between the load balancer and targets.
HTTP
Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
4000
1-65535
IP address type
Only targets with the indicated IP address type can be registered to this target group.
IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:
Account ID: 8020-4117-6590
Deepanshu Sehgal
Asia Pacific (Mumbai)
EC2 > Target groups > Create target group
VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-07df5afcc99f371e1 (project-one-vpc)
192.168.0.0/22
Create VPC
Protocol version
HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.
Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.
Health check protocol
HTTP
Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/health
Up to 1024 characters allowed.
Advanced health check settings
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:
Account ID: 8020-4117-6590
Deepanshu Sehgal
Asia Pacific (Mumbai)
EC2 > Target groups > Create target group
Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create
Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.
Available instances (1/1)
Filter instances
Instance ID
Name
State
Security groups
Zone
i-0ab40b2eb6b903a69
AppTierInstance
Running
App-SG
ap-south
1 selected
Ports for the selected instances
Ports for routing traffic to the selected instances.
4000
1-65535 (separate multiple ports with commas)
Include as pending below
1 selection is now pending below. Include more or register targets when ready.
Review targets
Targets (1)
Remove all pending
Show only pending
CloudShell
Feedback
Console Mobile App
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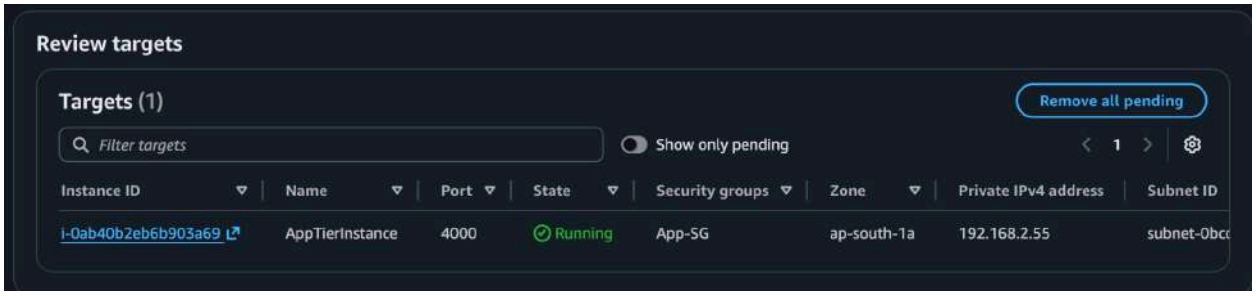
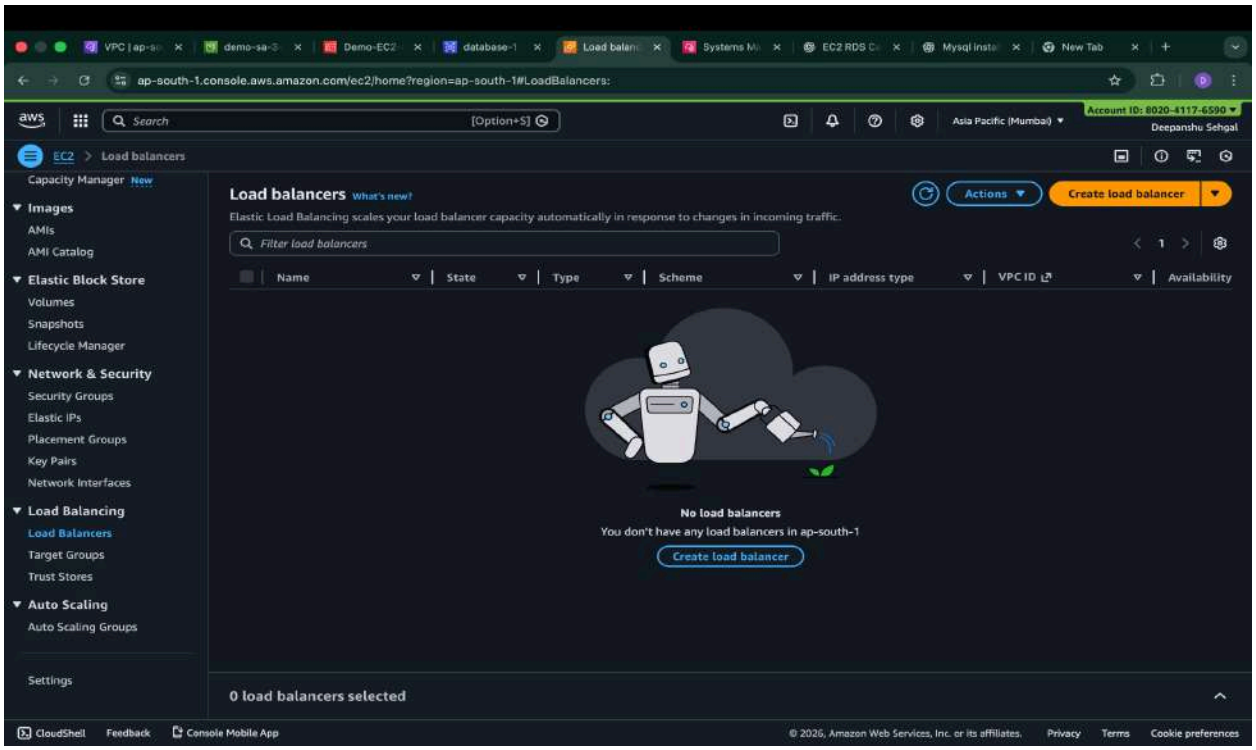
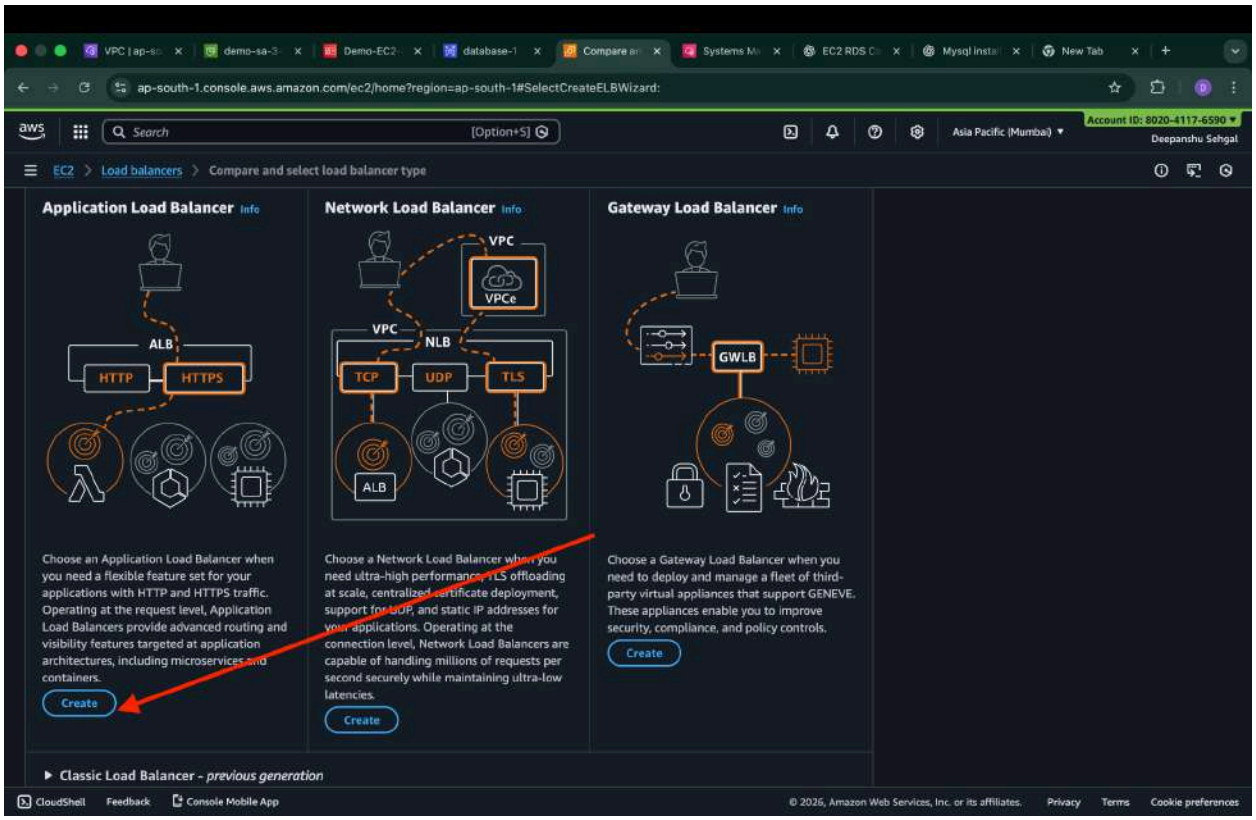
Privacy
Terms
Cookie preferences


Review targets
Targets (1)
Remove all pending
Filter targets
Show only pending
Instance ID
Name
Port
State
Security groups
Zone
Private IPv4 address
Subnet ID
i-0ab40b2eb6b903a69
AppTierInstance
4000
Running
App-SG
ap-south-1a
192.168.2.55

- **Create an internal load balancer.**





ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

Account ID: 8020-4117-6590 Deepanshu Sehgal

EC2 > Load balancers > Create Application Load Balancer

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

App-Internal-Lb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme Info

Scheme can't be changed after the load balancer is created.

- Internet-facing
  - Serves internet-facing traffic.
  - Has public IP addresses.
  - DNS name resolves to public IPs.
  - Requires a public subnet.
- Internal
  - Serves internal traffic.
  - Has private IP addresses.
  - DNS name resolves to private IPs.
  - Compatible with the **IPv4** and **Dualstack** IP address types.

Load balancer IP address type Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

- IPv4
  Includes only IPv4 addresses.
- Dualstack
  Includes IPv4 and IPv6 addresses.

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-07df5afcc99f371e1 (project-one-vpc)
192.168.0.0/22

Create VPC

IP pools Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the Amazon VPC IP Address Manager console.

Use IPAM pool for public IPv4 addresses

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

Account ID: 8020-4117-6590 Deepanshu Sehgal

EC2 > Load balancers > Create Application Load Balancer

Compatible with **Internet-facing** scheme, **IPv4** and **Dualstack** IP address types.

Availability Zones and subnets Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

- [x] **ap-south-1a (aps1-az1)**
  Subnet
  Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
  subnet-0bccf01cd54780743 project-one-vpc-subnet-App1-ap-south-1a
  IPv4 subnet CIDR: 192.168.2.0/26
- [x] **ap-south-1b (aps1-az3)**
  Subnet
  Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
  subnet-092a32cc5adafeaee project-one-vpc-subnet-App2-ap-south-1b
  IPv4 subnet CIDR: 192.168.2.64/26

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can create a new security group.

Security groups

Select up to 5 security groups

Internal-ALB_SG
sg-06e756f41c6bacabd VPC: vpc-07df5afcc99f371e1

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

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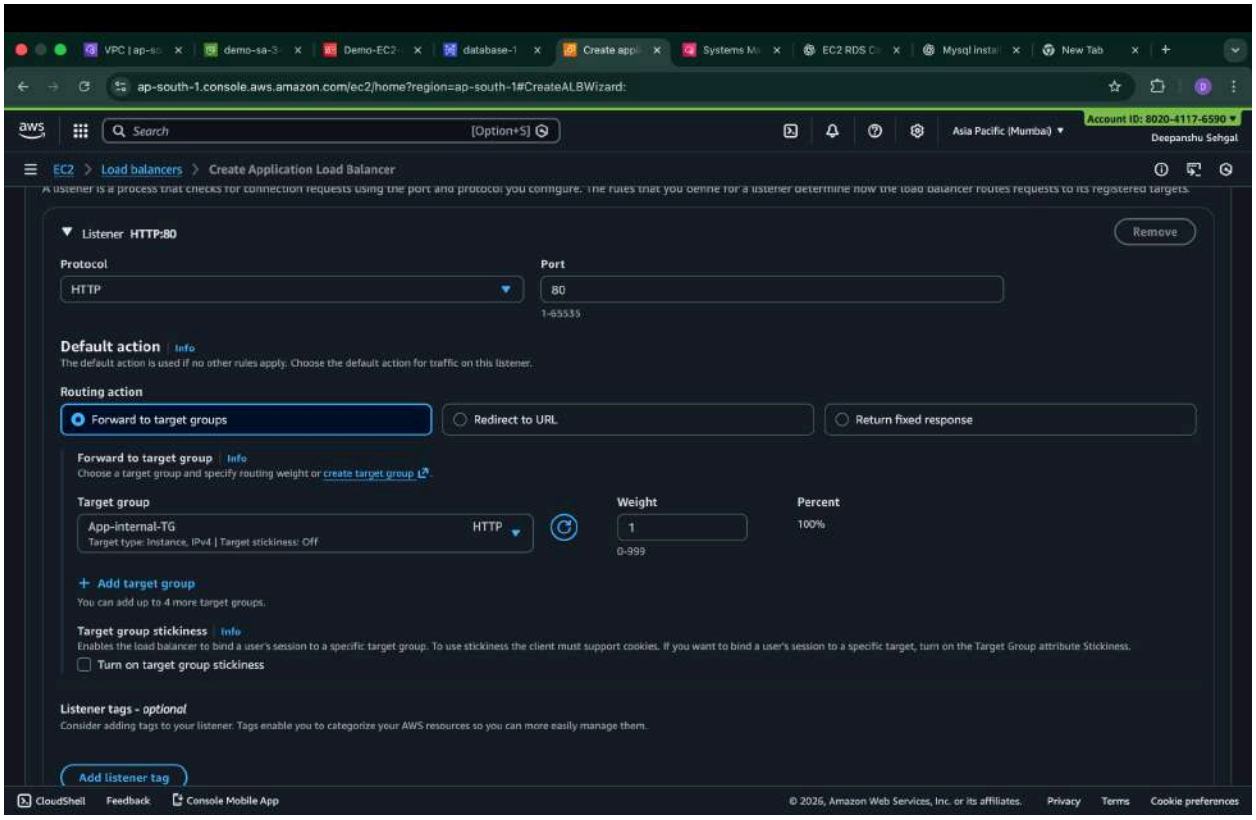


6. Web Tier Setup

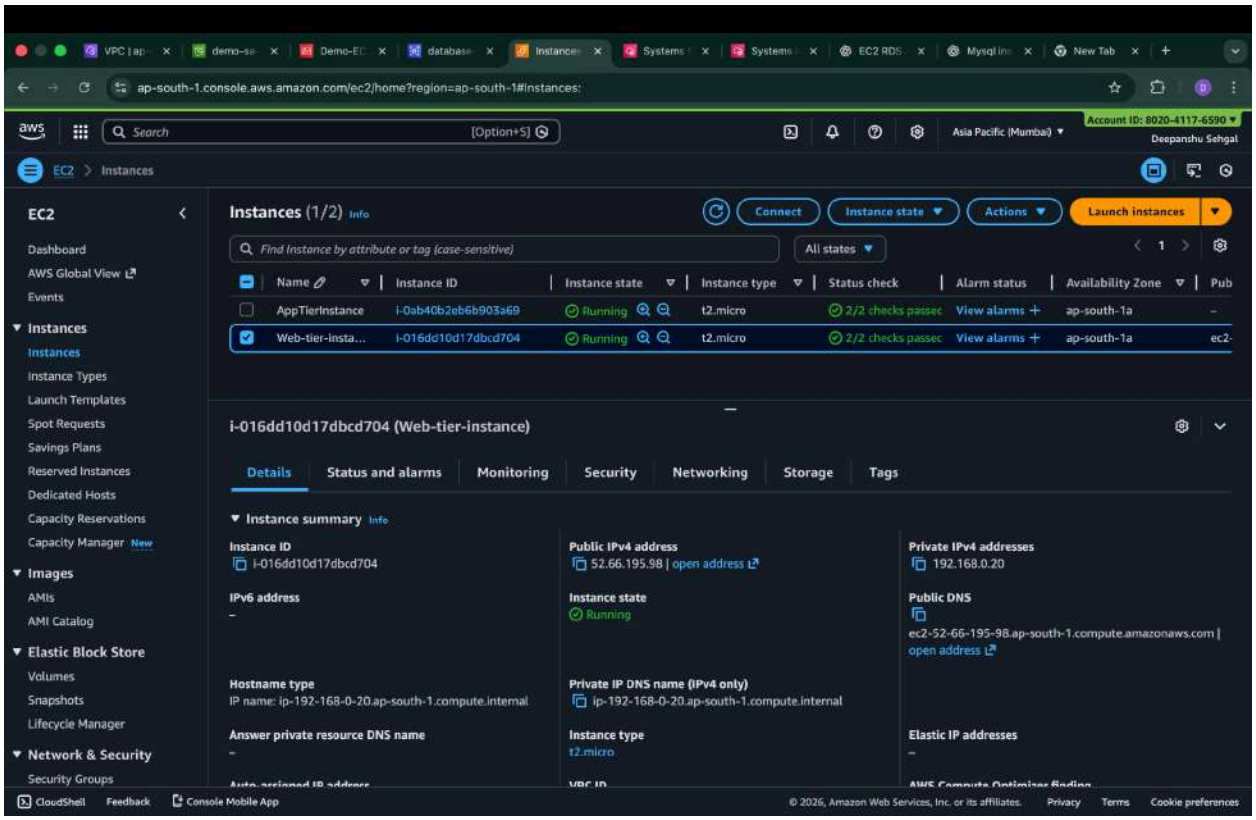
a. **EC2 Deployment:** Web tier EC2 instances launched in public subnets

Setup Steps

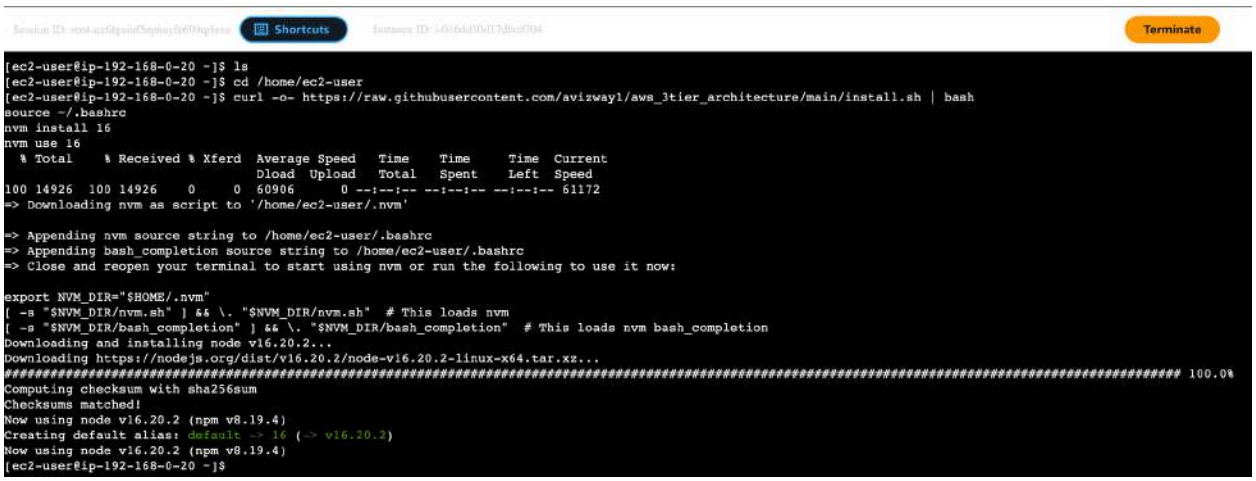
- Installed NVM
- Downloaded code from S3
- Built the application
- Installed Nginx
- Enabled and started the Nginx service

Access: Application became accessible via public IP

- **Now launch an EC2 instance for the Web Tier.**



- **Connect to the EC2 instance and install NVM (Node Version Manager).**



- **Download or copy the code from the S3 bucket.**

```
[ec2-user@ip-192-168-0-20 ~]$ aws s3 cp s3://demo-sa-3-tier-project/application-code/web-tier/ web-tier --recursive
download: s3://demo-sa-3-tier-project/application-code/web-tier/public/robots.txt to web-tier/public/robots.txt
download: s3://demo-sa-3-tier-project/application-code/web-tier/README.md to web-tier/README.md
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/App.js to web-tier/src/App.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Burger/index.js to web-tier/src/components/Burger/index.
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Burger/Burger.js to web-tier/src/components/Burger/Burge
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Burger/Burger.styled.js to web-tier/src/components/Burge
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/DatabaseDemo/DatabaseDemo.css to web-tier/src/components/
.css
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Menu/Menu.js to web-tier/src/components/Menu/Menu.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/App.css to web-tier/src/App.css
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Menu/index.js to web-tier/src/components/Menu/index.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Home/Home.js to web-tier/src/components/Home/Home.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/public/index.html to web-tier/public/index.html
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/Menu/Menu.styled.js to web-tier/src/components/Menu/Menu.
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/DatabaseDemo/DatabaseDemo.js to web-tier/src/components/D
js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/App.test.js to web-tier/src/App.test.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/hooks.js to web-tier/src/hooks.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/global.js to web-tier/src/global.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/components/index.js to web-tier/src/components/index.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/reportWebVitals.js to web-tier/src/reportWebVitals.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/index.js to web-tier/src/index.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/theme.js to web-tier/src/theme.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/package.json to web-tier/package.json
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/setupTests.js to web-tier/src/setupTests.js
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/index.css to web-tier/src/index.css
download: s3://demo-sa-3-tier-project/application-code/web-tier/src/assets/3TierArch.png to web-tier/src/assets/3TierArch.png
[ec2-user@ip-192-168-0-20 ~]$ ls
web-tier
[ec2-user@ip-192-168-0-20 ~]$
```

- **Change Dir to the web-tier directory.**

```
[ec2-user@ip-192-168-0-20 web-tier]$ ls
README.md  package.json  public  src
[ec2-user@ip-192-168-0-20 web-tier]$ npm install
```

- **Build the application.**

Session ID: root-uzfdgaiuf5ephayfe699qrlxea Shortcuts Instance ID: i-016dd10d17dbcd704

```
[ec2-user@ip-192-168-0-20 web-tier]$ npm run build

> aws-3tier-web-layer@0.1.0 build
> react-scripts build

Creating an optimized production build...
Compiled successfully.

File sizes after gzip:

  75.11 kB  build/static/js/main.ab3be8dd.js
  1.77 kB   build/static/js/453.4cfe9ff7.chunk.js
  493 B     build/static/css/main.b20b6ac4.css

The project was built assuming it is hosted at ./.
You can control this with the homepage field in your package.json.

The build folder is ready to be deployed.

Find out more about deployment here:

  https://cra.link/deployment

[ec2-user@ip-192-168-0-20 web-tier]$
```

- **Install the Nginx server to deploy the application on the web.**

```
[ec2-user@ip-192-168-0-20 web-tier]$ sudo amazon-linux-extras install nginx1 -y
sudo: amazon-linux-extras: command not found
[ec2-user@ip-192-168-0-20 web-tier]$ sudo yum install nginx -y
Amazon Linux 2023 Kernel Livepatch repository                                    209 kB/s |  30 kB     00:00
Dependencies resolved.
================================================================================================================
 Package                    Architecture        Version                          Repository                Size
================================================================================================================
Installing:
 nginx                      x86_64              1:1.28.0-1.amzn2023.0.2          amazonlinux               33 k
Installing dependencies:
 generic-logos-httpd        noarch              18.0.0-12.amzn2023.0.3           amazonlinux               19 k
 gperftools-libs            x86_64              2.9.1-1.amzn2023.0.3             amazonlinux              308 k
 libunwind                  x86_64              1.4.0-5.amzn2023.0.3             amazonlinux               66 k
 nginx-core                 x86_64              1:1.28.0-1.amzn2023.0.2          amazonlinux              686 k
 nginx-filesystem           noarch              1:1.28.0-1.amzn2023.0.2          amazonlinux              9.6 k
 nginx-mimetypes            noarch              2.1.49-3.amzn2023.0.3            amazonlinux               21 k

Transaction Summary
================================================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                                 1.8 MB/s |  66 kB     00:00
(2/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm                           6.2 MB/s | 308 kB     00:00
(3/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                     362 kB/s |  19 kB     00:00
(4/7): nginx-1.28.0-1.amzn2023.0.2.x86_64.rpm                                    1.1 MB/s |  33 kB     00:00
(5/7): nginx-filesystem-1.28.0-1.amzn2023.0.2.noarch.rpm                         427 kB/s | 9.6 kB     00:00
(6/7): nginx-core-1.28.0-1.amzn2023.0.2.x86_64.rpm                                17 MB/s | 686 kB     00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm                          768 kB/s |  21 kB     00:00
----------------------------------------------------------------------------------------------------------------
Total                                                                            8.5 MB/s | 1.1 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                        1/1
  Running scriptlet: nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch                                        1/7
  Installing       : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch                                        1/7
```

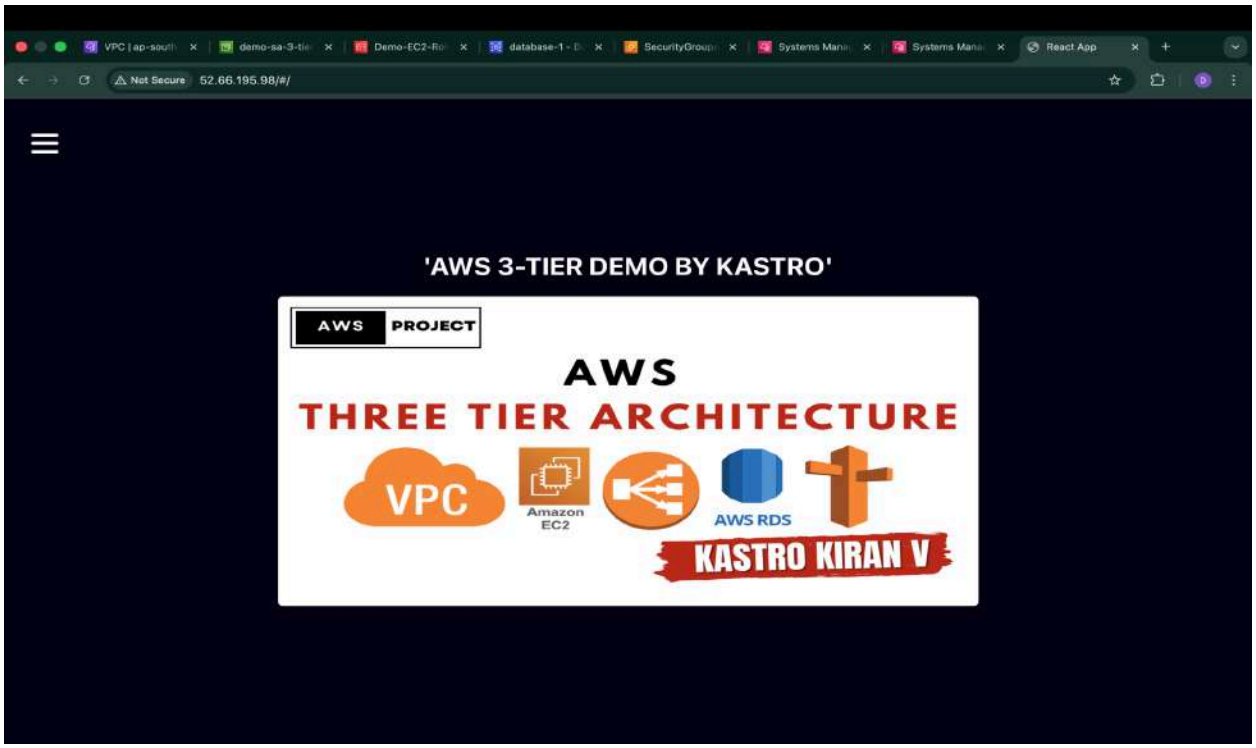
```
[ec2-user@ip-192-168-0-20 web-tier]$ cd /etc/nginx
[ec2-user@ip-192-168-0-20 nginx]$ ls
conf.d     fastcgi.conf          fastcgi_params          koi-utf  mime.types          nginx.conf          scgi_params          uwsgi_params          win-utf
default.d  fastcgi.conf.default  fastcgi_params.default  koi-win  mime.types.default  nginx.conf.default  scgi_params.default  uwsgi_params.default
[ec2-user@ip-192-168-0-20 nginx]$
```

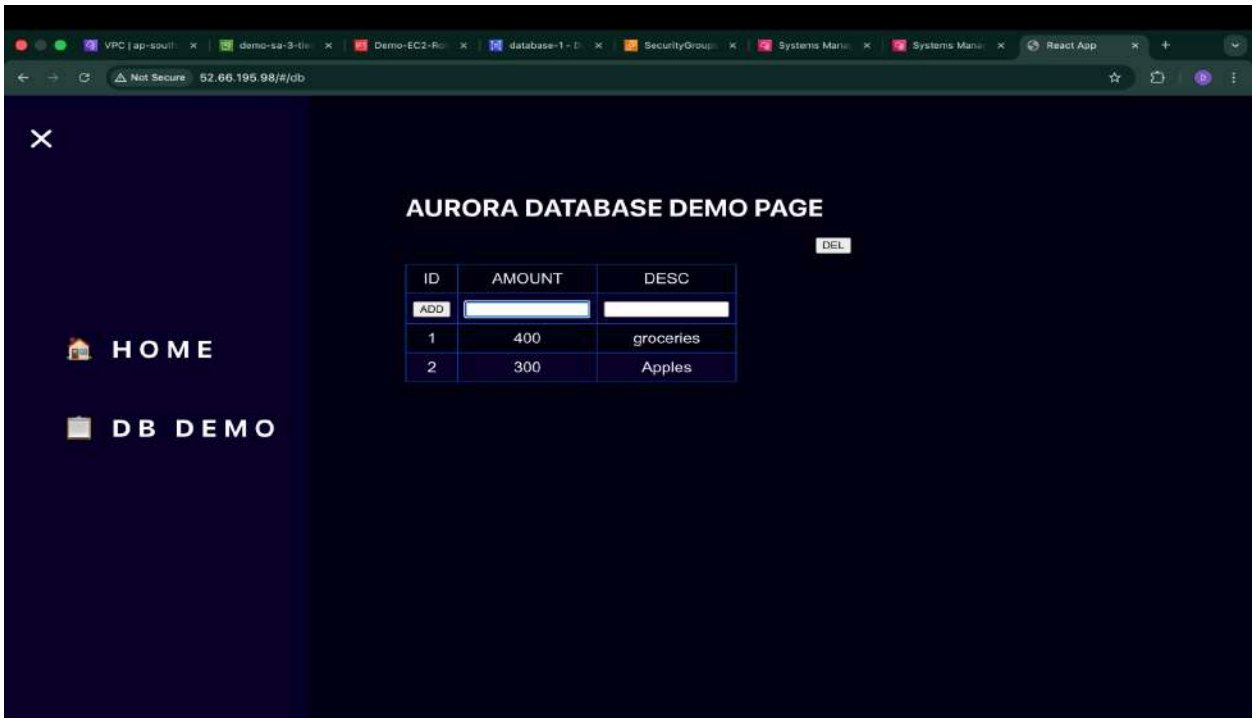
```
[ec2-user@ip-192-168-0-20 web-tier]$ cd /etc/nginx
[ec2-user@ip-192-168-0-20 nginx]$ ls
conf.d     fastcgi.conf          fastcgi_params          koi-utf  mime.types          nginx.conf          scgi_params          uwsgi_params          win-utf
default.d  fastcgi.conf.default  fastcgi_params.default  koi-win  mime.types.default  nginx.conf.default  scgi_params.default  uwsgi_params.default
[ec2-user@ip-192-168-0-20 nginx]$ sudo rm nginx.conf
[ec2-user@ip-192-168-0-20 nginx]$ ls
conf.d     fastcgi.conf          fastcgi_params          koi-utf  mime.types          nginx.conf.default  scgi_params.default  uwsgi_params.default
default.d  fastcgi.conf.default  fastcgi_params.default  koi-win  mime.types.default  scgi_params         uwsgi_params         win-utf
[ec2-user@ip-192-168-0-20 nginx]$ sudo aws s3 cp s3://demo-sa-3-tier-project/application-code/nginx.conf .
download: s3://demo-sa-3-tier-project/application-code/nginx.conf to ./nginx.conf
[ec2-user@ip-192-168-0-20 nginx]$ sudo service nginx restart
Redirecting to /bin/systemctl restart nginx.service
[ec2-user@ip-192-168-0-20 nginx]$
```

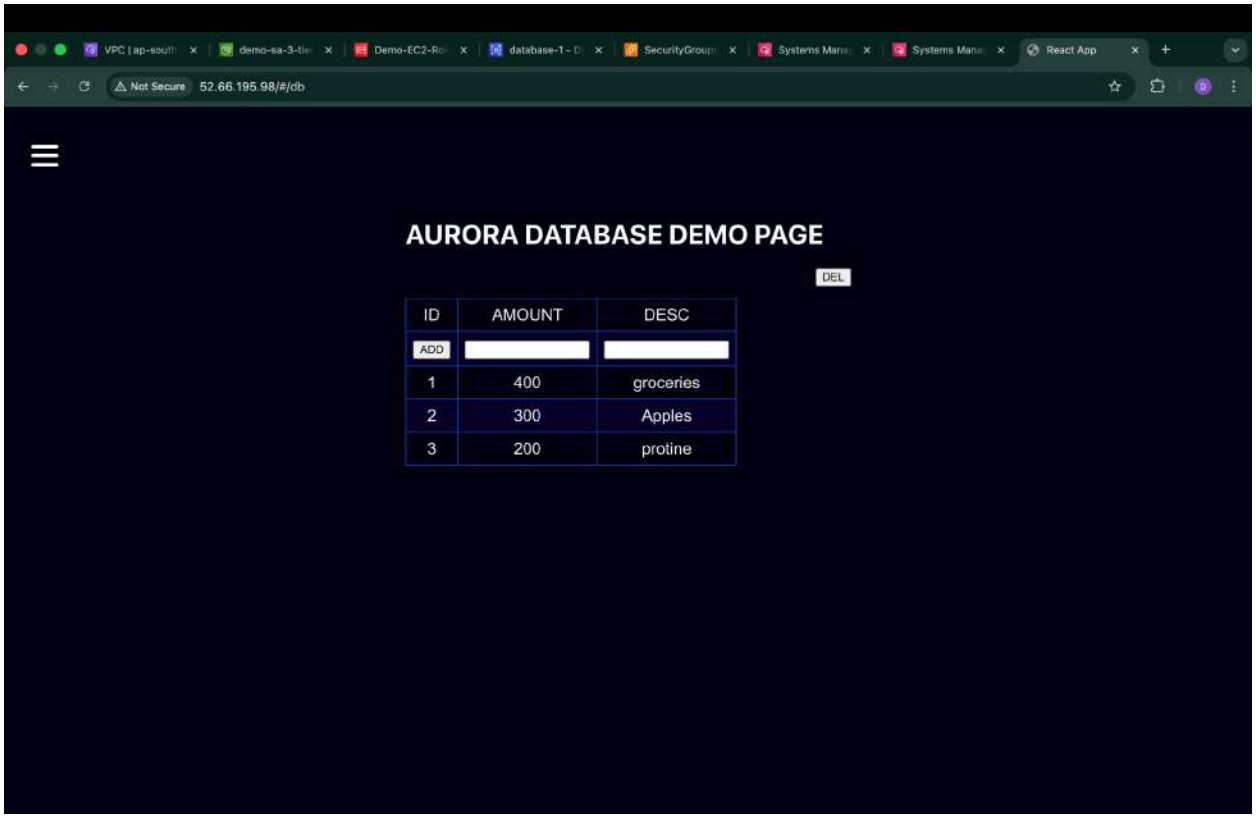
- **Enable and start the Nginx service.**

```
[ec2-user@ip-192-168-0-20 nginx]$ chmod -R 755 /home/ec2-user
[ec2-user@ip-192-168-0-20 nginx]$ sudo chkconfig nginx on
Note: Forwarding request to 'systemctl enable nginx.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-192-168-0-20 nginx]$
```

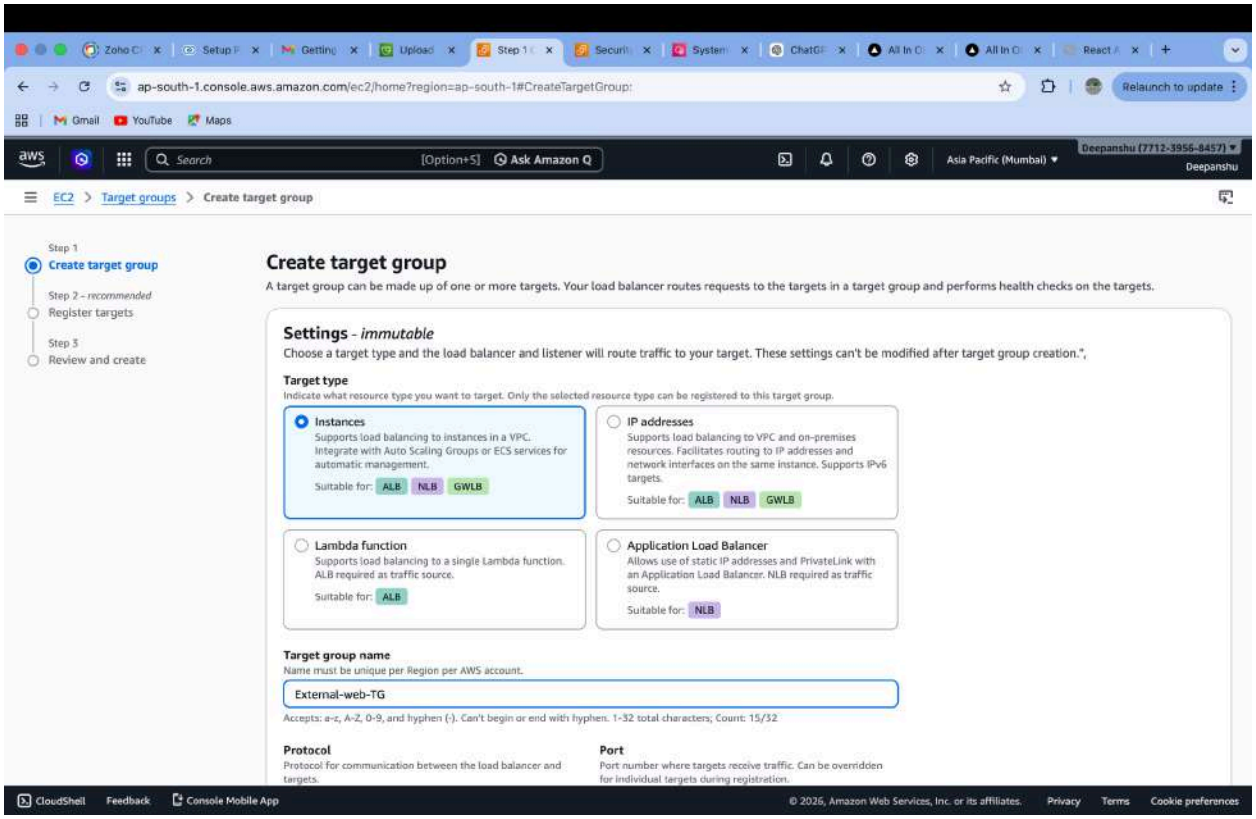
- **The application can now be accessed through the public IP address.**






VPC | ap-south
demo-sa-3-tier
Demo-EC2-Ro
database-1 - D
SecurityGroup
Systems Mana
Systems Mana
React App
Not Secure
52.66.195.98/#/db
AURORA DATABASE DEMO PAGE
DEL
ID
AMOUNT
DESC
ADD
1
400
groceries
2
300
Apples
3
200
protine

Route 53 Setup



ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - *recommended*
Register targets

Step 3
Review and create

Register targets - *recommended*

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/2)

Filter instances

Instance ID	Name	State	Security groups	Zone
i-076392f5dfc926440	Web-Tier-Instance	Running	Web-SG	ap-south
i-097e4ae6b82e7dab4	ApptierInstance	Running	App-SG	ap-south

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

EC2 > Target groups > Create target group

0 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Remove all pending

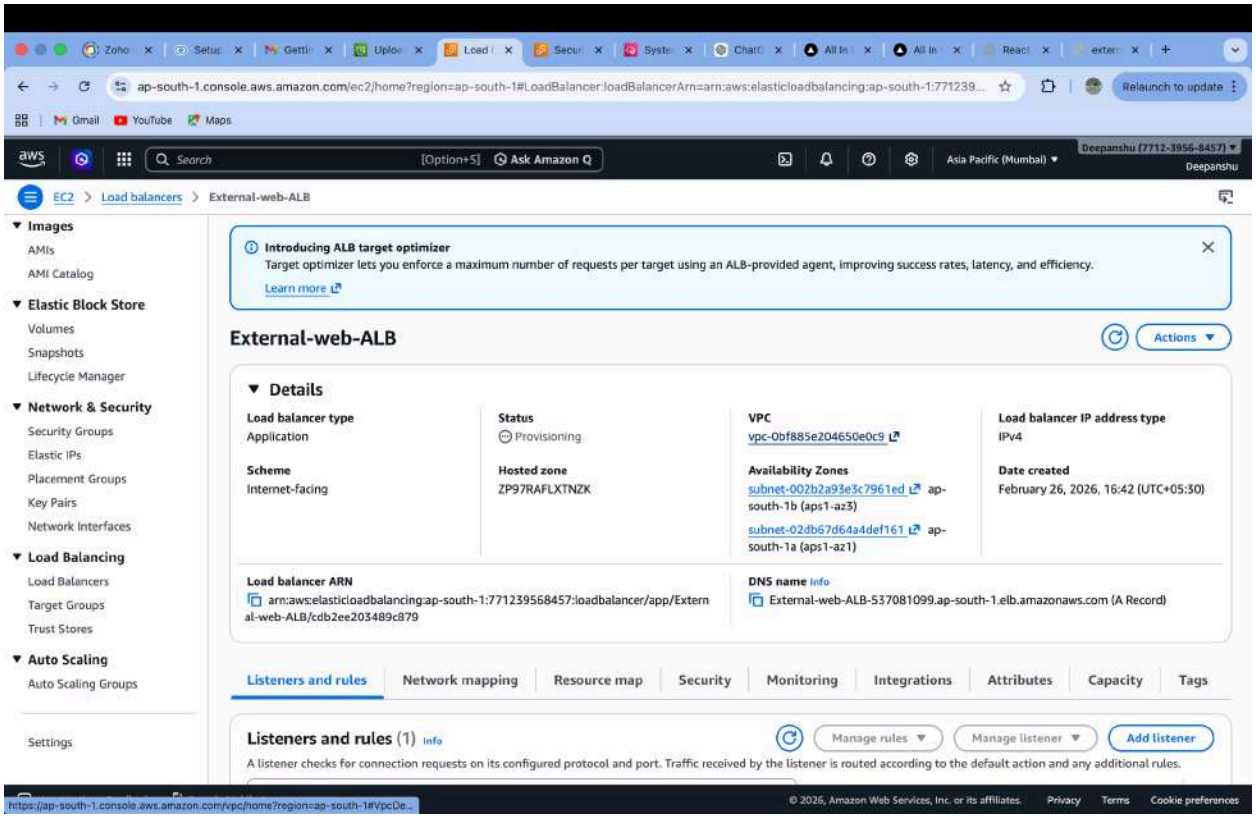
Filter targets

Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-076392f5dfc926440	Web-Tier-Instance	80	Running	Web-SG	ap-south-1a	192.168.0.40	subnet-02

1 pending

Cancel Previous Next


ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:ap-south-1:771239...
Relaunch to update
Gmail
YouTube
Maps
Search
[Option+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Deepanshu (7712-3956-8457)
Deepanshu
EC2 > Load balancers > External-web-ALB
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Introducing ALB target optimizer
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency.
Learn more
External-web-ALB
Actions
Details
Load balancer type
Application
Status
Provisioning
VPC
vpc-0bf885e204650e0c9
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
ZP97RAFLXTNZK
Availability Zones
subnet-002b2a93e3c7961ed ap-south-1b (aps1-az3)
subnet-02db67d64a4def161 ap-south-1a (aps1-az1)
Date created
February 26, 2026, 16:42 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:ap-south-1:771239568457:loadbalancer/app/External-web-ALB/cdb2ee203489c879
DNS name Info
External-web-ALB-537081099.ap-south-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes
Capacity
Tags
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.
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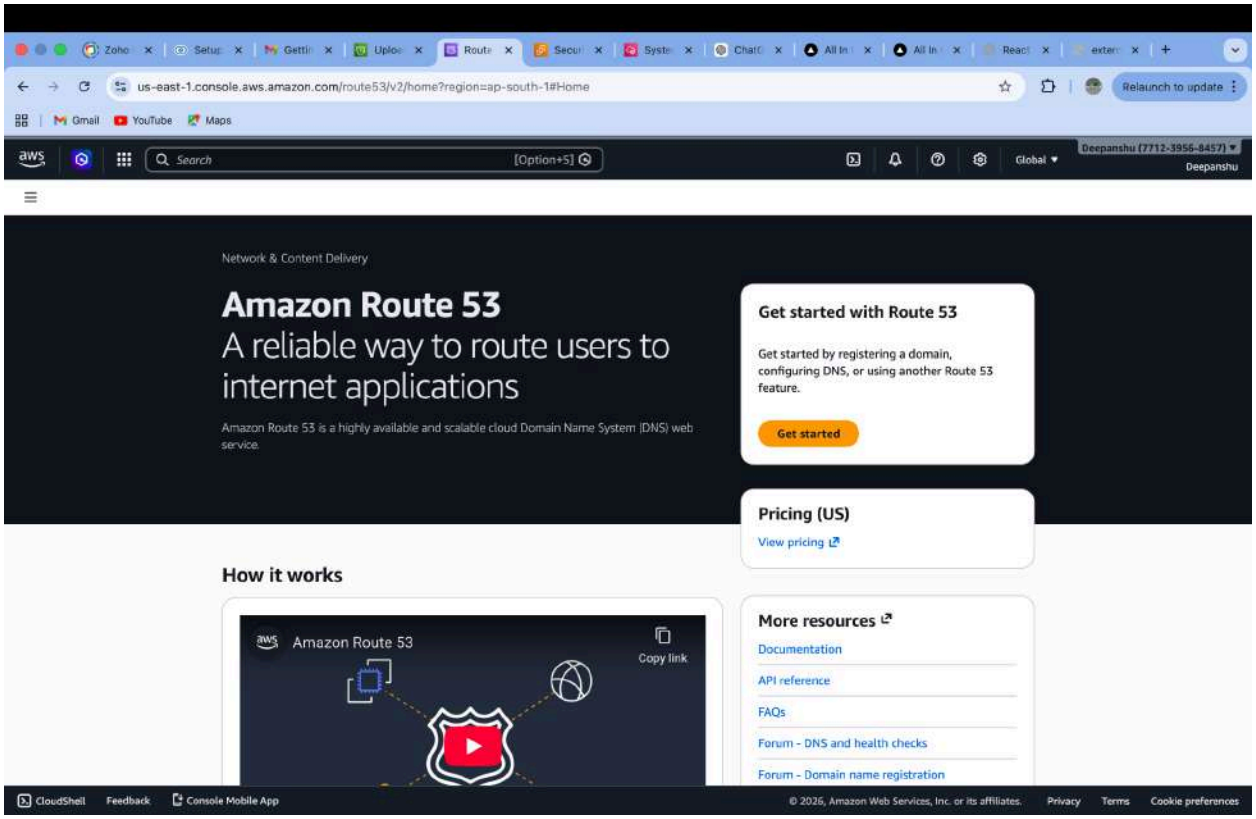
Cookie preferences


us-east-1.console.aws.amazon.com/route53/v2/home?region=ap-south-1#Home
Relaunch to update
Gmail
YouTube
Maps
Search
[Option+S]
Global
Deepanshu (7712-3956-8457)
Deepanshu
Network & Content Delivery
Amazon Route 53
A reliable way to route users to internet applications
Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service.
Get started with Route 53
Get started by registering a domain, configuring DNS, or using another Route 53 feature.
Get started
Pricing (US)
View pricing
How it works
Amazon Route 53
Copy link
More resources
Documentation
API reference
FAQs
Forum - DNS and health checks
Forum - Domain name registration
CloudShell
Feedback
Console Mobile App
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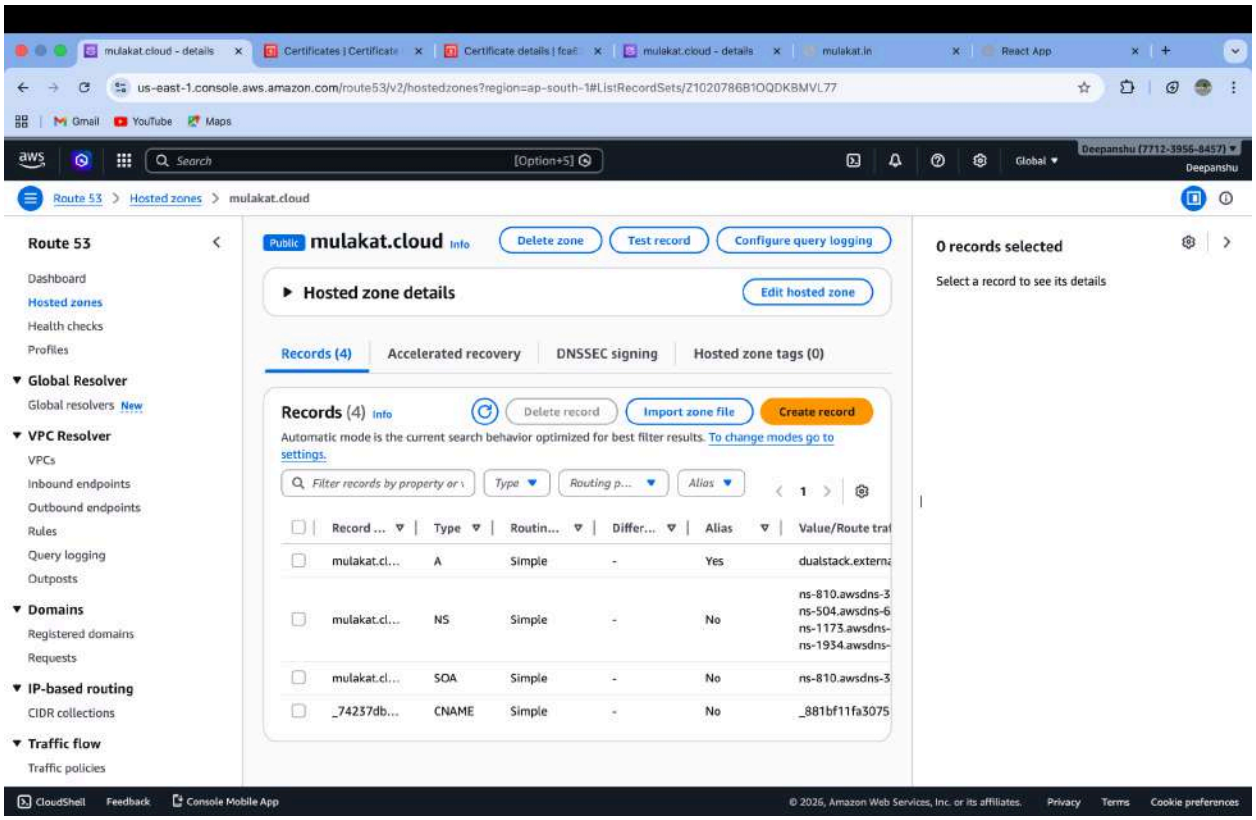
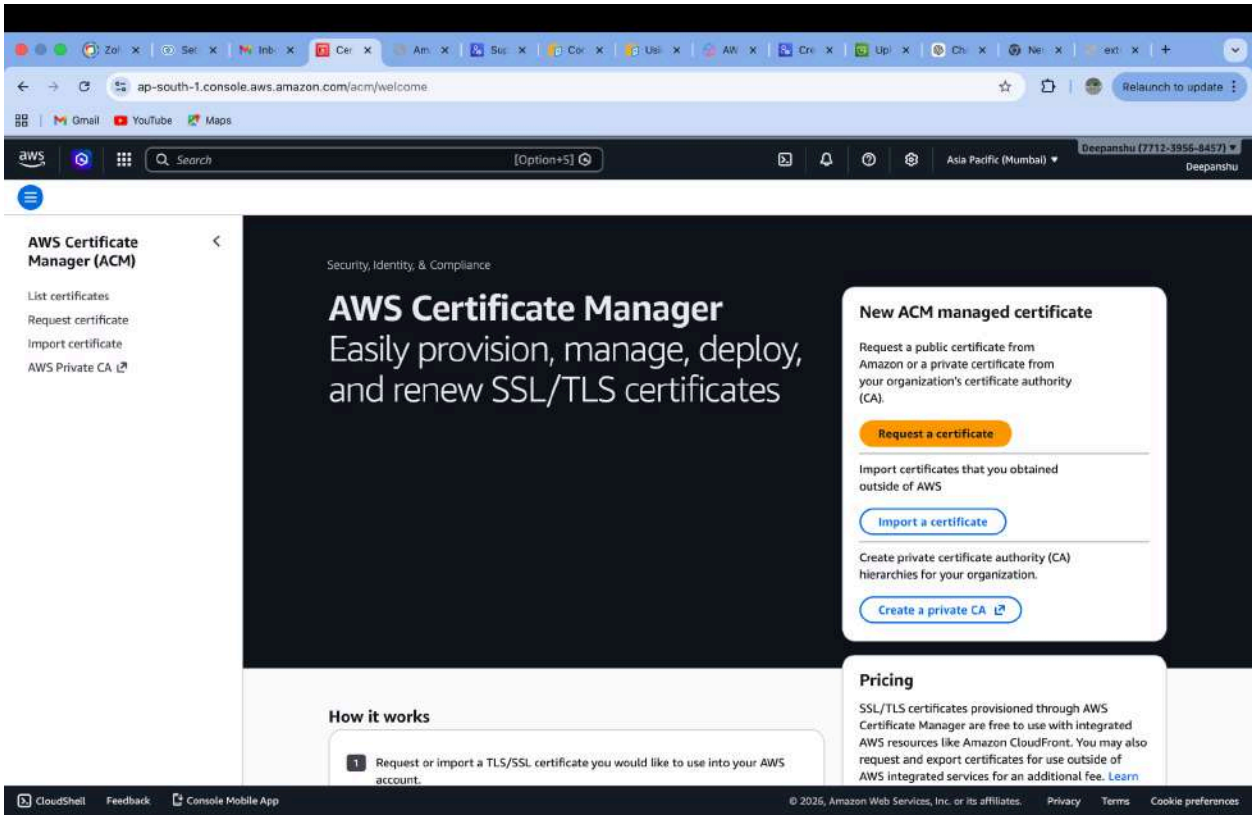
Terms
Cookie preferences

I am using Mulakat.cloud





ap-south-1.console.aws.amazon.com/acm/certificates/request/public

AWS Certificate Manager > Certificates > Request certificate > Request public certificate

Allow export Info

- Disable export
  Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.
- Enable export
  Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

Validation method Info

Select a method for validating domain ownership.

- DNS validation - recommended
  Choose this option if you are authorized to modify the DNS configuration for the domains in your certificate request.
- Email validation
  Choose this option if you do not have permission or cannot obtain permission to modify the DNS configuration for the domains in your certificate request.

Key algorithm Info

Select an encryption algorithm. Some algorithms may not be supported by all AWS services.

- RSA 2048
  RSA is the most widely used key type.
- ECDSA P 256
  Equivalent in cryptographic strength to RSA 3072.
- ECDSA P 384
  Equivalent in cryptographic strength to RSA 7680.

Tags Info

No tags associated with the resource.

ap-south-1.console.aws.amazon.com/acm/certificates/5ac55a78-a864-4f02-9797-aac36217da38

AWS Certificate Manager > Certificates > 5ac55a78-a864-4f02-9797-aac36217da38

AWS Certificate Manager (ACM)

- List certificates
- Request certificate
- Import certificate
- AWS Private CA

Successfully requested certificate with ID 5ac55a78-a864-4f02-9797-aac36217da38
A certificate request with a status of pending validation has been created. Further action is needed to complete the validation and approval of the certificate.

View certificate

Dismiss message

5ac55a78-a864-4f02-9797-aac36217da38

Delete

Certificate status

Identifier
5ac55a78-a864-4f02-9797-aac36217da38

Status
Pending validation Info

ARN
arn:aws:acm:ap-south-1:771239568457:certificate/5ac55a78-a864-4f02-9797-aac36217da38

Type
Amazon Issued

Domains (1)

Domain	Status	Renewal status	Type	CNAME name
Loading				

Details

ap-south-1.console.aws.amazon.com/acm/certificates/5ac55a78-a864-4f02-9797-aac36217da38

AWS Certificate Manager > Certificates > 5ac55a78-a864-4f02-9797-aac36217da38

AWS Certificate Manager (ACM)

- List certificates
- Request certificate
- Import certificate
- AWS Private CA

Identifier
5ac55a78-a864-4f02-9797-aac36217da38

Status
Pending validation Info

ARN
arn:aws:acm:ap-south-1:771239568457:certificate/5ac55a78-a864-4f02-9797-aac36217da38

Type
Amazon Issued

Domains (1)

Create records in Route 53 | Export to CSV

Renewal status	Type	CNAME name	
-	CNAME	_3e8c05c3afcffabf7ec84d6e4e58008c.mulakat.in.	_fc2e3e21d9a27ad0dfe7960e2eafbe14.jkddzztszr validations.aws.

Copied to clipboard

Details

In use No	**Serial number** N/A	**Requested at** February 26, 2026, 20:11:05 (UTC+5:30)	**Renewal eligibility** Ineligible
Domain name mulakat.in	**Public key info** RSA 2048	**Issued at** N/A	**Export option** Disabled
Number of additional names 0	**Signature algorithm** SHA-256 with RSA	**Not before** N/A	

us-east-1.console.aws.amazon.com/route53/v2/hostedzones?region=ap-south-1#ListRecordSets/Z1020786B1OQDKBMVL77

Route 53 > Hosted zones > mulakat.cloud

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles
- Global Resolver
  - Global resolvers New
- VPC Resolver
  - VPCs
  - Inbound endpoints
  - Outbound endpoints
  - Rules
  - Query logging
  - Outposts
- Domains
  - Registered domains
  - Requests
- IP-based routing
  - CIDR collections
- Traffic flow
  - Traffic policies

Public **mulakat.cloud** Info | Delete zone | Test record | Configure query logging

Hosted zone details | Edit hosted zone

Records (4) | Accelerated recovery | DNSSEC signing | Hosted zone tags (0)

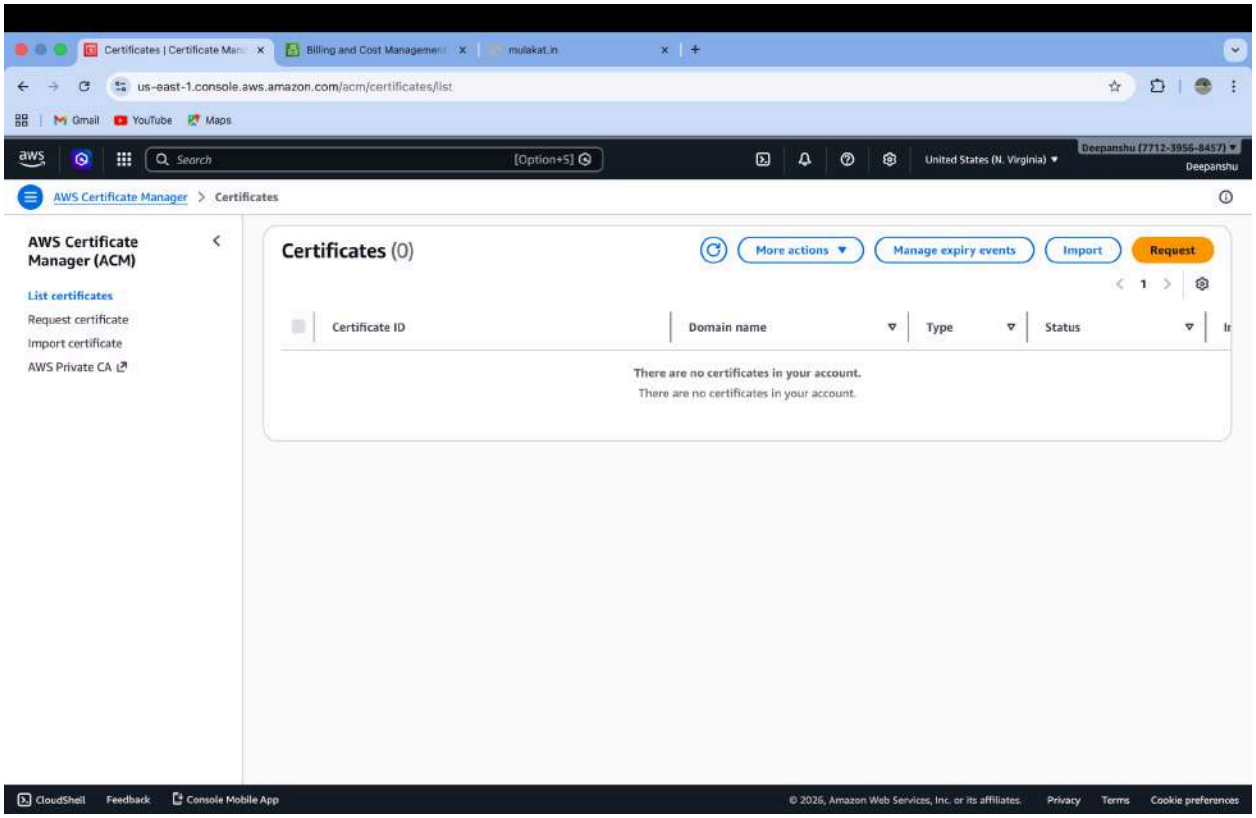
Records (4) Info | Delete record | Import zone file | Create record

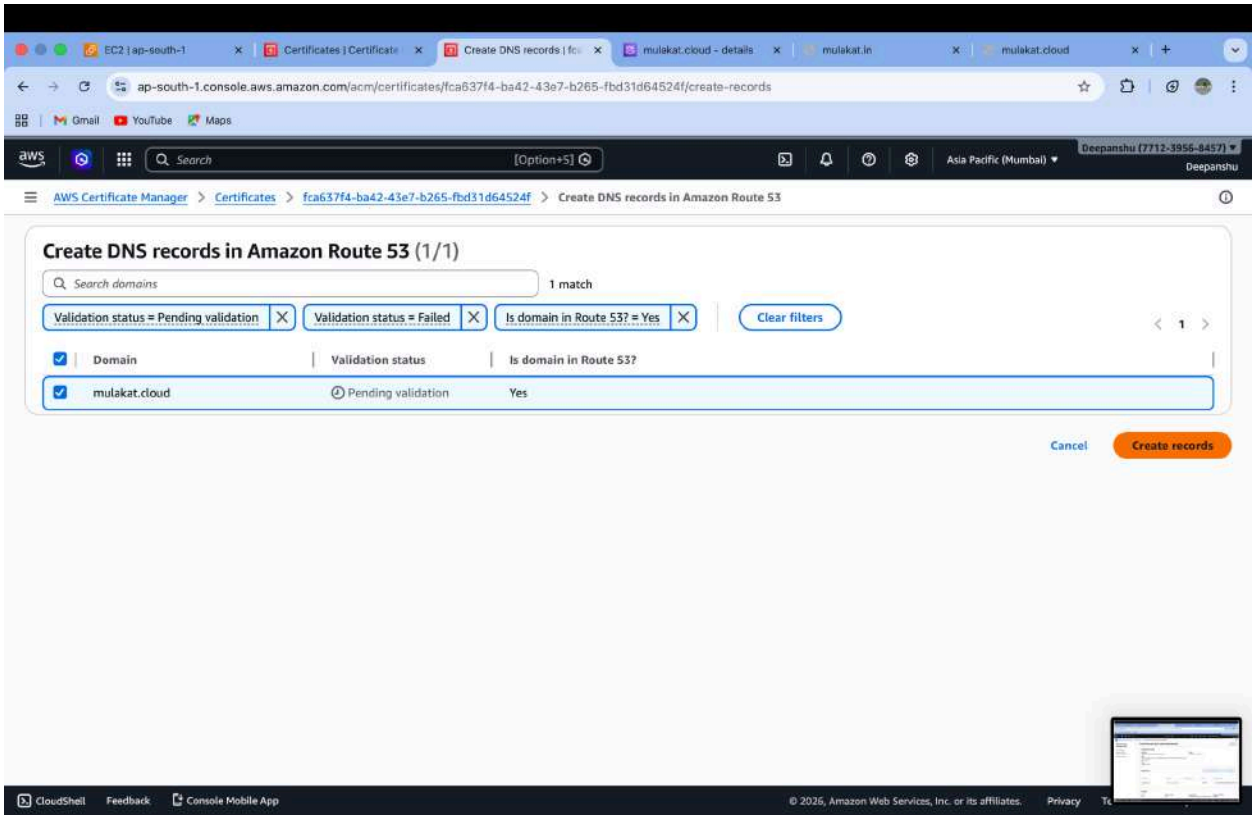
Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Record ...	Type	Routin...	Differ...	Alias	Value/Route traf
mulakat.cl...	A	Simple	-	Yes	dualstack.externa
mulakat.cl...	NS	Simple	-	No	ns-810.awsdns-3 ns-504.awsdns-6 ns-1173.awsdns- ns-1934.awsdns-
mulakat.cl...	SOA	Simple	-	No	ns-810.awsdns-3
_74237db...	CNAME	Simple	-	No	_881bf11fa3075

0 records selected

Select a record to see its details


Certificates | Certificate Man
Billing and Cost Management
mulakat.in
us-east-1.console.aws.amazon.com/acm/certificates/list
Gmail
YouTube
Maps
Search
[Option+5]
United States (N. Virginia)
Deepanshu (7712-3956-8457)
Deepanshu
AWS Certificate Manager > Certificates
AWS Certificate Manager (ACM)
List certificates
Request certificate
Import certificate
AWS Private CA
Certificates (0)
More actions
Manage expiry events
Import
Request
Certificate ID
Domain name
Type
Status
There are no certificates in your account.
There are no certificates in your account.
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


EC2 | ap-south-1
Certificates | Certificate
Create DNS records | fca
mulakat.cloud - details
mulakat.in
mulakat.cloud
ap-south-1.console.aws.amazon.com/acm/certificates/fca637f4-ba42-43e7-b265-fbd31d64524f/create-records
Gmail
YouTube
Maps
Search
[Option+5]
Asia Pacific (Mumbai)
Deepanshu (7712-3956-8457)
Deepanshu
AWS Certificate Manager > Certificates > fca637f4-ba42-43e7-b265-fbd31d64524f > Create DNS records in Amazon Route 53
Create DNS records in Amazon Route 53 (1/1)
Search domains
1 match
Validation status = Pending validation
Validation status = Failed
Is domain in Route 53? = Yes
Clear filters
Domain
Validation status
Is domain in Route 53?
mulakat.cloud
Pending validation
Yes
Cancel
Create records
CloudShell
Feedback
Console Mobile App
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Privacy

EC2 | ap-south-1 | Certificates | Certificate | Certificate details | fca6 | mulakat.cloud - details | mulakat.in | mulakat.cloud

ap-south-1.console.aws.amazon.com/acm/certificates/fca637f4-ba42-43e7-b265-fbd31d64524f

Gmail YouTube Maps

aws Search [Option+S] Asia Pacific (Mumbai) Deepanshu (7712-3956-8457) Deepanshu

AWS Certificate Manager > Certificates > fca637f4-ba42-43e7-b265-fbd31d64524f

AWS Certificate Manager (ACM)

- List certificates
- Request certificate
- Import certificate
- AWS Private CA

fca637f4-ba42-43e7-b265-fbd31d64524f

Delete

Certificate status

Identifier
fca637f4-ba42-43e7-b265-fbd31d64524f

Status
Pending validation Info

ARN
arn:aws:acm:ap-south-1:771239568457:certificate/fca637f4-ba42-43e7-b265-fbd31d64524f

Type
Amazon Issued

Domains (1)

Create records in Route 53 | Export to CSV

Domain	Status	Renewal status	Type	CNAME name
mulakat.cloud	Pending validation	-	CNAME	_74237db5fe78cb736ffd6c02933c5f

Details

In use	Serial number	Requested at	Renewal eligibility
No	N/A	February 27, 2026, 13:41:56 (UTC+5:30)	Ineligible

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Certificate details | fca6 | Certificates | Certificate | Certificate details | fca6 | mulakat.cloud - details | mulakat.in | React App

ap-south-1.console.aws.amazon.com/acm/certificates/fca637f4-ba42-43e7-b265-fbd31d64524f

Gmail YouTube Maps

aws Search [Option+S] Asia Pacific (Mumbai) Deepanshu (7712-3956-8457) Deepanshu

AWS Certificate Manager > Certificates > fca637f4-ba42-43e7-b265-fbd31d64524f

AWS Certificate Manager (ACM)

- List certificates
- Request certificate
- Import certificate
- AWS Private CA

fca637f4-ba42-43e7-b265-fbd31d64524f

Delete

Certificate status

Identifier
fca637f4-ba42-43e7-b265-fbd31d64524f

Status
Issued

ARN
arn:aws:acm:ap-south-1:771239568457:certificate/fca637f4-ba42-43e7-b265-fbd31d64524f

Type
Amazon Issued

Domains (1)

Create records in Route 53 | Export to CSV

Domain	Status	Renewal status	Type	CNAME name
mulakat.cloud	Success	-	CNAME	_74237db5fe78cb736ffd6c02933c5f

Details

In use	Serial number	Requested at	Renewal eligibility
Yes	0e:ed:69:71:9a:bf:f6:82:f9:30:80:ed:a9:3c:aa:4c	February 27, 2026, 13:41:56 (UTC+5:30)	Eligible

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Load balancer details | EC2 | Billing and Cost Management | mulakat.in

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:ap-south-1:771239568457:loadbalanc...

Gmail YouTube Maps

aws Search [Option+S] Asia Pacific (Mumbai) Deepanshu (7712-3956-8457) Deepanshu

EC2 > Load balancers > External-web-ALB

EC2

Dashboard
AWS Global View
Events
▼ Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
▼ Images
AMIs
AMI Catalog
▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
▼ Network & Security

Load balancer type
Application

Status
Active

VPC
vpc-0bf885e204650e0c9

Load balancer IP address type
IPv4

Scheme
Internet-facing

Hosted zone
ZP97RAFLXTNZK

Availability Zones
subnet-002b2a93e3c7961ed ap-south-1b (aps1-az3)
subnet-02db67d64a4def161 ap-south-1a (aps1-az1)

Date created
February 26, 2026, 16:42 (UTC+05:30)

Load balancer ARN
arn:aws:elasticloadbalancing:ap-south-1:771239568457:loadbalancer/app/External-web-ALB/cdb2ee203489c879

DNS name Info
External-web-ALB-537081099.ap-south-1.elb.amazonaws.com (A Record)

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Listeners and rules (1) Info

Manage rules | Manage listener | Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL
HTTP:80	• Forward to target group External-web-TG: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicab

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Load balancer details | E | Certificates | Certificate | Certificate details | fca | mulakat.cloud - details | mulakat.in | React App

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:ap-south-1:771239568457:loadb...

Gmail YouTube Maps

aws Search [Option+S] Asia Pacific (Mumbai) Deepanshu (7712-3956-8457) Deepanshu

EC2 > Load balancers > External-web-ALB

▼ Images
AMIs
AMI Catalog
▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
▼ Load Balancing
Load Balancers
Target Groups
Trust Stores
▼ Auto Scaling
Auto Scaling Groups

Settings

Internet-facing
ZP97RAFLXTNZK
subnet-002b2a93e3c7961ed ap-south-1b (aps1-az3)
subnet-02db67d64a4def161 ap-south-1a (aps1-az1)
February 26, 2026, 16:42 (UTC+05:30)

Load balancer ARN
arn:aws:elasticloadbalancing:ap-south-1:771239568457:loadbalancer/app/External-web-ALB/cdb2ee203489c879

DNS name Info
External-web-ALB-537081099.ap-south-1.elb.amazonaws.com (A Record)

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Listeners and rules (2) Info

Manage rules | Manage listener | Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL
HTTP:80	• Forward to target group External-web-TG: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicab
HTTPS:443	• Forward to target group External-web-TG: 1 (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-TLS13-1-2-...	mulakat.clou

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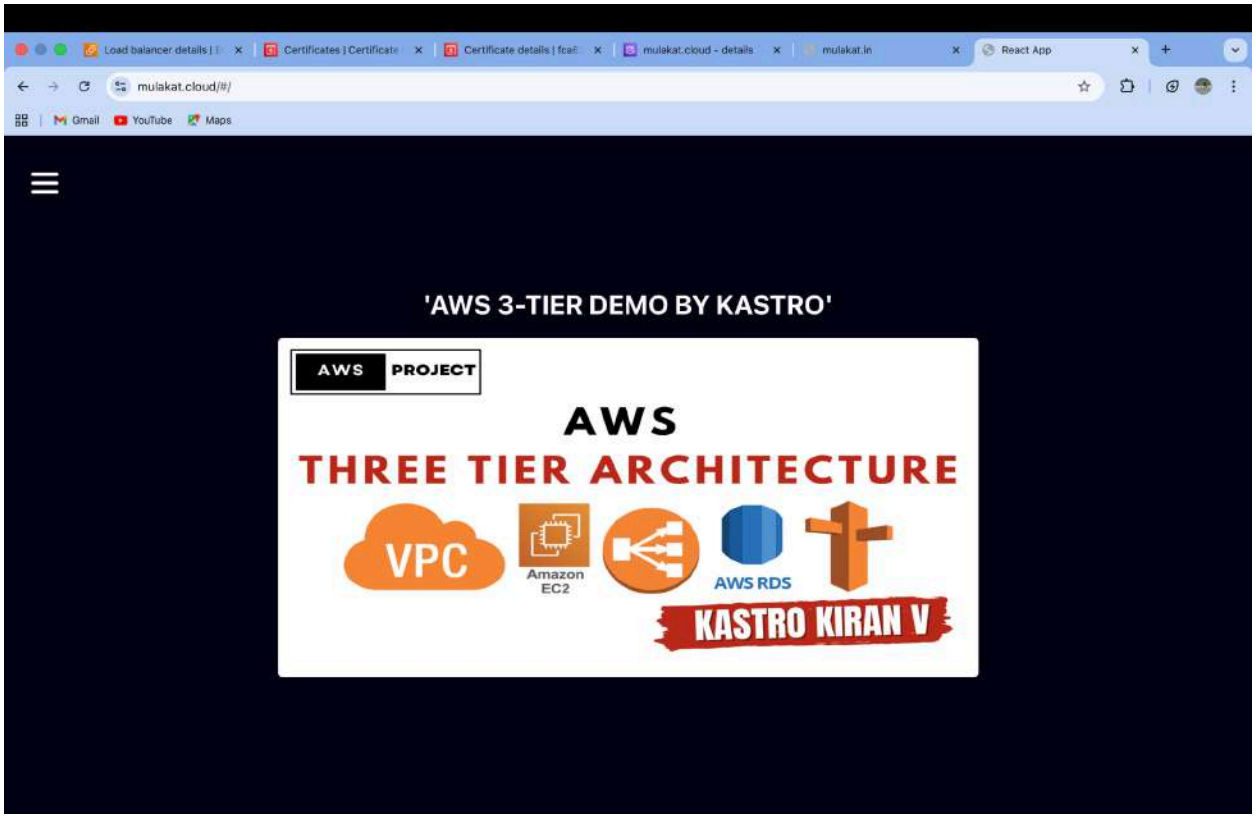
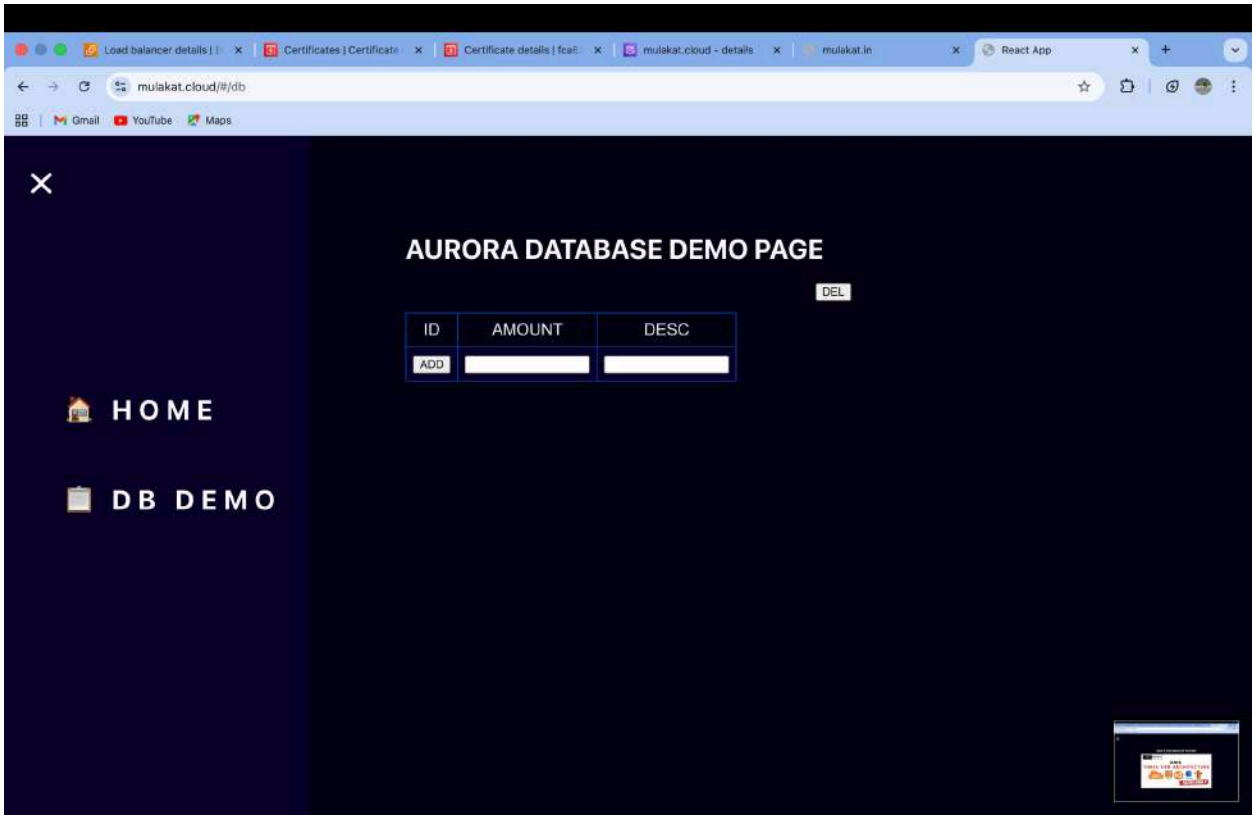

mulakat.cloud/#/
'AWS 3-TIER DEMO BY KASTRO'
AWS PROJECT
AWS
THREE TIER ARCHITECTURE
VPC
Amazon EC2
AWS RDS
KASTRO KIRAN V


mulakat.cloud/#/db
AURORA DATABASE DEMO PAGE
DEL
ID
AMOUNT
DESC
ADD
HOME
DB DEMO